

1510.06699 — formula report

1063 inline formulas (section omitted; all rendered), 279 display equations, 7 tables, 38 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00241 (A5) [conf 0.019]	047	■ 0.019 ● S -2	$\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i\gamma^5\right)}{\rho^2+\beta^2}$	$\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i\gamma^5\right)}{\rho^2+\beta^2}.$	$\bar{\psi}=\frac{\bar{\psi}(\rho+\beta i\gamma^5)\not{I}}{\rho^2+\beta^2}.$
1510.06699_ E00248 (B3)	049	■ 0.183 ● N -1	$L_{\mathrm{G}}=-\kappa^{-1}\left(\Lambda+a^0\mathcal{R}\right)+L_0\mathcal{R}^2,$	$L_{\mathrm{G}}=-\kappa^{-1}\left(\Lambda+a^0\mathcal{R}\right)+L_0\mathcal{R}^2,$	$L_{\mathrm{G}}=-\kappa^{-1}\left(\Lambda+a^0\mathcal{R}\right)+L_0\mathcal{R}^2,$
1510.06699_ E00210 (235)	039	■ 0.264 ● C +5	$\begin{aligned} & \hleft(s_{\mathrm{R}^2})_{ab}^c=4\alpha_1\left(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger}+\frac{1}{2}\mathcal{T}_{ab}^{\dagger c}\right)\mathcal{R}_{ab}^{\dagger} \\ & +4\alpha_2\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}_{[a}^{\dagger d}\delta_{b]}^e \\ & +4\alpha_3\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger d}_{[a}\delta_{b]}^e \\ & +4\alpha_4\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}_{abde}^{\dagger} \\ & +4\alpha_5\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger[d}_{[ab]} \\ & +4\alpha_6\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger de}_{ab}, \\ & h(s_{\mathcal{H}^2})_{ab}^c=\frac{2}{3}\xi\eta_{f[b}\delta_{a]}^c\left(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef}-\frac{1}{2}\mathcal{T}^{\dagger f}_{de}\mathcal{H}^{\dagger de}\right), \end{aligned}$	$\begin{aligned} & h(s_{\mathcal{R}^2})_{ab}^c=4\alpha_1\left(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger}+\frac{1}{2}\mathcal{T}_{ab}^{\dagger c}\right)\mathcal{R}^{\dagger} \\ & +4\alpha_2\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}_{[a}^{\dagger d}\delta_{b]}^e \\ & +4\alpha_3\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger d}_{[a}\delta_{b]}^e \\ & +4\alpha_4\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}_{abde}^{\dagger} \\ & +4\alpha_5\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger[d}_{[ab]} \\ & +4\alpha_6\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}_{de}^{\dagger c}\right)\mathcal{R}^{\dagger de}_{ab}, \\ & h(s_{\mathcal{H}^2})_{ab}^c=\frac{2}{3}\xi\eta_{f[b}\delta_{a]}^c\left(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef}-\frac{1}{2}\mathcal{T}^{\dagger f}_{de}\mathcal{H}^{\dagger de}\right), \end{aligned}$	$\begin{aligned} h(s_{\mathcal{R}^2})_{ab}^c &= 4\alpha_1(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{ab}^{\dagger c})\mathcal{R}^{\dagger} \\ & + 4\alpha_2(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{de}^{\dagger c})\mathcal{R}_{[a}^{\dagger d}\delta_{b]}^e \\ & + 4\alpha_3(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{de}^{\dagger c})\mathcal{R}^{\dagger d}_{[a}\delta_{b]}^e \\ & + 4\alpha_4(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{de}^{\dagger c})\mathcal{R}_{ab}^{\dagger de} \\ & + 4\alpha_5(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{de}^{\dagger c})\mathcal{R}^{\dagger[d}_{[ab]} \\ & + 4\alpha_6(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}_{de}^{\dagger c})\mathcal{R}^{\dagger de}_{ab}, \\ h(s_{\mathcal{H}^2})_{ab}^c &= \tfrac{2}{3}\xi\eta_{f[b}\delta_{a]}^c(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef} - \tfrac{1}{2}\mathcal{T}^{\dagger f}_{de}\mathcal{H}^{\dagger de}), \end{aligned}$
1510.06699_ E00192 (208)	036	■ 0.344 ● N +1	$\begin{aligned} & D_{\mu}^{\dagger}J^a=\partial_{\mu}J^a-w\left(V_{\mu}+\frac{1}{3}T_{\mu}\right)J^a+A_{b\mu}^{\dagger a}J^b \\ & =\partial_{\mu}^{\dagger}J^a+A_{b\mu}^{\dagger a}J^b \end{aligned}$	$\begin{aligned} D_{\mu}^{\dagger}J^a &= \partial_{\mu}J^a - w\left(V_{\mu} + \frac{1}{3}T_{\mu}\right)J^a + A_{b\mu}^{\dagger a}J^b \\ &= \partial_{\mu}^{\dagger}J^a + A_{b\mu}^{\dagger a}J^b \end{aligned}$	$\begin{aligned} D_{\mu}^{\dagger}J^a &= \partial_{\mu}J^a - w(V_{\mu} + \tfrac{1}{3}T_{\mu})J^a + A^{\dagger a}_{b\mu}J^b \\ &= \partial_{\mu}^{\dagger}J^a + A^{\dagger a}_{b\mu}J^b, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0205 (226)	038	0.356 C +5	$\begin{aligned} L_{\{\varphi\}} &= L_{\{\varphi\}}(\varphi, \partial\varphi, h, \partial h, A, V, \phi) \\ L_{\phi} &= \frac{1}{2} \nu \mathcal{D}_a^{\dagger} \phi \mathcal{D}^{\dagger a} \phi - \lambda \phi^4 \\ L_{\mathcal{R}^{\dagger}} &= -\frac{1}{2} a \mathcal{R}^{\dagger} \\ L_{\mathcal{T}^{\dagger 2}} &= \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}, \end{aligned}$	$\begin{aligned} L_{\varphi} &= L_{\varphi}(\varphi, \partial\varphi, h, \partial h, A, V, \phi) \\ L_{\phi} &= \frac{1}{2} \nu \mathcal{D}_a^{\dagger} \phi \mathcal{D}^{\dagger a} \phi - \lambda \phi^4 \\ L_{\mathcal{R}^{\dagger}} &= -\frac{1}{2} a \mathcal{R}^{\dagger} \\ L_{\mathcal{T}^{\dagger 2}} &= \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}, \end{aligned}$	$\begin{aligned} L_{\varphi} &= L_{\varphi}(\varphi, \partial\varphi, h, \partial h, A, V, \phi), \\ L_{\phi} &= \frac{1}{2} \nu \mathcal{D}_a^{\dagger} \phi \mathcal{D}^{\dagger a} \phi - \lambda \phi^4, \\ L_{\mathcal{R}^{\dagger}} &= -\frac{1}{2} a \mathcal{R}^{\dagger}, \\ L_{\mathcal{T}^{\dagger 2}} &= \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}, \end{aligned}$
1510.06699_ EQ0032 (33)	010	0.377 C +4	$D_{\{\mu\}}^* h_a^{\nu} = \partial_{\mu}^* h_a^{\nu} + \frac{1}{2} A_{\mu}^{cd} (\Sigma_{cd}^1)_a^{b} h_b^{\nu} = \partial_{\mu}^* h_a^{\nu} - A_{a\mu}^b h_b^{\nu},$	$D_{\mu}^* h_a^{\nu} = \partial_{\mu}^* h_a^{\nu} + \frac{1}{2} A_{\mu}^{cd} (\Sigma_{cd}^1)_a^{b} h_b^{\nu} = \partial_{\mu}^* h_a^{\nu} - A_{a\mu}^b h_b^{\nu},$	$D_{\mu}^* h_a^{\nu} = \partial_{\mu}^* h_a^{\nu} + \frac{1}{2} A_{\mu}^{cd} (\Sigma_{cd}^1)_a^{b} h_b^{\nu} = \partial_{\mu}^* h_a^{\nu} - A_{a\mu}^b h_b^{\nu},$
1510.06699_ EQ0010 (11)	008	0.398 W +0	$\begin{aligned} L_{\{\mathrm{KG}\}}^{\prime} &= \frac{1}{2} \gamma^{\prime \mu \nu} \partial_{\mu}^{\prime} \phi^{\prime} \partial_{\nu}^{\prime} \phi^{\prime} - \frac{1}{2} m^2 \phi^{\prime 2} \\ &= \frac{1}{2} e^{2(w-1)\rho} \eta^{\mu \nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 e^{2w\rho} \phi^2. \end{aligned}$	$\begin{aligned} L_{\mathrm{KG}}^{\prime} &= \frac{1}{2} \gamma^{\prime \mu \nu} \partial_{\mu}^{\prime} \phi^{\prime} \partial_{\nu}^{\prime} \phi^{\prime} - \frac{1}{2} m^2 \phi^{\prime 2} \\ &= \frac{1}{2} e^{2(w-1)\rho} \eta^{\mu \nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 e^{2w\rho} \phi^2. \end{aligned}$	$\begin{aligned} L_{\mathrm{KG}}^{\prime} &= \frac{1}{2} \gamma^{\prime \mu \nu} \partial_{\mu}^{\prime} \phi^{\prime} \partial_{\nu}^{\prime} \phi^{\prime} - \frac{1}{2} m^2 \phi^{\prime 2} \\ &= \frac{1}{2} e^{2(w-1)\rho} \eta^{\mu \nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 e^{2w\rho} \phi^2. \end{aligned}$
1510.06699_ EQ0176 (192)	032	0.428 W +0	$\begin{aligned} \mathcal{L}_{\mathrm{M}} &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^{\dagger} \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 \right. \\ &\quad \left. - a \phi^2 \mathcal{R}^{\dagger} + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right], (192) \end{aligned}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^{\dagger} \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 \right. \\ \left. - a \phi^2 \mathcal{R}^{\dagger} + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right], (192)$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^{\dagger} \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 - a \phi^2 \mathcal{R}^{\dagger} + \phi^2 L_{\mathcal{T}^{\dagger 2}}], (192)$
1510.06699_ EQ0229 (263b)	043	0.447 W +0	$\begin{aligned} \{ \}^{\{0\}} \mathcal{M}^{\{\mathrm{R}\}^{\{\prime a \, b\}} \{ \}_{\{c \, d\}}} &= e^{\{-2 \, \rho\}} \left\{ \{ \}^{\{0\}} \mathcal{M}^{\{\mathrm{R}\}^{\{a \, b\}} \{ \}_{\{c \, d\}}} + 2 \, \delta_d^{[a} ({}^0 \mathcal{D}_c - \mathcal{P}_c) \mathcal{P}^{b]} - 2 \, \delta_c^{[a} ({}^0 \mathcal{D}_d - \mathcal{P}_d) \mathcal{P}^{b]} - 2 \, \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\}, \\ {}^0 \mathcal{R}^{\prime a}{}_c &= e^{-2\rho} \{ {}^0 \mathcal{R}^a{}_c - 2 {}^0 \mathcal{D}_c \mathcal{P}^a - \delta_c^a {}^0 \mathcal{D}_b \mathcal{P}^b + 2 \mathcal{P}^a \mathcal{P}_c - 2 \delta_c^a \mathcal{P}^2 \}, \\ {}^0 \mathcal{R}' &= e^{-2\rho} \{ {}^0 \mathcal{R} - 6 {}^0 \mathcal{D}_a \mathcal{P}^a - 6 \mathcal{P}^2 \}, \end{aligned}$	$\begin{aligned} {}^0 \mathcal{R}'^{ab}{}_{cd} &= e^{-2\rho} \{ {}^0 \mathcal{R}^{ab}{}_{cd} + 2 \delta_d^{[a} ({}^0 \mathcal{D}_c - \mathcal{P}_c) \mathcal{P}^{b]} - 2 \delta_c^{[a} ({}^0 \mathcal{D}_d - \mathcal{P}_d) \mathcal{P}^{b]} - 2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \}, \\ {}^0 \mathcal{R}'^a{}_c &= e^{-2\rho} \{ {}^0 \mathcal{R}^a{}_c - 2 {}^0 \mathcal{D}_c \mathcal{P}^a - \delta_c^a {}^0 \mathcal{D}_b \mathcal{P}^b + 2 \mathcal{P}^a \mathcal{P}_c - 2 \delta_c^a \mathcal{P}^2 \}, \\ {}^0 \mathcal{R}' &= e^{-2\rho} \{ {}^0 \mathcal{R} - 6 {}^0 \mathcal{D}_a \mathcal{P}^a - 6 \mathcal{P}^2 \}, \end{aligned}$	$\begin{aligned} {}^0 \mathcal{R}'^{ab}{}_{cd} &= e^{-2\rho} \{ {}^0 \mathcal{R}^{ab}{}_{cd} + 2 \delta_d^{[a} ({}^0 \mathcal{D}_c - \mathcal{P}_c) \mathcal{P}^{b]} - 2 \delta_c^{[a} ({}^0 \mathcal{D}_d - \mathcal{P}_d) \mathcal{P}^{b]} - 2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \}, \\ {}^0 \mathcal{R}'^a{}_c &= e^{-2\rho} \{ {}^0 \mathcal{R}^a{}_c - 2 {}^0 \mathcal{D}_c \mathcal{P}^a - \delta_c^a {}^0 \mathcal{D}_b \mathcal{P}^b + 2 \mathcal{P}^a \mathcal{P}_c - 2 \delta_c^a \mathcal{P}^2 \}, \\ {}^0 \mathcal{R}' &= e^{-2\rho} \{ {}^0 \mathcal{R} - 6 {}^0 \mathcal{D}_a \mathcal{P}^a - 6 \mathcal{P}^2 \}, \end{aligned}$
Conf. MathPix confidence: ≥ 0.9 $0.5\text{--}0.9$ < 0.5 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E09067 (71)	015	■ 0.449 ● N +0	$\mathrm{L}_{\mathrm{D}}=h^{-1}\mathrm{L}_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\dot{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi\right)$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\dot{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi\right),$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\overset{\leftrightarrow}{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi),$
1510.06699_ E0105 (113)	021	■ 0.463 ● W +0	$\begin{aligned} \mathrm{L}_{\mathrm{T}}=&\mathrm{L}_{\mathrm{G}}\left(h,\partial h,\partial^2h,B,\partial B\right)\\ &+\mathrm{L}_{\mathrm{M}}\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2h,B,\partial B\right). \end{aligned}$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}\left(h,\partial h,\partial^2h,B,\partial B\right)\\ +\mathcal{L}_{\mathrm{M}}\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2h,B,\partial B\right).$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,\partial h,\partial^2h,B,\partial B)\\ +\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2h,B,\partial B).$
1510.06699_ E0116 (126)	022	■ 0.469 ● W +0	$\begin{aligned} R_{\sigma\mu\nu} &= 2\left(\partial_{[\mu}\Gamma_{ \sigma \nu]}^{\rho} + \Gamma_{\lambda[\mu}^{\rho}\Gamma_{ \sigma \nu]}^{\lambda}\right) - H_{\mu\nu}\delta_{\sigma}^{\rho}, \\ T^{*\lambda}_{\mu\nu} &= 2\Gamma_{[\nu\mu]}^{\lambda}, \\ H_{\mu\nu} &= 2\partial_{[\mu}B_{\nu]}, \end{aligned}$	$R_{\sigma\mu\nu}^{\rho} = 2\left(\partial_{[\mu}\Gamma_{ \sigma \nu]}^{\rho} + \Gamma_{\lambda[\mu}^{\rho}\Gamma_{ \sigma \nu]}^{\lambda}\right) - H_{\mu\nu}\delta_{\sigma}^{\rho},$ $T^{*\lambda}_{\mu\nu} = 2\Gamma_{[\nu\mu]}^{\lambda},$ $H_{\mu\nu} = 2\partial_{[\mu}B_{\nu]},$	$R_{\sigma\mu\nu}^{\rho} = 2(\partial_{[\mu}\Gamma_{ \sigma \nu]}^{\rho} + \Gamma_{\lambda[\mu}^{\rho}\Gamma_{ \sigma \nu]}^{\lambda}) - H_{\mu\nu}\delta_{\sigma}^{\rho},$ $T^{*\lambda}_{\mu\nu} = 2\Gamma_{[\nu\mu]}^{\lambda},$ $H_{\mu\nu} = 2\partial_{[\mu}B_{\nu]},$
1510.06699_ E09046 (48)	012	■ 0.479 ● N +0	$\{ \}^{\{0\}}\mathrm{D}_{\mu}^{\{*\}}\varphi\equiv\left(\partial_{\mu}^{\{*\}}+\frac{1}{2}\{ \}^{\{2\}}\{ \}^{\{0\}}\mathrm{A}^{\{*\}}\mathrm{a}\mathrm{b}\{ \}_{\mu}\mathrm{S}_{\mathrm{a}\mathrm{b}}\right)\varphi$	${}^0D_{\mu}^*\varphi\equiv\left(\partial_{\mu}^*+\frac{1}{2}{}^0A^{*ab}{}_{\mu}\Sigma_{ab}\right)\varphi,$	${}^0D_{\mu}^*\varphi\equiv(\partial_{\mu}^*+\frac{1}{2}{}^0A^{*ab}{}_{\mu}\Sigma_{ab})\varphi,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E08287 (229)	038	■ 0.487 ● W +1	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \h\left(\mathrm{t}_{\{\mathrm{mathcal}\{\mathrm{R}\}^{\{2\}}\}^{\{\dagger\}}\right. \\ & \left.\mathrm{right})^{\{a\}\{\}\{b\}} = \& \backslash\alpha_{\{1\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \\ & \backslash\mathrm{right}\}\left(4 \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a}\{\}\{b\} - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \right. \\ & \left. \backslash\mathrm{right}) \backslash\backslash \& + \backslash\alpha_{\{2\}} \backslash\left[2 \backslash\left(\backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{a}\right. \right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{b\}}^{\{\dagger\}} - \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d}\right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a}\{\}\{b\} \backslash\mathrm{right}) - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d}\right. \\ & \left. \backslash\mathrm{right}\right] \backslash\backslash \& + \backslash\alpha_{\{3\}} \backslash\left[2 \backslash\left(\backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \right. \right. \\ & \left. \backslash\mathrm{dagger} \mathrm{a} \mathrm{c}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{b\}}^{\{\dagger\}} - \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \\ & \left. \backslash\mathrm{dagger} \mathrm{d} \mathrm{c}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d}\} \backslash\mathrm{right}) - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{d} \mathrm{c}\right) \backslash\mathrm{right}\right] \backslash\backslash \& + \backslash\alpha_{\{4\}} \\ & \left. \backslash\left[4 \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d} \mathrm{e} \mathrm{a}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \right. \\ & \left. \mathrm{d} \mathrm{e} \mathrm{b}\}^{\{\dagger\}} - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d} \mathrm{e} \mathrm{f}\right. \\ & \left. \mathrm{c} \mathrm{d} \mathrm{e} \mathrm{f}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \backslash\mathrm{right}\right] \backslash\backslash \& + \\ & \left. \backslash\alpha_{\{5\}} \backslash\left[2 \backslash\left(\backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d} \mathrm{e}\right) - \right. \right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{e} \mathrm{c} \mathrm{d} \mathrm{a}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \right. \\ & \left. \mathrm{b}\}^{\{\dagger\}} - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{c} \mathrm{e} \mathrm{d} \mathrm{f}\right) \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \backslash\mathrm{right}\right] \backslash\backslash \& + \backslash\alpha_{\{6\}} \\ & \left. \backslash\left[4 \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{e} \mathrm{a} \mathrm{c} \mathrm{d}\right) \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \right. \\ & \left. \mathrm{d} \mathrm{e} \mathrm{b}\}^{\{\dagger\}} - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{\dagger\}} \mathrm{e} \mathrm{a} \mathrm{c} \mathrm{d}\right) \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{c\} \{d\}}^{\{\dagger\}} \backslash\mathrm{right}\right] \backslash\backslash \mathrm{h} \\ & \left. \backslash\left(\mathrm{t}_{\{\mathrm{mathcal}\{\mathrm{H}\}^{\{2\}}\}^{\{\dagger\}}\right) \mathrm{a}\{\}\{b\}} = \right. \\ & \left. \& \backslash\mathrm{frac}\{1\}{2} \backslash\mathrm{x}\backslash\left[4 \backslash\mathrm{mathcal}\{\mathrm{H}\}_{\{b\} \{c\}}^{\{\dagger\}} \right. \right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{H}\}^{\{\dagger\}} \mathrm{a} \mathrm{c}\right) - \backslash\delta_{\{b\}}^{\{a\}} \backslash\mathrm{mathcal}\{\mathrm{H}\}_{\{c\} \{d\}}^{\{\dagger\}} \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{H}\}^{\{\dagger\}} \mathrm{a} \mathrm{c} \mathrm{d}\right) + \backslash\mathrm{frac}\{4\}{3} \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\{b\}}^{\{\dagger\}} \backslash\left(\backslash\mathrm{mathcal}\{\mathrm{D}\}_{\{c\}}^{\{\dagger\}} \right. \right. \\ & \left. \backslash\mathrm{mathcal}\{\mathrm{H}\}^{\{\dagger\}} \mathrm{c} \mathrm{a}\right) - \backslash\mathrm{frac}\{1\}{2} \backslash\mathrm{mathcal}\{\mathrm{T}\}^{\{\dagger\}} \\ & \left. \backslash\mathrm{dagger} \mathrm{a}\}\{\}\{c\} \{d\} \backslash\mathrm{mathcal}\{\mathrm{H}\}^{\{\dagger\}} \mathrm{c} \mathrm{d}\right) \backslash\mathrm{right}) \\ & \left. \backslash\mathrm{right}\right], \backslash\mathrm{end}\{\mathrm{aligned}\} \end{aligned}$	$\begin{aligned} h\left(t_{\mathcal{R}^2}^\dagger\right)_b^a &= \alpha_1 \mathcal{R}^\dagger \left(4 \mathcal{R}^{\dagger a}_b - \delta_b^a \mathcal{R}^\dagger\right) \\ &+ \alpha_2 \left[2 \left(\mathcal{R}^{\dagger ca} \mathcal{R}_{cb}^\dagger - \mathcal{R}^{\dagger cd} \mathcal{R}_{cdb}^\dagger\right) - \delta_b^a \mathcal{R}_{cd}^\dagger \mathcal{R}^{\dagger cd}\right] \\ &+ \alpha_3 \left[2 \left(\mathcal{R}^{\dagger ac} \mathcal{R}_{cb}^\dagger - \mathcal{R}^{\dagger dc} \mathcal{R}_{cdb}^\dagger\right) - \delta_b^a \mathcal{R}_{cd}^\dagger \mathcal{R}^{\dagger dc}\right] \\ &+ \alpha_4 \left[4 \mathcal{R}^{\dagger cdea} \mathcal{R}_{cdeb}^\dagger - \delta_b^a \mathcal{R}^{\dagger cdef} \mathcal{R}_{cdef}^\dagger\right] \\ &+ \alpha_5 \left[2 \left(\mathcal{R}^{\dagger acde} - \mathcal{R}^{\dagger ecda}\right) \mathcal{R}_{cdeb}^\dagger - \delta_b^a \mathcal{R}^{\dagger cedf} \mathcal{R}_{cdef}^\dagger\right] \\ &+ \alpha_6 \left[4 \mathcal{R}^{\dagger eacd} \mathcal{R}_{cdeb}^\dagger - \delta_b^a \mathcal{R}^{\dagger efcd} \mathcal{R}_{cdef}^\dagger\right] \\ h\left(t_{\mathcal{H}^2}^\dagger\right)_b^a &= \frac{1}{2} \xi \left[4 \mathcal{H}_{bc}^\dagger \mathcal{H}^{\dagger ac} - \delta_b^a \mathcal{H}_{cd}^\dagger \mathcal{H}^{\dagger cd} + \frac{4}{3} \mathcal{D}_b^\dagger \left(\mathcal{D}_c^\dagger \mathcal{H}^{\dagger ca} - \frac{1}{2} \mathcal{T}^{\dagger a}_{cd} \mathcal{H}^{\dagger cd}\right)\right], \end{aligned}$	
1510.06699_ E08117 (128)	022	■ 0.497 ● N +0	$\widetilde{R}_{\sigma\mu\nu}^{\rho} \equiv R_{\sigma\mu\nu}^{\rho} + H_{\mu\nu} \delta_{\sigma}^{\rho}.$	$\widetilde{R}^{\rho}_{\sigma\mu\nu} \equiv R^{\rho}_{\sigma\mu\nu} + H_{\mu\nu} \delta_{\sigma}^{\rho}.$	
1510.06699_ E08249 (B4)	049	■ 0.500 ● N +0	$\mathrm{L}_{\{0\}} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{2\}} = \alpha_{\{1\}} \{\}\{0\} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{2\}} + \alpha_{\{2\}} \{\}\{0\} \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{a\} \{b\}}^{\{\}\{0\}} \{\}\{0\} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{a\} \{b\}} + \alpha_{\{3\}} \{\}\{0\} \backslash\mathrm{mathcal}\{\mathrm{R}\}_{\{a\} \{b\} \{c\} \{d\}}^{\{\}\{0\}} \{\}\{0\} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\{a\} \{b\} \{c\} \{d\}} \backslash\mathrm{text}\{,\} \}$	$L_0 \mathcal{R}^2 = \alpha_1 {}^0 \mathcal{R}^2 + \alpha_2 {}^0 \mathcal{R}_{ab} {}^0 \mathcal{R}^{ab} + \alpha_3 {}^0 \mathcal{R}_{abcd} {}^0 \mathcal{R}^{abcd},$	$L_0 \mathcal{R}^2 = \alpha_1 {}^0 \mathcal{R}^2 + \alpha_2 {}^0 \mathcal{R}_{ab} {}^0 \mathcal{R}^{ab} + \alpha_3 {}^0 \mathcal{R}_{abcd} {}^0 \mathcal{R}^{abcd},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0261 (C12)	050	■0.600 ●C +3	$\begin{aligned} R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R&=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}}\\ T^{\lambda}{}_{\mu\nu}&=\frac{\kappa}{2\sqrt{-g}}\left(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}\right), \end{aligned}$	$R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}},$ $T^{\lambda}{}_{\mu\nu}=\frac{\kappa}{2\sqrt{-g}}(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}),$	$R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}},$ $T^{\lambda}{}_{\mu\nu}=\frac{\kappa}{2\sqrt{-g}}(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}),$
1510.06699_ EQ0011 (12)	008	■0.614 ●C +4	$L_{\mathrm{D}}\!=\!\frac{1}{2}\,i\!\left[\!\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi\!-\!(\partial_{\mu}\bar{\psi})\,\gamma^{\mu}\psi\right]\!-\!m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi\!-\!m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{1}{2}i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-(\partial_{\mu}\bar{\psi})\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi-m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{1}{2}i[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-(\partial_{\mu}\bar{\psi})\gamma^{\mu}\psi]-m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi-m\bar{\psi}\psi.$
1510.06699_ EQ0183 (199b)	034	■0.615 ●C +4	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^{\mu} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^{\mu} \\ \widehat{A}_\mu^{\dagger ab} &\equiv A_\mu^{ab} + (\mathcal{V}^a b_\mu^b - \mathcal{V}^b b_\mu^a) \\ \widehat{V}_\mu &\equiv V_\mu + \frac{1}{3} T_\mu + \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi$ $\widehat{h}_a{}^{\mu} \equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^{\mu}$ $\widehat{A}_\mu^{\dagger ab} \equiv A_\mu^{ab} + (\mathcal{V}^a b_\mu^b - \mathcal{V}^b b_\mu^a)$ $\widehat{V}_\mu \equiv V_\mu + \frac{1}{3} T_\mu + \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right)$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi,$ $\widehat{h}_a{}^{\mu} \equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^{\mu},$ $\widehat{A}_\mu^{\dagger ab} \equiv A_\mu^{ab} + (\mathcal{V}^a b_\mu^b - \mathcal{V}^b b_\mu^a),$ $\widehat{V}_\mu \equiv V_\mu + \frac{1}{3} T_\mu + \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00208 (232)	038	■ 0.629 ● W +1	$\begin{aligned} & \hleft(\tau_{\phi}^{\dagger})^a_b = \frac{1}{2} \nu \left[\frac{4}{3} \mathcal{D}^{\dagger a} \phi \mathcal{D}^{\dagger}_b \phi - \frac{1}{3} \delta^a_b \mathcal{D}^{\dagger c} \phi \mathcal{D}^{\dagger}_c \phi + \frac{2}{3} \phi \left(\delta^a_b \mathcal{D}^{\dagger}_c \mathcal{D}^{\dagger c} \phi - \mathcal{D}^{\dagger}_b \mathcal{D}^{\dagger a} \phi \right) \right] + \lambda \delta^a_b \phi^4 \\ & \hleft(\tau_{\mathcal{R}}^{\dagger})^a_b = -a \phi^2 \left(\mathcal{R}^{\dagger a}_b - \frac{1}{2} \delta^a_b \mathcal{R}^{\dagger} \right) \\ & \hleft(\tau_{\mathcal{T}^2}^{\dagger})^a_b = \beta_1 \left[\phi^2 \left(4 \mathcal{T}^{\dagger cda} \mathcal{T}^{\dagger}_{cdb} - 2 \mathcal{T}^{\dagger ade} \mathcal{T}^{\dagger}_{bde} - \delta^a_b \mathcal{T}^{\dagger cde} \mathcal{T}^{\dagger}_{cde} \right) + 4 \mathcal{D}^{\dagger}_c \left(\phi^2 \mathcal{T}^{\dagger ca}_b \right) \right] \\ & \quad + \beta_2 \left[\phi^2 \left(2 \mathcal{T}^{\dagger dca} \mathcal{T}^{\dagger}_{cdb} + 2 \mathcal{T}^{\dagger adc} \mathcal{T}^{\dagger}_{cdb} - 2 \mathcal{T}^{\dagger acd} \mathcal{T}^{\dagger}_{cbd} - \delta^a_b \mathcal{T}^{\dagger dce} \mathcal{T}^{\dagger}_{cde} \right) + 4 \mathcal{D}^{\dagger}_c \left(\phi^2 \mathcal{T}^{\dagger [c}_b{}^{a]} \right) \right] \end{aligned}$	$\begin{aligned} h(\tau_{\phi}^{\dagger})^a_b &= \tfrac{1}{2} \nu [\tfrac{4}{3} \mathcal{D}^{\dagger a} \phi \mathcal{D}^{\dagger}_b \phi - \tfrac{1}{3} \delta^a_b \mathcal{D}^{\dagger c} \phi \mathcal{D}^{\dagger}_c \phi + \tfrac{2}{3} \phi (\delta^a_b \mathcal{D}^{\dagger}_c \mathcal{D}^{\dagger c} \phi - \mathcal{D}^{\dagger}_b \mathcal{D}^{\dagger a} \phi)] + \lambda \delta^a_b \phi^4 \\ h(\tau_{\mathcal{R}}^{\dagger})^a_b &= -a \phi^2 (\mathcal{R}^{\dagger a}_b - \tfrac{1}{2} \delta^a_b \mathcal{R}^{\dagger}), \\ h(\tau_{\mathcal{T}^2}^{\dagger})^a_b &= \beta_1 [\phi^2 (4 \mathcal{T}^{\dagger cda} \mathcal{T}^{\dagger}_{cdb} - 2 \mathcal{T}^{\dagger ade} \mathcal{T}^{\dagger}_{bde} - \delta^a_b \mathcal{T}^{\dagger cde} \mathcal{T}^{\dagger}_{cde}) + 4 \mathcal{D}^{\dagger}_c (\phi^2 \mathcal{T}^{\dagger ca}_b)] \\ &\quad + \beta_2 [\phi^2 (2 \mathcal{T}^{\dagger dca} \mathcal{T}^{\dagger}_{cdb} + 2 \mathcal{T}^{\dagger adc} \mathcal{T}^{\dagger}_{cdb} - 2 \mathcal{T}^{\dagger acd} \mathcal{T}^{\dagger}_{cbd} - \delta^a_b \mathcal{T}^{\dagger dce} \mathcal{T}^{\dagger}_{cde}) + 4 \mathcal{D}^{\dagger}_c (\phi^2 \mathcal{T}^{\dagger [c}_b{}^{a]})]. \end{aligned}$	
1510.06699_ E00153 (167)	029	■ 0.631 ● W +0	$\begin{aligned} & \mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) \\ & \quad + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, A, \partial A, V, \partial V), \end{aligned}$	$\begin{aligned} \mathcal{L}_{\mathrm{T}} &= \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) \\ &\quad + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, A, \partial A, V, \partial V), \end{aligned}$	
1510.06699_ E00066 (70)	015	■ 0.647 ● N +0	$L_{\mathrm{D}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - \mu \phi \bar{\psi} \psi,$	$L_{\mathrm{D}} = \tfrac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - \mu \phi \bar{\psi} \psi,$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0097 (105b)	020	■ 0.649 ● W +2	$\begin{aligned} \widehat{\lambda} &\equiv \left(\frac{\phi}{\phi_0}\right) \lambda \\ \widehat{p}_a &\equiv \widehat{h}_a{}^\mu p_\mu = \left(\frac{\phi}{\phi_0}\right)^{-1} p_a \\ \widehat{v}^a &\equiv \widehat{b}^a{}_\mu \frac{dx^\mu}{d\widehat{\lambda}} = v^a, \\ \widehat{e} &\equiv \left(\frac{\phi}{\phi_0}\right) e, \end{aligned}$	$\begin{aligned} \widehat{\lambda} &\equiv \left(\frac{\phi}{\phi_0}\right) \lambda \\ \widehat{p}_a &\equiv \widehat{h}_a{}^\mu p_\mu = \left(\frac{\phi}{\phi_0}\right)^{-1} p_a \\ \widehat{v}^a &\equiv \widehat{b}^a{}_\mu \frac{dx^\mu}{d\widehat{\lambda}} = v^a, \\ \widehat{e} &\equiv \left(\frac{\phi}{\phi_0}\right) e, \end{aligned}$	
1510.06699_ EQ0085 (91b)	018	■ 0.651 ● C +5	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	
1510.06699_ EQ0063 (67a)	014	■ 0.657 ● N +0	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	
1510.06699_ EQ0227 (261)	042	■ 0.664 ● C +10	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00074 (79)	016	0.671 C +5	$\begin{aligned} \mathrm{L}_{\mathrm{M}} = & h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 \right. \\ & \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right] \end{aligned}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 \right. \\ \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 \\ - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}}], \quad (79)$
1510.06699_ E00211 (238)	039	0.675 C +9	$\begin{aligned} & h \left(\sigma_{\phi} \right)_{ab}{}^c = -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi \\ & h \left(\sigma_{\mathcal{R}} \right)_{ab}{}^c = -a \left(\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab} \right) \phi^2 \\ & h \left(\sigma_{\mathcal{T}^2} \right)_{ab}{}^c = -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 \left(\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c \right) \end{aligned}$	$\begin{aligned} h \left(\sigma_{\phi} \right)_{ab}{}^c &= -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi \\ h \left(\sigma_{\mathcal{R}} \right)_{ab}{}^c &= -a \left(\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab} \right) \phi^2 \\ h \left(\sigma_{\mathcal{T}^2} \right)_{ab}{}^c &= -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 \left(\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c \right) \end{aligned}$	$\begin{aligned} h(\sigma_{\phi})_{ab}{}^c &= -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi, \\ h(\sigma_{\mathcal{R}})_{ab}{}^c &= -a (\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab}) \phi^2, \\ h(\sigma_{\mathcal{T}^2})_{ab}{}^c &= -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 (\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c) \end{aligned}$
1510.06699_ E00216 (248)	039	0.686 N +0	$\begin{aligned} & \partial_{\phi} L_{\mathcal{R}^2} = 0, \\ & \partial_{\phi} L_{\mathcal{H}^2} = 0, \end{aligned}$	$\begin{aligned} \partial_{\phi} L_{\mathcal{R}^2} &= 0, \\ \partial_{\phi} L_{\mathcal{H}^2} &= 0, \end{aligned}$	$\begin{aligned} \partial_{\phi} L_{\mathcal{R}^2} &= 0, \\ \partial_{\phi} L_{\mathcal{H}^2} &= 0, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0213 (239)	039	■ 0.687 ● C +16	$\begin{aligned} & \hleft(j_{\mathcal{R}^2})_a = & -12\alpha_1\mathcal{D}_a^\dagger\mathcal{R}^\dagger \\ & -4\alpha_2\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}_b^{\dagger c}\right] \\ & -4\alpha_3\left[\mathcal{D}_b^\dagger\left(\mathcal{R}^{\dagger b}{}_a+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}^{\dagger c}{}_b\right] \\ & -8\alpha_4\left[\mathcal{D}_b^\dagger\mathcal{R}_a^{\dagger b}-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{abcd}^\dagger\right] \\ & -2\alpha_5\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\mathcal{R}^{\dagger b}{}_a\right)-\mathcal{T}^{\dagger bcd}\left(\mathcal{R}_{cbad}^\dagger+\mathcal{R}_{cabd}^\dagger\right)\right] \\ & -8\alpha_6\left[\mathcal{D}_b^\dagger\mathcal{R}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{cdab}^\dagger\right] \\ & h(j_{\mathcal{H}^2})_a = -2\xi\left[\mathcal{D}_b^\dagger\mathcal{H}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger}{}_{acd}\mathcal{H}^{\dagger cd}\right] \end{aligned}$	$\begin{aligned} h(j_{\mathcal{R}^2})_a = & -12\alpha_1\mathcal{D}_a^\dagger\mathcal{R}^\dagger \\ & -4\alpha_2[\mathcal{D}_b^\dagger(\mathcal{R}_a^{\dagger b}+\tfrac{1}{2}\delta_a^b\mathcal{R}^\dagger)-\tfrac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}_b^{\dagger c}] \\ & -4\alpha_3[\mathcal{D}_b^\dagger(\mathcal{R}^{\dagger b}{}_a+\tfrac{1}{2}\delta_a^b\mathcal{R}^\dagger)-\tfrac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}^{\dagger c}{}_b] \\ & -8\alpha_4[\mathcal{D}_b^\dagger\mathcal{R}_a^{\dagger b}-\tfrac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{abcd}^\dagger] \\ & -2\alpha_5[\mathcal{D}_b^\dagger(\mathcal{R}_a^{\dagger b}+\mathcal{R}^{\dagger b}{}_a)-\mathcal{T}^{\dagger bcd}(\mathcal{R}_{cbad}^\dagger+\mathcal{R}_{cabd}^\dagger)] \\ & -8\alpha_6[\mathcal{D}_b^\dagger\mathcal{R}^{\dagger b}{}_a-\tfrac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{cdab}^\dagger], \tag{241} \\ h(j_{\mathcal{H}^2})_a = & -2\xi[\mathcal{D}_b^\dagger\mathcal{H}^{\dagger b}{}_a-\tfrac{1}{2}\mathcal{T}^{\dagger}{}_{acd}\mathcal{H}^{\dagger cd}], \tag{242} \end{aligned}$	
1510.06699_ EQ0201 (219)	037	■ 0.691 ● W +1	$\begin{aligned} & \{ \}^{\{0\}}\Gamma^{\dagger}\{\lambda\}_{\mu\nu} = & \frac{1}{2}g^{\lambda\rho}(\partial_\mu^\dagger g_{\nu\rho}+\partial_\nu^\dagger g_{\mu\rho}-\partial_\rho^\dagger g_{\mu\nu}) \\ & = {}^0\Gamma^\lambda{}_{\mu\nu}-\delta_\nu^\lambda\left(V_\mu+\frac{1}{3}T_\mu\right)-\delta_\mu^\lambda\left(V_\nu+\frac{1}{3}T_\nu\right) \\ & +g_{\mu\nu}\left(V^\lambda+\frac{1}{3}T^\lambda\right) \end{aligned}$	$\begin{aligned} {}^0\Gamma^{\dagger\lambda}{}_{\mu\nu} = & \tfrac{1}{2}g^{\lambda\rho}(\partial_\mu^\dagger g_{\nu\rho}+\partial_\nu^\dagger g_{\mu\rho}-\partial_\rho^\dagger g_{\mu\nu}) \\ = {}^0\Gamma^\lambda{}_{\mu\nu} - \delta_\nu^\lambda(V_\mu + \tfrac{1}{3}T_\mu) - \delta_\mu^\lambda(V_\nu + \tfrac{1}{3}T_\nu) \\ & +g_{\mu\nu}(V^\lambda + \tfrac{1}{3}T^\lambda), \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00147 (161b)	028	■ 0.692 ● C +4	$\begin{aligned} & \mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0. \end{aligned}$	$\mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0,$ $\mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0,$ $\mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0.$	$\mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0,$ $\mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0,$ $\mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0.$
1510.06699_ E00115 (124)	022	■ 0.692 ● N +0	$\nabla_{\sigma} g_{\mu\nu} = -2 B_{\sigma} g_{\mu\nu}$	$\nabla_{\sigma} g_{\mu\nu} = -2 B_{\sigma} g_{\mu\nu},$	$\nabla_{\sigma} g_{\mu\nu} = -2 B_{\sigma} g_{\mu\nu},$
1510.06699_ E00226 (260)	042	■ 0.694 ● N +1	$\mathcal{L}_{\mathbf{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathbf{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathbf{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}}],$
1510.06699_ E00218 (252)	040	■ 0.699 ● C +4	$\begin{aligned} & \nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi} \\ & + \phi \left(a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac} \right) = 0 \end{aligned}$	$\nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi}$ $+ \phi \left(a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac} \right) = 0$	$\nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi}$ $+ \phi (a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}) = 0.$
1510.06699_ E00137 (151a)	026	■ 0.704 ● W +2	$\begin{aligned} & R^{\dagger ab}{}_{\mu\nu} \equiv \partial_{\mu} A^{\dagger ab}{}_{\nu} - \partial_{\nu} A^{\dagger ab}{}_{\mu} + A^{\dagger a}{}_{c\mu} A^{\dagger cb}{}_{\nu} - A^{\dagger a}{}_{c\nu} A^{\dagger cb}{}_{\mu} \\ & = R^{ab}{}_{\mu\nu} + 4 b^{[b}{}_{[\nu} D_{\mu]} \mathcal{V}^{a]} + 4 \mathcal{V}^{[a} V_{[\mu} b^{b]}{}_{\nu]} - 2 \mathcal{V}^e \mathcal{V}_e b^{[a}{}_{\mu} b^{b]}{}_{\nu} + 2 \mathcal{V}^{[a} T^{b]}{}_{\mu\nu} \end{aligned}$	$R^{\dagger ab}{}_{\mu\nu} \equiv \partial_{\mu} A^{\dagger ab}{}_{\nu} - \partial_{\nu} A^{\dagger ab}{}_{\mu} + A^{\dagger a}{}_{c\mu} A^{\dagger cb}{}_{\nu} - A^{\dagger a}{}_{c\nu} A^{\dagger cb}{}_{\mu}$ $= R^{ab}{}_{\mu\nu} + 4 b^{[b}{}_{[\nu} D_{\mu]} \mathcal{V}^{a]} + 4 \mathcal{V}^{[a} V_{[\mu} b^{b]}{}_{\nu]} - 2 \mathcal{V}^e \mathcal{V}_e b^{[a}{}_{\mu} b^{b]}{}_{\nu} + 2 \mathcal{V}^{[a} T^{b]}{}_{\mu\nu}$	$R^{\dagger ab}{}_{\mu\nu} \equiv \partial_{\mu} A^{\dagger ab}{}_{\nu} - \partial_{\nu} A^{\dagger ab}{}_{\mu} + A^{\dagger a}{}_{c\mu} A^{\dagger cb}{}_{\nu} - A^{\dagger a}{}_{c\nu} A^{\dagger cb}{}_{\mu},$ $= R^{ab}{}_{\mu\nu} + 4 b^{[b}{}_{[\nu} D_{\mu]} \mathcal{V}^{a]} + 4 \mathcal{V}^{[a} V_{[\mu} b^{b]}{}_{\nu]} - 2 \mathcal{V}^e \mathcal{V}_e b^{[a}{}_{\mu} b^{b]}{}_{\nu} + 2 \mathcal{V}^{[a} T^{b]}{}_{\mu\nu},$
1510.06699_ E00022 (23)	009	■ 0.723 ● N +2	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \left[\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x) \right],$	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \left[\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x) \right],$	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} [\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x)],$
1510.06699_ E00013 (14)	008	■ 0.734 ● N +0	$L_{\mathbf{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$	$L_{\mathbf{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$	$L_{\mathbf{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00182 (198)	033	■ 0.744 ● W +0	$\begin{aligned} \mathcal{L}_{\mathrm{T}} = & \mathcal{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) \\ & + \mathcal{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) . \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) \\ + \mathcal{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) \\ + \mathcal{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .$
1510.06699_ E00012 (13)	008	■ 0.749 ● N +2	$\begin{aligned} \mathrm{L}_{\mathrm{D}}^{\prime} & = \frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime} - m \bar{\psi}^{\prime} \psi^{\prime} \\ & = \frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - m e^{2 w \rho} \bar{\psi} \psi \end{aligned}$	$L_{\mathrm{D}}^{\prime} = \frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime} - m \bar{\psi}^{\prime} \psi^{\prime} \\ = \frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - m e^{2 w \rho} \bar{\psi} \psi$	$L_{\mathrm{D}}^{\prime} = \frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime} - m \bar{\psi}^{\prime} \psi^{\prime} \\ = \frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - m e^{2 w \rho} \bar{\psi} \psi .$
1510.06699_ E00155 (169a)	029	■ 0.758 ● C +3	$\begin{aligned} \mathrm{t}_{\mathrm{a} \mathrm{b}}^{\prime} & = \mathrm{t}_{\mathrm{a} \mathrm{b}} + 2 \theta \left(s_{\mathrm{a} \mathrm{c}}^{\mathrm{c}} \mathcal{P}_{\mathrm{b}} - s_{\mathrm{c} \mathrm{b} \mathrm{a}} \mathcal{P}^{\mathrm{c}} \right) \\ \tau_{\mathrm{a} \mathrm{b}}^{\prime} & = \tau_{\mathrm{a} \mathrm{b}} + 2 \theta \left(\sigma_{\mathrm{a} \mathrm{c}}^{\mathrm{c}} \mathcal{P}_{\mathrm{b}} - \sigma_{\mathrm{c} \mathrm{b} \mathrm{a}} \mathcal{P}^{\mathrm{c}} \right) \end{aligned}$	$t_{ab}^{\prime} = t_{ab} + 2 \theta \left(s_{ac}^{\mathrm{c}} \mathcal{P}_b - s_{cba} \mathcal{P}^c \right) \\ \tau_{ab}^{\prime} = \tau_{ab} + 2 \theta \left(\sigma_{ac}^{\mathrm{c}} \mathcal{P}_b - \sigma_{cba} \mathcal{P}^c \right)$	$t_{ab}^{\prime} = t_{ab} + 2 \theta \left(s_{ac}^{\mathrm{c}} \mathcal{P}_b - s_{cba} \mathcal{P}^c \right), \\ \tau_{ab}^{\prime} = \tau_{ab} + 2 \theta \left(\sigma_{ac}^{\mathrm{c}} \mathcal{P}_b - \sigma_{cba} \mathcal{P}^c \right),$
1510.06699_ E00108 (116)	021	■ 0.766 ● N +0	$\hat{\mathrm{e}}_{\mathrm{a}} \cdot \hat{\mathrm{e}}_{\mathrm{b}} = \eta_{a b} = g_{\mu \nu} h_{\mathrm{a}}^{\mu} h_{\mathrm{b}}^{\nu} .$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab} = g_{\mu \nu} h_a^{\mu} h_b^{\nu} .$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab} = g_{\mu \nu} h_a^{\mu} h_b^{\nu} .$
1510.06699_ E00123 (134)	023	■ 0.768 ● S +0	$\begin{aligned} \mathrm{h}_{\mathrm{a}}^{\prime} & \left(\mathrm{h}_{\mathrm{a}}^{\prime} \right)^{\prime} \backslash \mathrm{mu} \} (x) \& = \mathrm{e}^{\{-\rho(x)\}} \\ \mathrm{h}_{\mathrm{a}}\} & \left. \right\}^{\wedge} \{ \backslash \mathrm{mu} \} (x), \backslash \backslash \mathrm{A}^{\wedge} \{ \mathrm{prime} \mathrm{a} \mathrm{b} \} \left\} _{\backslash \mathrm{mu} \} (x) \& = \mathrm{A}^{\wedge} \{ \mathrm{a} \mathrm{b} \} \left\} _{\backslash \mathrm{mu} \} (x), \end{aligned}$	$h_a^{\prime \mu}(x) = e^{-\rho(x)} h_a^{\mu}(x), \\ A^{\prime ab}{}_{\mu}(x) = A^{ab}{}_{\mu}(x),$	$h_a^{\prime \mu}(x) = e^{-\rho(x)} h_a^{\mu}(x), \\ A^{\prime ab}{}_{\mu}(x) = A^{ab}{}_{\mu}(x),$
1510.06699_ E00222 (256)	041	■ 0.771 ● C +3	$\begin{aligned} \mathrm{L}_{\mathrm{M}} & = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu \left(\mathcal{D}_a \phi \right) \left(\mathcal{D}^a \phi \right) - \lambda \phi^4 \right. \\ & \quad \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^2} \right] , \end{aligned}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu \left(\mathcal{D}_a \phi \right) \left(\mathcal{D}^a \phi \right) - \lambda \phi^4 \right. \\ \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^2} \right] , \quad (256$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu \left(\mathcal{D}_a \phi \right) \left(\mathcal{D}^a \phi \right) - \lambda \phi^4 \right. \\ \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^2} \right] , \quad (256$
1510.06699_ E00114 (123)	022	■ 0.786 ● N +1	$\begin{aligned} \Gamma_{\nu \mu}^{\lambda} & = e_a^{\lambda} \left(\partial_{\mu}^* e_{\nu}^a + A_{b \mu}^a e_{\nu}^b \right) , \\ A_{b \mu}^a & = e_{\lambda}^a \left(\partial_{\mu}^* e_b^{\lambda} + \Gamma_{\nu \mu}^{\lambda} e_b^{\nu} \right) . \end{aligned}$	$\Gamma_{\nu \mu}^{\lambda} = e_a^{\lambda} \left(\partial_{\mu}^* e_{\nu}^a + A_{b \mu}^a e_{\nu}^b \right) , \\ A_{b \mu}^a = e_{\lambda}^a \left(\partial_{\mu}^* e_b^{\lambda} + \Gamma_{\nu \mu}^{\lambda} e_b^{\nu} \right) .$	$\Gamma_{\nu \mu}^{\lambda} = e_a^{\lambda} \left(\partial_{\mu}^* e_{\nu}^a + A_{b \mu}^a e_{\nu}^b \right) , \\ A_{b \mu}^a = e_{\lambda}^a \left(\partial_{\mu}^* e_b^{\lambda} + \Gamma_{\nu \mu}^{\lambda} e_b^{\nu} \right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00168 (184c)	031	0.814 C +25	$\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b_\mu^b - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0 \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, \end{aligned}$	$\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b_\mu^b - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0 \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, \end{aligned}$	$\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b_\mu^b - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0, (184c) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, (184d) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, (184e) \end{aligned}$
1510.06699_ E00103 (111)	020	0.819 N +0	$L_{\{\mathrm{G}\}} = L_{0_{\{\mathrm{R}\}}} * 2 + L_{\{\mathrm{H}\}^2}$	$L_G = L_{0_{\mathcal{R}*2}} + L_{\mathcal{H}^2},$	$L_G = L_{0_{\mathcal{R}*2}} + L_{\mathcal{H}^2},$
1510.06699_ E00257 (C7)	050	0.842 N +1	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$
1510.06699_ E00209 (234)	039	0.860 N +2	$\begin{aligned} & \left(s_{\mathcal{R}^2}\right)_{ab}{}^c + \left(s_{\mathcal{H}^2}\right)_{ab}{}^c \\ & + \left(\sigma_\varphi\right)_{ab}{}^c + \left(\sigma_\phi\right)_{ab}{}^c + \left(\sigma_{\mathcal{R}}\right)_{ab}{}^c + \left(\sigma_{\mathcal{T}^2}\right)_{ab}{}^c = 0, \end{aligned}$	$\begin{aligned} & \left(s_{\mathcal{R}^2}\right)_{ab}{}^c + \left(s_{\mathcal{H}^2}\right)_{ab}{}^c \\ & + \left(\sigma_\varphi\right)_{ab}{}^c + \left(\sigma_\phi\right)_{ab}{}^c + \left(\sigma_{\mathcal{R}}\right)_{ab}{}^c + \left(\sigma_{\mathcal{T}^2}\right)_{ab}{}^c = 0, \end{aligned}$	$\begin{aligned} & \left(s_{\mathcal{R}^2}\right)_{ab}{}^c + \left(s_{\mathcal{H}^2}\right)_{ab}{}^c \\ & + \left(\sigma_\varphi\right)_{ab}{}^c + \left(\sigma_\phi\right)_{ab}{}^c + \left(\sigma_{\mathcal{R}}\right)_{ab}{}^c + \left(\sigma_{\mathcal{T}^2}\right)_{ab}{}^c = 0, \end{aligned}$
1510.06699_ E00196 (213)	036	0.861 N +0	$\begin{aligned} & R^{\dagger \rho}{}_{\sigma \mu \nu} = 2 \left(\partial_{[\mu} \Gamma^\rho{}_{ \sigma \nu]} + \Gamma^\rho{}_{\lambda [\mu} \Gamma^\lambda{}_{ \sigma \nu]} \right) + H^\dagger_{\mu \nu} \delta_\sigma^\rho, \\ & T^{\dagger \lambda}{}_{\mu \nu} = 2 \Gamma^\lambda{}_{[\nu \mu]}, \\ & H^\dagger_{\mu \nu} = \partial_\mu \left(V_\nu + \frac{1}{3} T_\nu \right) - \partial_\nu \left(V_\mu + \frac{1}{3} T_\mu \right). \end{aligned}$	$\begin{aligned} & R^{\dagger \rho}{}_{\sigma \mu \nu} = 2 \left(\partial_{[\mu} \Gamma^\rho{}_{ \sigma \nu]} + \Gamma^\rho{}_{\lambda [\mu} \Gamma^\lambda{}_{ \sigma \nu]} \right) + H^\dagger_{\mu \nu} \delta_\sigma^\rho, \\ & T^{\dagger \lambda}{}_{\mu \nu} = 2 \Gamma^\lambda{}_{[\nu \mu]}, \\ & H^\dagger_{\mu \nu} = \partial_\mu \left(V_\nu + \frac{1}{3} T_\nu \right) - \partial_\nu \left(V_\mu + \frac{1}{3} T_\mu \right). \end{aligned}$	$\begin{aligned} & R^{\dagger \rho}{}_{\sigma \mu \nu} = 2 \left(\partial_{[\mu} \Gamma^\rho{}_{ \sigma \nu]} + \Gamma^\rho{}_{\lambda [\mu} \Gamma^\lambda{}_{ \sigma \nu]} \right) + H^\dagger_{\mu \nu} \delta_\sigma^\rho, \\ & T^{\dagger \lambda}{}_{\mu \nu} = 2 \Gamma^\lambda{}_{[\nu \mu]}, \\ & H^\dagger_{\mu \nu} = \partial_\mu (V_\nu + \tfrac{1}{3} T_\nu) - \partial_\nu (V_\mu + \tfrac{1}{3} T_\mu). \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0102 (110b)	020	■ 0.862 ● C +3	$\begin{aligned} & \mathcal{R}^{ab}_{cd} = {}^0\mathcal{R}^{*ab}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_d - \mathcal{K}^{*a}_{ed}\mathcal{K}^{*eb}_c \\ & \mathcal{R}^a_c = {}^0\mathcal{R}^{*a}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_b - \mathcal{K}^{*a}_{eb}\mathcal{K}^{*eb}_c, \\ & \mathcal{R} = {}^0\mathcal{R}^* + \frac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \frac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}. \end{aligned}$	$\begin{aligned} \mathcal{R}^{ab}_{cd} &= {}^0\mathcal{R}^{*ab}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_d - \mathcal{K}^{*a}_{ed}\mathcal{K}^{*eb}_c \\ \mathcal{R}^a_c &= {}^0\mathcal{R}^{*a}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_b - \mathcal{K}^{*a}_{eb}\mathcal{K}^{*eb}_c, \\ \mathcal{R} &= {}^0\mathcal{R}^* + \frac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \frac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}. \end{aligned}$	$\mathcal{R}^{ab}_{cd} = {}^0\mathcal{R}^{*ab}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_d - \mathcal{K}^{*a}_{ed}\mathcal{K}^{*eb}_c$ $\mathcal{R}^a_c = {}^0\mathcal{R}^{*a}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}_c + \mathcal{K}^{*a}_{ec}\mathcal{K}^{*eb}_b - \mathcal{K}^{*a}_{eb}\mathcal{K}^{*eb}_c,$ $\mathcal{R} = {}^0\mathcal{R}^* + \frac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \frac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}.$
1510.06699_ EQ0238 (A2)	047	■ 0.875 ● N +0	$\bar{\psi} \mathscr{J} = J^\mu \bar{\psi} \gamma_\mu = \bar{\psi} \gamma^\mu \psi \bar{\psi} \gamma_\mu.$	$\bar{\psi} \not{J} = J^\mu \bar{\psi} \gamma_\mu = \bar{\psi} \gamma^\mu \psi \bar{\psi} \gamma_\mu.$	$\bar{\psi} \not{J} = J^\mu \bar{\psi} \gamma_\mu = \bar{\psi} \gamma^\mu \psi \bar{\psi} \gamma_\mu.$
1510.06699_ EQ0234 (268)	044	■ 0.882 ● C +5	$\begin{array}{l} \mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu \left({}^0\mathcal{D}_a \phi \right) \left({}^0\mathcal{D}^a \phi \right) - \lambda \phi^4 \right. \\ \left. - a \phi^2 {}^0\mathcal{R} \right]. \end{array}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu \left({}^0\mathcal{D}_a \phi \right) \left({}^0\mathcal{D}^a \phi \right) - \lambda \phi^4 - a \phi^2 {}^0\mathcal{R} \right].$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a {}^0\overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu ({}^0\mathcal{D}_a \phi) ({}^0\mathcal{D}^a \phi) - \lambda \phi^4 - a \phi^2 {}^0\mathcal{R}]. \quad (268)$
1510.06699_ EQ0180 (196)	033	■ 0.900 ● C +5	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^\mu \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^\mu, \quad \widehat{A}^{\dagger ab}_\mu \equiv A^{\dagger ab}_\mu,$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^\mu \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^\mu, \quad \widehat{A}^{\dagger ab}_\mu \equiv A^{\dagger ab}_\mu,$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^\mu \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^\mu, \quad \widehat{A}^{\dagger ab}_\mu \equiv A^{\dagger ab}_\mu,$
1510.06699_ EQ0190 (206)	035	■ 0.901 ● W +1	$\begin{aligned} & \mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger) \\ & + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger), \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger),$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger),$
1510.06699_ EQ0145 (159)	027	■ 0.908 ● N +1	${}^0D^\dagger_\mu \varphi \equiv \left(\partial^\dagger_\mu + \frac{1}{2} {}^0A^{\dagger ab}_\mu \Sigma_{ab} \right) \varphi,$	${}^0D^\dagger_\mu \varphi \equiv \left(\partial^\dagger_\mu + \frac{1}{2} {}^0A^{\dagger ab}_\mu \Sigma_{ab} \right) \varphi,$	${}^0D^\dagger_\mu \varphi \equiv \left(\partial^\dagger_\mu + \frac{1}{2} {}^0A^{\dagger ab}_\mu \Sigma_{ab} \right) \varphi,$
1510.06699_ EQ0275 (D14)	053	■ 0.909 ● W +1	$\begin{aligned} & \mathcal{D}_a \mathcal{D}^a \mathcal{B}^c - \mathcal{R}^{ac} \mathcal{B}_a + m^2 \mathcal{B}^c \\ & + 2\kappa \left[(\delta^c_b \mathcal{D}_a - \kappa h \sigma_{ab}{}^c) (h \sigma^{abd} \mathcal{B}_d) \right] = 0 \end{aligned}$	$\mathcal{D}_a \mathcal{D}^a \mathcal{B}^c - \mathcal{R}^{ac} \mathcal{B}_a + m^2 \mathcal{B}^c + 2\kappa \left[(\delta^c_b \mathcal{D}_a - \kappa h \sigma_{ab}{}^c) (h \sigma^{abd} \mathcal{B}_d) \right] = 0$	$\mathcal{D}_a \mathcal{D}^a \mathcal{B}^c - \mathcal{R}^{ac} \mathcal{B}_a + m^2 \mathcal{B}^c + 2\kappa [(\delta^c_b \mathcal{D}_a - \kappa h \sigma_{ab}{}^c) (h \sigma^{abd} \mathcal{B}_d)] = 0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ EQ0273 (D12)	053	■ 0.910 ● N +0	$\mathrm{L}_{\psi}=\frac{1}{2}\mathrm{i}\,\overline{\psi}\,\gamma^a\,\stackrel{\leftrightarrow}{\mathcal{D}}_a\,\psi\stackrel{\leftrightarrow}{\mathcal{D}}_a\,\psi-\mu\phi\overline{\psi}\psi$	$L_\psi=\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi,$	$L_\psi=\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi,$
1510.06699_ EQ0058 (62b)	014	■ 0.910 ● C +3	$\begin{aligned} t^a{}_b+\tau^a{}_b&=0\\ s_{ab}{}^c+\sigma_{ab}{}^c&=0\\ j^a+\zeta^a&=0 \end{aligned}$	$t^a{}_b+\tau^a{}_b=0$ $s_{ab}{}^c+\sigma_{ab}{}^c=0$ $j^a+\zeta^a=0$	$t^a{}_b+\tau^a{}_b=0,$ $s_{ab}{}^c+\sigma_{ab}{}^c=0,$ $j^a+\zeta^a=0,$
1510.06699_ EQ0125 (138)	023	■ 0.912 ● N +1	$\mathcal{T}^{\prime a}{}_{bc}=\mathrm{e}^{\{-\rho\}}\left(\mathcal{T}^a{}_{bc}+\mathcal{P}_b\delta_c^a-\mathcal{P}_c\delta_b^a\right)$	$\mathcal{T}^{\prime a}{}_{bc}=e^{-\rho}\left(\mathcal{T}^a{}_{bc}+\mathcal{P}_b\delta_c^a-\mathcal{P}_c\delta_b^a\right),$	$\mathcal{T}^{\prime a}{}_{bc}=e^{-\rho}(\mathcal{T}^a{}_{bc}+\mathcal{P}_b\delta_c^a-\mathcal{P}_c\delta_b^a),$
1510.06699_ EQ0161 (176)	030	■ 0.929 ● N +1	$\begin{aligned} D_{\mathrm{c}}^{\dagger}(\mathrm{h}\,j^{\dagger\mathrm{c}})&=0,\\ \mathrm{h}t^{\dagger\mathrm{c}}_{\mathrm{c}}&=0. \end{aligned}$	$D_c^\dagger(hj^{\dagger c})=0,$ $ht^{\dagger c}_c=0.$	$D_c^\dagger(hj^{\dagger c})=0,$ $ht^{\dagger c}_c=0.$
1510.06699_ EQ0112 (120)	022	■ 0.934 ● K +0	$J^a+\delta J^a=\left(J^\mu+\delta J^\mu\right)e^a{}_\mu(x+\delta x)$	$J^a+\delta J^a=\left(J^\mu+\delta J^\mu\right)e^a{}_\mu(x+\delta x).$	$J^a+\delta J^a=\left(J^\mu+\delta J^\mu\right)e^a{}_\mu(x+\delta x).$
1510.06699_ EQ0240 (A4)	047	■ 0.943 ● N +0	$\overline{\psi}\mathscr{J}=\overline{\psi}\left(\overline{\psi}\psi+\overline{\psi}i\gamma^5\psi i\gamma^5\right).$	$\bar{\psi}\mathscr{J}=\bar{\psi}\left(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5\right).$	$\bar{\psi}\not{J}=\bar{\psi}(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5).$
1510.06699_ EQ0264 (D3)	051	■ 0.943 ● W +0	$\begin{aligned} L_{\mathrm{T}}\,\,\&=L_{\mathrm{M}}+L_{\mathrm{G}}\\ \&=L_{\varphi}+\frac{1}{2}\nu\mathcal{D}_a^*\phi\mathcal{D}^{*a}\phi-\lambda\phi^4-\frac{1}{2}a\phi^2\mathcal{R}-\frac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab}. \end{aligned}$	$L_T=L_M+L_G$ $=L_\varphi+\frac{1}{2}\nu\mathcal{D}_a^*\phi\mathcal{D}^{*a}\phi-\lambda\phi^4-\frac{1}{2}a\phi^2\mathcal{R}-\frac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab}.$	$L_T=L_M+L_G$ $=L_\varphi+\frac{1}{2}\nu\mathcal{D}_a^*\phi\mathcal{D}^{*a}\phi-\lambda\phi^4-\frac{1}{2}a\phi^2\mathcal{R}-\frac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab}.$
1510.06699_ EQ0019 (20)	009	■ 0.944 ● W +2	$\begin{aligned} D_{\mu}^*\varphi(x)\,\,\&\equiv\left[\partial_\mu+\frac{1}{2}A^{ab}{}_\mu(x)\Sigma_{ab}+wB_\mu(x)\right]\varphi(x)\\ &=[D_\mu+wB_\mu(x)]\varphi(x) \end{aligned}$	$D_\mu^*\varphi(x)\equiv\left[\partial_\mu+\frac{1}{2}A^{ab}{}_\mu(x)\Sigma_{ab}+wB_\mu(x)\right]\varphi(x)$ $=[D_\mu+wB_\mu(x)]\varphi(x)$	$D_\mu^*\varphi(x)\equiv[\partial_\mu+\frac{1}{2}A^{ab}{}_\mu(x)\Sigma_{ab}+wB_\mu(x)]\varphi(x)$ $=[D_\mu+wB_\mu(x)]\varphi(x),$
1510.06699_ EQ0167 (183)	031	■ 0.946 ● W +0	$\begin{aligned} \mathcal{L}_{\mathrm{T}}=&\mathcal{L}_{\mathrm{G}}\left(h,\partial h,\partial^2h,A^\dagger,\partial A^\dagger\right)\\ &+\mathcal{L}_{\mathrm{M}}\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A^\dagger,\partial A^\dagger\right), \end{aligned}$	$\mathcal{L}_T=\mathcal{L}_G\left(h,\partial h,\partial^2h,A^\dagger,\partial A^\dagger\right)$ $+\mathcal{L}_M\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A^\dagger,\partial A^\dagger\right),$	$\mathcal{L}_T=\mathcal{L}_G(h,\partial h,\partial^2h,A^\dagger,\partial A^\dagger)$ $+\mathcal{L}_M(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A^\dagger,\partial A^\dagger),$
1510.06699_ EQ0038 (39)	011	■ 0.946 ● N +0	$\begin{aligned} R^a{}_{\mu\nu}\equiv&\partial_\mu A^{ab}{}_\nu-\partial_\nu A^{ab}{}_\mu+A^a{}_{c\mu}A^{cb}{}_\nu-A^a{}_{c\nu}A^{cb}{}_\mu,\\ H_{\mu\nu}\equiv&\partial_\mu B_\nu-\partial_\nu B_\mu. \end{aligned}$	$R^{ab}{}_{\mu\nu}\equiv\partial_\mu A^{ab}{}_\nu-\partial_\nu A^{ab}{}_\mu+A^a{}_{c\mu}A^{cb}{}_\nu-A^a{}_{c\nu}A^{cb}{}_\mu,$ $H_{\mu\nu}\equiv\partial_\mu B_\nu-\partial_\nu B_\mu.$	$R^{ab}{}_{\mu\nu}\equiv\partial_\mu A^{ab}{}_\nu-\partial_\nu A^{ab}{}_\mu+A^a{}_{c\mu}A^{cb}{}_\nu-A^a{}_{c\nu}A^{cb}{}_\mu,$ $H_{\mu\nu}\equiv\partial_\mu B_\nu-\partial_\nu B_\mu.$
1510.06699_ EQ0124 (136)	023	■ 0.952 ● W +1	$\begin{aligned} h_a{}^{\prime\mu}\,\,\&=e^{\{-\rho\}}h_a{}^\mu\\ A^{\prime ab}{}_\mu&=A^{ab}{}_\mu+\theta\left(b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a\right), \end{aligned}$	$h_a^{\prime\mu}=e^{-\rho}h_a{}^\mu$ $A^{\prime ab}{}_\mu=A^{ab}{}_\mu+\theta\left(b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a\right),$	$h_a^{\prime\mu}=e^{-\rho}h_a{}^\mu$ $A^{\prime ab}{}_\mu=A^{ab}{}_\mu+\theta(b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ EQ0221 (255a)	041	■ 0.954 ● W +1	$\begin{aligned} & \mathcal{T}^{\prime a}{}_{bc} = e^{-\rho} \left\{ \mathcal{T}^a{}_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \right\}, \\ & \mathcal{T}'_b = e^{-\rho} \{ \mathcal{T}_b + 3(1-\theta) \mathcal{P}_b \}. \end{aligned}$	$\mathcal{T}'^a{}_{bc} = e^{-\rho} \left\{ \mathcal{T}^a{}_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \right\},$ $\mathcal{T}'_b = e^{-\rho} \{ \mathcal{T}_b + 3(1-\theta) \mathcal{P}_b \}.$	$\mathcal{T}'^a{}_{bc} = e^{-\rho} \{ \mathcal{T}^a{}_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \},$ $\mathcal{T}'_b = e^{-\rho} \{ \mathcal{T}_b + 3(1-\theta) \mathcal{P}_b \}.$
1510.06699_ EQ0086 (92)	018	■ 0.955 ● N +2	$\begin{aligned} & \mathcal{D}_a \chi \equiv (\mathcal{D}_a + w \mathcal{B}_a) \chi \\ & \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \left(\widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a \right) \widehat{\chi} \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^* \widehat{\chi} \end{aligned}$	$\mathcal{D}_a^* \chi \equiv (\mathcal{D}_a + w \mathcal{B}_a) \chi$ $= \left(\frac{\phi}{\phi_0} \right)^{1-w} \left(\widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a \right) \widehat{\chi} \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^* \widehat{\chi}$	$\mathcal{D}_a^* \chi \equiv (\mathcal{D}_a + w \mathcal{B}_a) \chi$ $= \left(\frac{\phi}{\phi_0} \right)^{1-w} \left(\widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a \right) \widehat{\chi} \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^* \widehat{\chi},$
1510.06699_ EQ0068 (72)	015	■ 0.956 ● N +0	$\mathcal{L}_{\mathrm{D}} = h^{-1} \left(\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi \right).$	$\mathcal{L}_{\mathrm{D}} = h^{-1} \left(\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi \right).$	$\mathcal{L}_{\mathrm{D}} = h^{-1} (\tfrac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi).$
1510.06699_ EQ0093 (101)	019	■ 0.959 ● N +0	$v^{c0} \mathcal{D}_c^* p_a = e \mu^2 \phi^0 \mathcal{D}_a^* \phi,$	$v^{c0} \mathcal{D}_c^* p_a = e \mu^2 \phi^0 \mathcal{D}_a^* \phi,$	$v^{c0} \mathcal{D}_c^* p_a = e \mu^2 \phi^0 \mathcal{D}_a^* \phi,$
1510.06699_ EQ0026 (27)	010	■ 0.964 ● N +0	$h_{\mathrm{a}}^{\prime \mu}(x') = \frac{\partial x^{\prime \mu}}{\partial x^{\nu}} e^{-\rho(x)} \Lambda_{\mathrm{a}}{}^b(x) h_b{}^{\nu}(x).$	$h_a^{\prime \mu}(x') = \frac{\partial x^{\prime \mu}}{\partial x^{\nu}} e^{-\rho(x)} \Lambda_a{}^b(x) h_b{}^{\nu}(x).$	$h_a^{\prime \mu}(x') = \frac{\partial x^{\prime \mu}}{\partial x^{\nu}} e^{-\rho(x)} \Lambda_a{}^b(x) h_b{}^{\nu}(x).$
1510.06699_ EQ0008 (9)	008	■ 0.965 ● K +0	$L_{\mathrm{M}}(\varphi'(x'), \partial'_\mu \varphi'(x')) = e^{-4\rho} L_{\mathrm{M}}(\varphi(x), \partial_\mu \varphi(x)),$	$L_{\mathrm{M}}(\varphi'(x'), \partial'_\mu \varphi'(x')) = e^{-4\rho} L_{\mathrm{M}}(\varphi(x), \partial_\mu \varphi(x)),$	$L_{\mathrm{M}}(\varphi'(x'), \partial'_\mu \varphi'(x')) = e^{-4\rho} L_{\mathrm{M}}(\varphi(x), \partial_\mu \varphi(x)),$
1510.06699_ EQ0057 (61b)	014	■ 0.971 ● C +3	$\begin{aligned} & t^a{}_\mu + \tau^a{}_\mu = 0 \\ & s_{ab}{}^\mu + \sigma_{ab}{}^\mu = 0 \\ & j^\mu + \zeta^\mu = 0 \end{aligned}$	$t^a{}_\mu + \tau^a{}_\mu = 0$ $s_{ab}{}^\mu + \sigma_{ab}{}^\mu = 0$ $j^\mu + \zeta^\mu = 0$	$t^a{}_\mu + \tau^a{}_\mu = 0,$ $s_{ab}{}^\mu + \sigma_{ab}{}^\mu = 0,$ $j^\mu + \zeta^\mu = 0.$
1510.06699_ EQ0250 (B5)	049	■ 0.973 ● K +0	$L_{\mathrm{G}} = -\kappa^{-1} \left(\Lambda + a^0 \mathcal{R} \right) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$	$L_{\mathrm{G}} = -\kappa^{-1} \left(\Lambda + a^0 \mathcal{R} \right) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$	$L_{\mathrm{G}} = -\kappa^{-1} (\Lambda + a^0 \mathcal{R}) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$
1510.06699_ EQ0152 (166)	028	■ 0.974 ● K +0	$L_{\mathrm{G}} = L_{\mathcal{R}^\dagger 2} + L_{\mathcal{H}^\dagger 2},$	$L_{\mathrm{G}} = L_{\mathcal{R}^\dagger 2} + L_{\mathcal{H}^\dagger 2},$	$L_{\mathrm{G}} = L_{\mathcal{R}^\dagger 2} + L_{\mathcal{H}^\dagger 2},$
1510.06699_ EQ0228 (262)	043	■ 0.976 ● N +0	${}^0\mathcal{D}'_c \varphi' = e^{(w-1)\rho} \left[{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi \right],$	${}^0\mathcal{D}'_c \varphi' = e^{(w-1)\rho} \left[{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi \right],$	${}^0\mathcal{D}'_c \varphi' = e^{(w-1)\rho} [{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi],$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00220 (254b)	041	■0.979 ●C +4	$\begin{aligned} & \backslash begin{aligned} \backslash mathcal{R}^{\wedge\{\prime a\ b\}}{ }_{\{c\ d\}} \& =e^{\{-2\ \rho\}}\backslash left\{\backslash mathcal{R}^{\wedge\{a\ b\}}{ }_{\{c\ d\}}+2\ \theta\backslash theta\ \backslash delta_{\{d\}^{\{a\}}\backslash left(\backslash mathcal{D}_{\{c\}}-\backslash theta\ \backslash mathcal{P}_{\{c\}}\backslash right)\ \backslash mathcal{P}^{\wedge\{b\}}\}-2\ \backslash theta\ \backslash delta_{\{c\}^{\{a\}}\backslash left(\backslash mathcal{D}_{\{d\}}-\backslash theta\ \backslash mathcal{P}_{\{d\}}\backslash right)\ \backslash mathcal{P}^{\wedge\{b\}}\}-2\ \backslash theta\ \backslash mathcal{P}^{\wedge\{[a\}\ \backslash mathcal{T}^{\wedge\{b\}}\}}{ }_{\{c\ d\}}-2\ \backslash theta^{\{2\}}\backslash delta_{\{c\}^{\{a\}}\ \backslash delta_{\{d\}^{\{b\}}}\ \backslash mathcal{P}^{\wedge\{e\}}\ \backslash mathcal{P}_{\{e\}}\backslash right\},\ \backslash\backslash\ \backslash mathcal{R}^{\wedge\{\prime a\}}{ }_{\{c\}}\& =e^{\{-2\ \rho\}}\backslash left\{\backslash mathcal{R}^{\wedge\{a\}}{ }_{\{c\}}-2\ \backslash theta\ \backslash mathcal{D}_{\{c\}}\ \backslash mathcal{P}^{\wedge\{a\}}-\backslash theta\ \backslash delta_{\{c\}^{\{a\}}\ \backslash mathcal{D}_{\{b\}}\ \backslash mathcal{P}^{\wedge\{b\}}+2\ \backslash theta^{\{2\}}\ \backslash mathcal{P}^{\wedge\{a\}}\ \backslash mathcal{P}_{\{c\}}-2\ \backslash theta^{\{2\}}\backslash delta_{\{c\}^{\{a\}}\ \backslash mathcal{P}^{\wedge\{2\}}+\backslash theta\ \backslash mathcal{P}^{\wedge\{b\}}\ \backslash mathcal{T}^{\wedge\{a\}}{ }_{\{c\}}\}-\backslash theta\ \backslash mathcal{P}^{\wedge\{a\}}\ \backslash mathcal{T}^{\wedge\{c\}}\backslash right\},\ \backslash\backslash\ \backslash mathcal{R}^{\wedge\{\prime a\}}{ }_{\{c\}}\& =e^{\{-2\ \rho\}}\backslash left\{\backslash mathcal{R}^{\wedge\{a\}}{ }_{\{c\}}-6\ \backslash theta\backslash left(\backslash mathcal{D}_{\{c\}}^{\wedge\{a\}}+\backslash frac{1\}{3}\ \backslash mathcal{T}_{\{c\}}\backslash right)\ \backslash mathcal{P}^{\wedge\{a\}}-6\ \backslash theta^{\{2\}}\ \backslash mathcal{P}^{\wedge\{2\}}\backslash right\},\ \backslash end{aligned} \end{aligned}$	$\begin{aligned} \mathcal{R}^{\prime ab}_{\ cd} &= e^{-2\rho}\left\{\mathcal{R}^{ab}_{\ cd} + 2\theta\delta_d^{[a}(\mathcal{D}_c - \theta\mathcal{P}_c)\mathcal{P}^{b]} - 2\theta\delta_c^{[a}(\mathcal{D}_d - \theta\mathcal{P}_d)\mathcal{P}^{b]} - 2\theta\mathcal{P}^{[a}\mathcal{T}^{b]}_{cd} - 2\theta^2\delta_c^{[a}\delta_d^{b]}\mathcal{P}^e\mathcal{P}_e\right\}, \\ \mathcal{R}^{\prime a}_{\ c} &= e^{-2\rho}\left\{\mathcal{R}^a_{\ c} - 2\theta\mathcal{D}_c\mathcal{P}^a - \theta\delta_c^a\mathcal{D}_b\mathcal{P}^b + 2\theta^2\mathcal{P}^a\mathcal{P}_c - 2\theta^2\delta_c^a\mathcal{P}^2 + \theta\mathcal{P}^b\mathcal{T}^a_{\ cb} - \theta\mathcal{P}^a\mathcal{T}_c\right\}, \\ \mathcal{R}' &= e^{-2\rho}\left\{\mathcal{R} - 6\theta\left(\mathcal{D}_a + \frac{1}{3}\mathcal{T}_a\right)\mathcal{P}^a - 6\theta^2\mathcal{P}^2\right\}, \end{aligned}$	$\begin{aligned} \mathcal{R}^{\prime ab}_{\ cd} &= e^{-2\rho}\{\mathcal{R}^{ab}_{\ cd} + 2\theta\delta_d^{[a}(\mathcal{D}_c - \theta\mathcal{P}_c)\mathcal{P}^{b]} - 2\theta\delta_c^{[a}(\mathcal{D}_d - \theta\mathcal{P}_d)\mathcal{P}^{b]} - 2\theta\mathcal{P}^{[a}\mathcal{T}^{b]}_{cd} - 2\theta^2\delta_c^{[a}\delta_d^{b]}\mathcal{P}^e\mathcal{P}_e\}, \\ \mathcal{R}^{\prime a}_{\ c} &= e^{-2\rho}\{\mathcal{R}^a_{\ c} - 2\theta\mathcal{D}_c\mathcal{P}^a - \theta\delta_c^a\mathcal{D}_b\mathcal{P}^b + 2\theta^2\mathcal{P}^a\mathcal{P}_c - 2\theta^2\delta_c^a\mathcal{P}^2 + \theta\mathcal{P}^b\mathcal{T}^a_{\ cb} - \theta\mathcal{P}^a\mathcal{T}_c\}, \\ \mathcal{R}' &= e^{-2\rho}\{\mathcal{R} - 6\theta(\mathcal{D}_a + \tfrac{1}{3}\mathcal{T}_a)\mathcal{P}^a - 6\theta^2\mathcal{P}^2\}, \end{aligned}$
1510.06699_ E00120 (131)	023	■0.979 ●W +1	$\begin{aligned} & \backslash begin{aligned} { }^{\wedge\{0\}} \backslash Gamma^{\wedge\{\ast\}}_{\backslash lambda b d a}{ }_{\backslash mu \backslash nu} \\ & \& =\frac{1\}{2} g^{\wedge\{\backslash lambda b d a\}}\backslash left(\backslash partial_{\backslash mu}{ }^{\wedge\{\ast\}} g_{\backslash nu \backslash rho}+\backslash partial_{\backslash nu}{ }^{\wedge\{\ast\}} g_{\backslash mu \backslash rho}-\backslash partial_{\backslash rho}{ }^{\wedge\{\ast\}} g_{\backslash mu \backslash nu}\backslash right)\ \backslash\backslash \& = { }^{\wedge\{0\}} \backslash Gamma^{\wedge\{\backslash lambda b d a\}}{ }_{\backslash mu \backslash nu}+\backslash delta_{\backslash nu}^{\backslash lambda} B_{\backslash mu}+\backslash delta_{\backslash mu}^{\backslash lambda} B_{\backslash nu}-g_{\backslash mu \backslash nu} B^{\backslash lambda} \\ & \backslash end{aligned} \end{aligned}$	$\begin{aligned} {}^0\Gamma^{*\lambda}_{\mu\nu} &= \frac{1}{2}g^{\lambda\rho}\left(\partial_{\mu}^*g_{\nu\rho} + \partial_{\nu}^*g_{\mu\rho} - \partial_{\rho}^*g_{\mu\nu}\right) \\ &= {}^0\Gamma^{\lambda}_{\mu\nu} + \delta_{\nu}^{\lambda}B_{\mu} + \delta_{\mu}^{\lambda}B_{\nu} - g_{\mu\nu}B^{\lambda} \end{aligned}$	$\begin{aligned} {}^0\Gamma^{*\lambda}_{\mu\nu} &= \tfrac{1}{2}g^{\lambda\rho}(\partial_{\mu}^*g_{\nu\rho} + \partial_{\nu}^*g_{\mu\rho} - \partial_{\rho}^*g_{\mu\nu}) \\ &= {}^0\Gamma^{\lambda}_{\mu\nu} + \delta_{\nu}^{\lambda}B_{\mu} + \delta_{\mu}^{\lambda}B_{\nu} - g_{\mu\nu}B^{\lambda}, \end{aligned}$
1510.06699_ E00110 (118)	022	■0.979 ●N +1	$\begin{aligned} & D_{\backslash mu}{ }^{\wedge\{\ast\}} J^{\wedge\{a\}}=\backslash partial_{\backslash mu} J^{\wedge\{a\}}+w B_{\backslash mu} J^{\wedge\{a\}}+A^{\wedge\{a\}}_{\backslash b}{ }_{\backslash mu} J^{\wedge\{b\}}=\backslash partial_{\backslash mu}{ }^{\wedge\{\ast\}} J^{\wedge\{a\}}+A^{\wedge\{a\}}_{\backslash b}{ }_{\backslash mu} J^{\wedge\{b\}} \end{aligned}$	$D_{\mu}^*J^a = \partial_{\mu}J^a + wB_{\mu}J^a + A^a_{\ b\mu}J^b = \partial_{\mu}^*J^a + A^a_{\ b\mu}J^b,$	$D_{\mu}^*J^a = \partial_{\mu}J^a + wB_{\mu}J^a + A^a_{\ b\mu}J^b = \partial_{\mu}^*J^a + A^a_{\ b\mu}J^b,$
1510.06699_ E00233 (267)	044	■0.982 ●C +5	$\begin{aligned} & \backslash begin{aligned} { }^{\wedge\{0\}} \backslash mathcal{C}_{\{a\ b\ c\ d\}}=& { }^{\wedge\{0\}} \backslash mathcal{R}_{\{a\ b\ c\ d\}} \& -\backslash frac{1\}{2}\backslash left(\backslash eta_{\{a\ c\}}{ }^{\wedge\{0\}} \backslash mathcal{R}_{\{b\ d\}}-\backslash eta_{\{a\ d\}}{ }^{\wedge\{0\}} \backslash mathcal{R}_{\{b\ c\}}-\backslash eta_{\{b\ c\}}{ }^{\wedge\{0\}} \backslash mathcal{R}_{\{a\ d\}}+\backslash eta_{\{b\ d\}}{ }^{\wedge\{0\}} \backslash mathcal{R}_{\{a\ c\}}\backslash right)\ \backslash\backslash \& +\backslash frac{1\}{6}\backslash left(\backslash eta_{\{a\ c\}}\ \backslash eta_{\{b\ d\}}-\backslash eta_{\{a\ d\}}\ \backslash eta_{\{b\ c\}}\backslash right)^{\wedge\{0\}} \backslash mathcal{R}. \backslash end{aligned} \end{aligned}$	$\begin{aligned} {}^0\mathcal{C}_{abcd} &= {}^0\mathcal{R}_{abcd} - \frac{1}{2}\left(\eta_{ac}{}^0\mathcal{R}_{bd} - \eta_{ad}{}^0\mathcal{R}_{bc} - \eta_{bc}{}^0\mathcal{R}_{ad} + \eta_{bd}{}^0\mathcal{R}_{ac}\right) \\ &\quad + \frac{1}{6}\left(\eta_{ac}\eta_{bd} - \eta_{ad}\eta_{bc}\right){}^0\mathcal{R}. \end{aligned}$	$\begin{aligned} {}^0\mathcal{C}_{abcd} &= {}^0\mathcal{R}_{abcd} - \tfrac{1}{2}(\eta_{ac}{}^0\mathcal{R}_{bd} - \eta_{ad}{}^0\mathcal{R}_{bc} - \eta_{bc}{}^0\mathcal{R}_{ad} + \eta_{bd}{}^0\mathcal{R}_{ac}) \\ &\quad + \tfrac{1}{6}(\eta_{ac}\eta_{bd} - \eta_{ad}\eta_{bc}){}^0\mathcal{R}. \end{aligned} \tag{267}$
1510.06699_ E00184 (200)	034	■0.982 ●N +0	$\backslash phi\ v^{\{b\}}{ }^{\wedge\{0\}} \backslash mathcal{D}_{\{b\}}^{\wedge\{\dagger\}} v_{\{a\}}=\backslash left(\backslash delta_{\{a\}^{\wedge\{b\}}}-v_{\{a\}}\ v^{\{b\}}\backslash right)^{\wedge\{0\}} \backslash mathcal{D}_{\{b\}}^{\wedge\{\dagger\}} \backslash phi$	$\phi v^{b\,0}\mathcal{D}_b^\dagger v_a = (\delta_a^b - v_a v^b)^0\mathcal{D}_b^\dagger \phi.$	$\phi\,v^{b\,0}\mathcal{D}_b^\dagger v_a = (\delta_a^b - v_a v^b)^0\mathcal{D}_b^\dagger \phi.$
1510.06699_ E00243 (A7)	047	■0.983 ●K +0	$\backslash nexists \backslash not \backslash partial \backslash psi = J^{\wedge\{\mu\}} \backslash partial_{\backslash mu} \backslash psi - i \backslash sigma^{\wedge\{\mu\ \nu\}} J_{\backslash mu} \backslash partial_{\backslash nu} \backslash psi$	$\not\partial\psi = J^\mu\partial_\mu\psi - i\sigma^{\mu\nu}J_\mu\partial_\nu\psi.$	$\not\!\partial\psi = J^\mu\partial_\mu\psi - i\sigma^{\mu\nu}J_\mu\partial_\nu\psi.$
1510.06699_ E00118 (129)	023	■0.984 ●N +0	$\backslash left[\backslash nabla_{\backslash mu}{ }^{\wedge\{\ast\}}, \backslash nabla_{\backslash nu}{ }^{\wedge\{\ast\}} \backslash right] J^{\wedge\{\rho\}} = \backslash widetilde{R}^{\wedge\{\rho\}}{ }_{\{ \backslash sigma \ \mu \ \nu \}} J^{\wedge\{\backslash sigma\}} + w H_{\backslash mu \backslash nu} J^{\wedge\{\rho\}} - T^{\wedge\{\ast\}}_{\backslash mu \backslash nu} \backslash nabla_{\backslash sigma}{ }^{\wedge\{\ast\}} V^{\wedge\{\rho\}}$	$[\nabla_{\mu}^*, \nabla_{\nu}^*]J^{\rho} = \widetilde{R}^{\rho}_{\ \sigma\mu\nu}J^{\sigma} + wH_{\mu\nu}J^{\rho} - T^{*\sigma}_{\ \mu\nu}\nabla_{\sigma}^*V^{\rho}.$	$[\nabla_{\mu}^*, \nabla_{\nu}^*]J^{\rho} = \widetilde{R}^{\rho}_{\ \sigma\mu\nu}J^{\sigma} + wH_{\mu\nu}J^{\rho} - T^{*\sigma}_{\ \mu\nu}\nabla_{\sigma}^*V^{\rho}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ EQ0111 (119)	022	■ 0.984 ● N +0	$\Delta_{\mu}^{\ast}\equiv\partial_{\mu}^{\ast}+\Gamma_{\rho\mu}^{\sigma}\mathbf{X}^{\rho}_{\sigma}+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\ast}+D_{\mu}^{\ast}-\partial_{\mu}^{\ast},$	$\Delta_{\mu}^{\ast}\equiv\partial_{\mu}^{\ast}+\Gamma^{\sigma}_{\rho\mu}\mathbf{X}^{\rho}_{\sigma}+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\ast}+D_{\mu}^{\ast}-\partial_{\mu}^{\ast},$	
1510.06699_ EQ0018 (19)	009	■ 0.985 ● N +0	$\partial_{\mu}^{\prime}\varphi^{\prime}(x^{\prime})=\frac{\partial x^{\nu}}{\partial x^{\prime\mu}}e^{w\rho(x)}\left[\mathcal{S}(\Lambda(x))\partial_{\nu}\varphi(x)+\partial_{\nu}\mathcal{S}(\Lambda(x))\varphi(x)+w\partial_{\nu}\rho\mathcal{S}(\Lambda(x))\varphi(x)\right].$	$\partial_{\mu}^{\prime}\varphi^{\prime}(x^{\prime})=\frac{\partial x^{\nu}}{\partial x^{\prime\mu}}e^{w\rho(x)}\left[\mathcal{S}(\Lambda(x))\partial_{\nu}\varphi(x)+\partial_{\nu}\mathcal{S}(\Lambda(x))\varphi(x)+w\partial_{\nu}\rho\mathcal{S}(\Lambda(x))\varphi(x)\right].$	
1510.06699_ EQ0033 (34)	010	■ 0.985 ● N +0	$D_{\mu}^{\ast}b^a_{\nu}=\partial_{\mu}^{\ast}b^a_{\nu}+A^a_{b\mu}b^b_{\nu}.$	$D_{\mu}^{\ast}b^a_{\nu}=\partial_{\mu}^{\ast}b^a_{\nu}+A^a_{b\mu}b^b_{\nu}.$	
1510.06699_ EQ0175 (188)	032	■ 0.985 ● N +0	$\tau^{\dagger a}_b=\tau^a_b+2\sigma_{cb}{}^a\mathcal{V}^c.$	$\tau^{\dagger a}_b=\tau^a_b+2\sigma_{cb}{}^a\mathcal{V}^c.$	
1510.06699_ EQ0080 (85)	017	■ 0.986 ● N +1	$\mathcal{D}_{[a}\mathcal{F}_{bc]}-\mathcal{T}^{*d}{}_{[ab}\mathcal{F}_{c]d}=0.$	$\mathcal{D}_{[a}\mathcal{F}_{bc]}-\mathcal{T}^{*d}{}_{[ab}\mathcal{F}_{c]d}=0.$	
1510.06699_ EQ0188 (204)	035	■ 0.987 ● W +0	$\begin{array}{l} \mathcal{L}_{\mathcal{M}}=\mathcal{L}_{\mathcal{M}}^{\mathcal{L}}=\mathcal{L}_{\mathcal{M}}^{\mathcal{L}}\left[\frac{1}{2}i\bar{\psi}\gamma^{a0}\overleftrightarrow{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_a^{\dagger}\phi\right)\left({}^0\mathcal{D}^{\dagger a}\phi\right)-\lambda\phi^4\right.\\ \left.-a\phi^{20}\mathcal{R}^{\dagger}\right].(204) \end{array}$	$\mathcal{L}_{\mathcal{M}}=h^{-1}\left[\frac{1}{2}i\bar{\psi}\gamma^{a0}\overleftrightarrow{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_a^{\dagger}\phi\right)\left({}^0\mathcal{D}^{\dagger a}\phi\right)-\lambda\phi^4\right.\\ \left.-a\phi^{20}\mathcal{R}^{\dagger}\right].(204)$	$\mathcal{L}_{\mathcal{M}}=h^{-1}[\frac{1}{2}i\bar{\psi}\gamma^a{}^0\overleftrightarrow{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^{\dagger}\phi)({}^0\mathcal{D}^{\dagger a}\phi)-\lambda\phi^4-a\phi^{20}\mathcal{R}^{\dagger}].(204)$
1510.06699_ EQ0106 (114)	021	■ 0.987 ● N +0	$\hat{\boldsymbol{\mathrm{e}}}_{\mathfrak{a}}=\mathfrak{h}_{\mathfrak{a}}\{\}_{\mu}\boldsymbol{\mathrm{e}}_{\mu},\quad\boldsymbol{\mathrm{e}}_{\mu}=\mathfrak{b}^{\mathfrak{a}}_{\mu}\hat{\boldsymbol{\mathrm{e}}}_{\mathfrak{a}}$	$\hat{e}_a=h_a{}^{\mu}e_{\mu},\quad e_{\mu}=b^a{}_{\mu}\hat{e}_a,$	$\hat{e}_a=h_a{}^{\mu}e_{\mu},\quad e_{\mu}=b^a{}_{\mu}\hat{e}_a,$
1510.06699_ EQ0154 (168b)	029	■ 0.988 ● C +3	$\begin{aligned} t^a{}_b+\tau^a{}_b&=0\\ s_{ab}{}^c+\sigma_{ab}{}^c&=0\\ j^a+\zeta^a&=0 \end{aligned}$	$\begin{aligned} t^a{}_b+\tau^a{}_b&=0\\ s_{ab}{}^c+\sigma_{ab}{}^c&=0\\ j^a+\zeta^a&=0 \end{aligned}$	$\begin{aligned} t^a{}_b+\tau^a{}_b&=0,\\ s_{ab}{}^c+\sigma_{ab}{}^c&=0,\\ j^a+\zeta^a&=0, \end{aligned}$
1510.06699_ EQ0204 (224)	038	■ 0.988 ● C +6	$L_{\varphi}+L_{\phi}+\phi^2L_{\mathcal{R}\dagger}+\phi^2L_{\mathcal{T}\dagger 2}$	$L_{\varphi}+L_{\phi}+\phi^2L_{\mathcal{R}\dagger}+\phi^2L_{\mathcal{T}\dagger 2}$	$L_{\varphi}+L_{\phi}+\phi^2L_{\mathcal{R}\dagger}+\phi^2L_{\mathcal{T}\dagger 2},$ where
1510.06699_ EQ0073 (77)	016	■ 0.990 ● N +2	$\begin{aligned} h\tau^a{}_b&=\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b\psi-\delta_b^aL_{\mathcal{D}},\\ h\sigma_{abc}&=\frac{1}{4}i\bar{\psi}\gamma_{[a}\gamma_b\gamma_{c]}\psi \end{aligned}$	$\begin{aligned} h\tau^a{}_b&=\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b\psi-\delta_b^aL_{\mathcal{D}},\\ h\sigma_{abc}&=\frac{1}{4}i\bar{\psi}\gamma_{[a}\gamma_b\gamma_{c]}\psi \end{aligned}$	$\begin{aligned} h\tau^a{}_b&=\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b\psi-\delta_b^aL_{\mathcal{D}},\\ h\sigma_{abc}&=\frac{1}{4}i\bar{\psi}\gamma_{[a}\gamma_b\gamma_{c]}\psi, \end{aligned}$
1510.06699_ EQ0163 (179)	030	■ 0.990 ● N +0	$\zeta^{\dagger a}\equiv\zeta^a-2\sigma^{ab}{}_b,$	$\zeta^{\dagger a}\equiv\zeta^a-2\sigma^{ab}{}_b,$	$\zeta^{\dagger a}\equiv\zeta^a-2\sigma^{ab}{}_b,$
1510.06699_ EQ0247 (B2)	048	■ 0.991 ● K +0	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(h,\partial h,A,\partial A)+\mathcal{L}_{\mathcal{M}}(\varphi,\partial\varphi,h,A),$	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(h,\partial h,A,\partial A)+\mathcal{L}_{\mathcal{M}}(\varphi,\partial\varphi,h,A),$	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(h,\partial h,A,\partial A)+\mathcal{L}_{\mathcal{M}}(\varphi,\partial\varphi,h,A),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ EQ0189 (205)	035	■ 0.991 ● W +0	$\begin{aligned} \mathrm{L}_{\mathrm{T}} = & \mathrm{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) \\ & + \mathrm{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V). \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V).$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V).$
1510.06699_ EQ0191 (207)	036	■ 0.992 ● N +0	$\delta J^a = -\left[A^{\dagger a}{}_{b\mu} - w\left(V_\mu + \frac{1}{3}T_\mu\right)\delta_b^a\right]J^b\delta x^\mu.$	$\delta J^a = -[A^{\dagger a}{}_{b\mu} - w(V_\mu + \frac{1}{3}T_\mu)\delta_b^a]J^b\delta x^\mu.$	$\delta J^a = -[A^{\dagger a}{}_{b\mu} - w(V_\mu + \frac{1}{3}T_\mu)\delta_b^a]J^b\delta x^\mu.$
1510.06699_ EQ0015 (16)	008	■ 0.993 ● N +0	$\begin{aligned} L_{\mathrm{EM}}^{\prime} = & -\frac{1}{4}F_{\mu\nu}^{\prime}F^{\prime\mu\nu} - J^{\prime\mu}A_{\mu}^{\prime} \\ = & -\frac{1}{4}e^{2(w-1)\rho}F_{\mu\nu}F^{\mu\nu} - e^{(w+w_J)\rho}J^{\mu}A_{\mu}, \end{aligned}$	$L'_{\mathrm{EM}} = -\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - J'^{\mu}A'_{\mu} = -\frac{1}{4}e^{2(w-1)\rho}F_{\mu\nu}F^{\mu\nu} - e^{(w+w_J)\rho}J^{\mu}A_{\mu},$	$L'_{\mathrm{EM}} = -\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - J'^{\mu}A'_{\mu} = -\frac{1}{4}e^{2(w-1)\rho}F_{\mu\nu}F^{\mu\nu} - e^{(w+w_J)\rho}J^{\mu}A_{\mu},$
1510.06699_ EQ0065 (69b)	015	■ 0.993 ● N +1	$\begin{aligned} & \left(\mathrm{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\sigma_{ab}{}^c\right) + h\tau_{[ab]} + \frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi = 0, \\ & \left(\mathrm{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\tau^c{}_d\right) - h\left(\sigma_{ab}{}^c\mathcal{R}^{ab}{}_{cd} - \tau^c{}_b\mathcal{T}^{*b}{}_{cd} + \zeta^c\mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^*\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^*\varphi = 0, \\ & \left(\mathrm{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\zeta^c\right) - h\tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi = 0. \end{aligned}$	$\begin{aligned} & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\sigma_{ab}{}^c\right) + h\tau_{[ab]} + \frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi = 0, \\ & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\tau^c{}_d\right) - h\left(\sigma_{ab}{}^c\mathcal{R}^{ab}{}_{cd} - \tau^c{}_b\mathcal{T}^{*b}{}_{cd} + \zeta^c\mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^*\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^*\varphi = 0, \\ & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\zeta^c\right) - h\tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi = 0. \end{aligned}$	$\begin{aligned} & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\sigma_{ab}{}^c\right) + h\tau_{[ab]} + \frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi = 0, \\ & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\tau^c{}_d\right) - h\left(\sigma_{ab}{}^c\mathcal{R}^{ab}{}_{cd} - \tau^c{}_b\mathcal{T}^{*b}{}_{cd} + \zeta^c\mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^*\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^*\varphi = 0, \\ & \left(\mathcal{D}_{\mathrm{c}}^* + \mathcal{T}_{\mathrm{c}}^*\right)\left(h\zeta^c\right) - h\tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi + \frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi = 0. \end{aligned}$
1510.06699_ EQ0049 (52)	013	■ 0.993 ● C +3	$\begin{aligned} & \mathrm{D}_{[a}^*\mathcal{R}^{de}{}_{bc]} - \mathcal{T}^{*f}{}_{[ab}\mathcal{R}^{de}{}_{c]f} = 0, \\ & \mathcal{D}_{[a}^*\mathcal{T}^{*d}{}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab}\delta_{c]}^d = 0, \\ & \mathcal{D}_{[a}^*\mathcal{H}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{H}_{c]e} = 0. \end{aligned}$	$\begin{aligned} & \mathcal{D}_{[a}^*\mathcal{T}^{*d}{}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab}\delta_{c]}^d = 0, \\ & \mathcal{D}_{[a}^*\mathcal{H}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{H}_{c]e} = 0. \end{aligned}$	$\begin{aligned} & \mathcal{D}_{[a}^*\mathcal{R}^{de}{}_{bc]} - \mathcal{T}^{*f}{}_{[ab}\mathcal{R}^{de}{}_{c]f} = 0, \\ & \mathcal{D}_{[a}^*\mathcal{T}^{*d}{}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab}\delta_{c]}^d = 0, \\ & \mathcal{D}_{[a}^*\mathcal{H}_{bc]} - \mathcal{T}^{*e}{}_{[ab}\mathcal{H}_{c]e} = 0. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0217 (250)	040	■ 0.995 ● W +1	$\begin{aligned} \delta_{\phi} L_{\phi} &= -\nu D_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) - 4\lambda \phi^3. \\ \partial_{\phi} L_{\mathcal{R}} &= -a \phi^2 \mathcal{R}^{\dagger}, \\ \partial_{\phi} L_{\mathcal{T}^2} &= 2\beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2\beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}. \end{aligned}$	$\delta_{\phi} L_{\phi} = -\nu D_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) - 4\lambda \phi^3.$ $\partial_{\phi} L_{\mathcal{R}} = -a \phi^2 \mathcal{R}^{\dagger},$ $\partial_{\phi} L_{\mathcal{T}^2} = 2\beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2\beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}.$	$\delta_{\phi} L_{\phi} = -\nu D_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) - 4\lambda \phi^3.$ $\partial_{\phi} L_{\mathcal{R}} = -a \phi^2 \mathcal{R}^{\dagger},$ $\partial_{\phi} L_{\mathcal{T}^2} = 2\beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2\beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}.$
1510.06699_ EQ0126 (139)	024	■ 0.995 ● C +9	$\begin{aligned} \mathcal{R}^{'ab}_{cd} &= e^{-2\rho} \left\{ \mathcal{R}^{ab}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\}, \\ \mathcal{T}^{'a}_{bc} &= e^{-\rho} \left\{ \mathcal{T}^a_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \right\}. \end{aligned}$	$\mathcal{R}^{'ab}_{cd} = e^{-2\rho} \left\{ \mathcal{R}^{ab}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\},$ $\mathcal{T}^{'a}_{bc} = e^{-\rho} \left\{ \mathcal{T}^a_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \right\}.$	$\mathcal{R}^{'ab}_{cd} = e^{-2\rho} \{ \mathcal{R}^{ab}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \},$ $\mathcal{T}^{'a}_{bc} = e^{-\rho} \{ \mathcal{T}^a_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta_{c]}^a \}.$
1510.06699_ EQ0021 (22)	009	■ 0.996 ● N +0	$D_{\mu}^{\ast\prime}\varphi'(x')=\frac{\partial x^{\nu}}{\partial x'^{\mu}}e^{w\rho(x)}\mathsf{S}(\Lambda(x))D_{\nu}^{\ast}\varphi(x).$	$D_{\mu}^{\ast\prime}\varphi'(x')=\frac{\partial x^{\nu}}{\partial x'^{\mu}}e^{w\rho(x)}\mathsf{S}(\Lambda(x))D_{\nu}^{\ast}\varphi(x).$	$D_{\mu}^{\ast\prime}\varphi'(x')=\frac{\partial x^{\nu}}{\partial x'^{\mu}}e^{w\rho(x)}\mathsf{S}(\Lambda(x))D_{\nu}^{\ast}\varphi(x).$
1510.06699_ EQ0165 (181)	030	■ 0.996 ● N +0	$\mathcal{T}^{\natural a}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\natural}b^a{}_{\nu}-D_{\nu}^{\natural}b^a{}_{\mu}\right),$	$\mathcal{T}^{\natural a}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\natural}b^a{}_{\nu}-D_{\nu}^{\natural}b^a{}_{\mu}\right),$	$\mathcal{T}^{\natural a}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\natural}b^a{}_{\nu}-D_{\nu}^{\natural}b^a{}_{\mu}\right),$
1510.06699_ EQ0215 (246)	039	■ 0.997 ● N -1	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$
1510.06699_ EQ0242 (A6)	047	■ 0.997 ● N -1	$L_{\mathrm{D}}=\frac{\Re\left[i\bar{\psi}\left(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5\right),\mathbb{J}\partial\psi\right]}{\left(\bar{\psi}\psi\right)^2+\left(\bar{\psi}i\gamma^5\psi\right)^2}-m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{\Re\left[i\bar{\psi}\left(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5\right),\mathbb{J}\partial\psi\right]}{\left(\bar{\psi}\psi\right)^2+\left(\bar{\psi}i\gamma^5\psi\right)^2}-m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{\Re[i\bar{\psi}(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5)\not{J}\not{\partial}\psi]}{(\psi\psi)^2+(\psi i\gamma^5\psi)^2}-m\bar{\psi}\psi.$
1510.06699_ EQ0239 (A3)	047	■ 0.997 ● N +2	$\begin{aligned} \psi \bar{\psi} &= \frac{1}{4} \left(\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} \right. \\ &\quad \left. + \bar{\psi} \gamma^5 \psi \gamma^5 \right), \quad (\end{aligned}$	$\psi \bar{\psi} = \frac{1}{4} \left(\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} \right. \\ \left. + \bar{\psi} \gamma^5 \psi \gamma^5 \right)$	$\psi \bar{\psi} = \frac{1}{4} (\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} + \bar{\psi} \gamma^5 \psi \gamma^5), \quad ($
1510.06699_ EQ0056 (60)	013	■ 0.997 ● N +0	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,A,\partial A,\partial B)+\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A,\partial A,B),$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,A,\partial A,\partial B)+\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A,\partial A,B),$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,A,\partial A,\partial B)+\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A,\partial A,B),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00077 (82)	017	0.997 K +0	$\widehat{\mathrm{mathcal{F}}}_{\{a\,b\}^*}=\mathrm{mathcal{F}}_{\{a\,b\}}-\mathrm{mathcal{T}}^{\wedge{*}\,\,c}_{\{\,\,\}_{\{a\,b\}}\,\,\mathrm{mathcal{A}}_{\{c\}}}$	$\widehat{\mathcal{F}}_{ab}^* = \mathcal{F}_{ab} - \mathcal{T}^{*c}_{ab}\mathcal{A}_c,$	$\widehat{\mathcal{F}}_{ab}^* = \mathcal{F}_{ab} - \mathcal{T}^{*c}_{ab}\mathcal{A}_c,$
1510.06699_ E00159 (173)	029	0.997 N +1	$\begin{aligned} &\mathrm{mathcal{D}}_{\{c\}^{\dagger}}^{\dagger}\!\!\left(h\,s_{\{a\,b\}^{\dagger}\{c\}}\right)+h\,t_{\{[a\,b]\}^{\dagger}}^{\dagger}\,\&=0,\,\,\backslash\backslash\\ &\mathrm{mathcal{D}}_{\{c\}^{\dagger}}^{\dagger}\!\!\left(h\,t^{\dagger}_{\{d\}}\mathrm{mathcal{R}}^{\dagger}_{\{a\,b\}^{\dagger}}\right)-h\!\left(s_{\{a\,b\}^{\dagger}\{c\}}\mathrm{mathcal{R}}^{\dagger}_{\{a\,b\}^{\dagger}}\right.\\ &\left.\}_{\{c\,d\}}-t^{\wedge{*}\,\,c}_{\{d\}}\mathrm{mathcal{T}}^{\wedge{*}\,\,b}_{\{b\}}\mathrm{mathcal{H}}_{\{c\,d\}}\right)+j^{\dagger c}\mathrm{mathcal{H}}_{\{c\,d\}}\right)\,\&=0,\\ &\end{aligned}$	$\mathcal{D}_c^\dagger\left(hs_{ab}^c\right)+ht_{[ab]}^\dagger=0,$ $\mathcal{D}_c^\dagger\left(ht^{\dagger c}_d\right)-h\left(s_{ab}^c\mathcal{R}^{\dagger ab}_{cd}-t^{\dagger c}_b\mathcal{T}^{\dagger b}_{cd}+j^{\dagger c}\mathcal{H}_{cd}\right)=0,$	$\mathcal{D}_c^\dagger(hs_{ab}^c)+ht_{[ab]}^\dagger=0,$ $\mathcal{D}_c^\dagger(ht^{\dagger c}_d)-h(s_{ab}^c\mathcal{R}^{\dagger ab}_{cd}-t^{\dagger c}_b\mathcal{T}^{\dagger b}_{cd}+j^{\dagger c}\mathcal{H}_{cd})=0,$
1510.06699_ E00047 (49)	012	0.998 N +1	$\mathrm{mathcal{D}}_{\{a\}^*}^{\wedge{*}\,\,c}\,\varphi=\left(\{\,\,\}^{\wedge{*}\,\,0}\mathrm{mathcal{D}}_{\{a\}^*}^{\wedge{*}\,\,c}+\frac{1}{2}\mathrm{mathcal{K}}^{\wedge{*}\,\,bc}_{\{a\}}\Sigma_{bc}\right)\mathrm{mathcal{K}}^{\wedge{*}\,\,c}_{\{b\,c\}}\Sigma_{\{a\}}\mathrm{mathcal{K}}_{\{b\,c\}}\right)\,\varphi$	$\mathcal{D}_a^*\varphi=\left({}^0\mathcal{D}_a^*+\frac{1}{2}\mathcal{K}^{*bc}_a\Sigma_{bc}\right)\varphi,$	$\mathcal{D}_a^*\varphi=\left({}^0\mathcal{D}_a^*+\frac{1}{2}\mathcal{K}^{*bc}_a\Sigma_{bc}\right)\varphi,,$
1510.06699_ E00002 (2)	003	0.998 N +1	$\begin{aligned} &\&L_{\mathrm{mathcal{R}}^{\wedge{2}}}=\alpha_{\{1\}}\mathrm{mathcal{R}}^{\wedge{2}}+\alpha_{\{2\}}\mathrm{mathcal{R}}_{\{a\,b\}}^{\wedge{2}}+\alpha_{\{3\}}\mathrm{mathcal{R}}_{\{a\,b\}}^{\wedge{3}}+\alpha_{\{4\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{4}}+\alpha_{\{5\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{5}}+\alpha_{\{6\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{6}}\\ &\mathrm{mathcal{R}}_{\{a\,b\}}^{\wedge{2}}+\alpha_{\{3\}}\mathrm{mathcal{R}}_{\{a\,b\}}^{\wedge{3}}+\alpha_{\{4\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{4}}+\alpha_{\{5\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{5}}+\alpha_{\{6\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{6}}\\ &\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{4}}+\alpha_{\{5\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{5}}+\alpha_{\{6\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{6}}\\ &\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{6}}+\alpha_{\{6\}}\mathrm{mathcal{R}}_{\{a\,b\,c\,d\}}^{\wedge{6}}\\ &\mathrm{mathcal{R}}_{\{c\,d\,a\,b\}}^{\wedge{2}},\,\,\&\&L_{\mathrm{mathcal{T}}^{\wedge{2}}}=\beta_{\{1\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{2\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{3\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}\\ &\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{2\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{3\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}\\ &\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{3\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}\\ &\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}}+\beta_{\{3\}}\mathrm{mathcal{T}}_{\{a\,b\,c\}}^{\wedge{2}},\,\,\end{aligned}$	$L_{\mathcal{R}^2}=\alpha_1\mathcal{R}^2+\alpha_2\mathcal{R}_{ab}\mathcal{R}^{ab}+\alpha_3\mathcal{R}_{ab}\mathcal{R}^{ba}+\alpha_4\mathcal{R}_{abcd}\mathcal{R}^{abcd}+\alpha_5\mathcal{R}_{abcd}\mathcal{R}^{acbd}+\alpha_6\mathcal{R}_{abcd}\mathcal{R}^{cdab},$ $L_{\mathcal{T}^2}=\beta_1\mathcal{T}_{abc}\mathcal{T}^{abc}+\beta_2\mathcal{T}_{abc}\mathcal{T}^{bac}+\beta_3\mathcal{T}_a\mathcal{T}^a,$	$L_{\mathcal{R}^2}=\alpha_1\mathcal{R}^2+\alpha_2\mathcal{R}_{ab}\mathcal{R}^{ab}+\alpha_3\mathcal{R}_{ab}\mathcal{R}^{ba}+\alpha_4\mathcal{R}_{abcd}\mathcal{R}^{abcd}+\alpha_5\mathcal{R}_{abcd}\mathcal{R}^{acbd}+\alpha_6\mathcal{R}_{abcd}\mathcal{R}^{cdab},$ $L_{\mathcal{T}^2}=\beta_1\mathcal{T}_{abc}\mathcal{T}^{abc}+\beta_2\mathcal{T}_{abc}\mathcal{T}^{bac}+\beta_3\mathcal{T}_a\mathcal{T}^a,$
1510.06699_ E00064 (68b)	014	0.998 W +2	$\begin{aligned} &\left(\mathrm{mathcal{D}}_{\{c\}^*}^{\wedge{*}\,\,c}\right)\!\!\left(h\,s_{\{a\,b\}^{\dagger}\{c\}}\right)+h\,t_{\{[a\,b]\}^{\dagger}}^{\dagger}\,\&=0,\,\,\backslash\backslash\left(\mathrm{mathcal{D}}_{\{c\}^*}^{\wedge{*}\,\,c}\right)\!\!\left(h\,t^{\dagger}_{\{d\}}\mathrm{mathcal{R}}^{\dagger}_{\{a\,b\}^{\dagger}}\right)-h\!\left(s_{\{a\,b\}^{\dagger}\{c\}}\mathrm{mathcal{R}}^{\dagger}_{\{a\,b\}^{\dagger}}\right.\\ &\left.\}_{\{c\,d\}}-t^{\wedge{*}\,\,c}_{\{d\}}\mathrm{mathcal{T}}^{\wedge{*}\,\,b}_{\{b\}}\mathrm{mathcal{H}}_{\{c\,d\}}\right)+j^{\dagger c}\mathrm{mathcal{H}}_{\{c\,d\}}\right)\,\&=0,\,\,\backslash\backslash\left(\mathrm{mathcal{D}}_{\{c\}^*}^{\wedge{*}\,\,c}\right)\!\!\left(h\,j^{\dagger c}\right)-h\,t^{\dagger c}_c=0.\\ &\end{aligned}$	$\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(hs_{ab}^c\right)+ht_{[ab]}=0,$ $\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(ht^c_d\right)-h\left(s_{ab}^c\mathcal{R}^{ab}_{cd}-t^c_b\mathcal{T}^{*b}_{cd}+j^c\mathcal{H}_{cd}\right)=0,$ $\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(hj^c\right)-ht^c_c=0.$	$\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(hs_{ab}^c\right)+ht_{[ab]}=0,$ $\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(ht^c_d\right)-h\left(s_{ab}^c\mathcal{R}^{ab}_{cd}-t^c_b\mathcal{T}^{*b}_{cd}+j^c\mathcal{H}_{cd}\right)=0,$ $\left(\mathcal{D}_c^*+\mathcal{T}_c^*\right)\left(hj^c\right)-ht^c_c=0.$
1510.06699_ E00079 (84)	017	0.998 K +0	$\left(\mathrm{mathcal{D}}_{\{a\}^*}^{\wedge{*}\,\,c}+\mathrm{mathcal{T}}_{\{a\}^*}^{\wedge{*}\,\,c}\right)\mathrm{mathcal{F}}^{\wedge{*}\,\,c}_{\{a\,c\}}-\frac{1}{2}\mathrm{mathcal{T}}^{\wedge{*}\,\,c}_{\{a\,b\}}\mathrm{mathcal{F}}^{\wedge{*}\,\,c}_{\{c\}}\}_{\{a\,b\}}\mathrm{mathcal{F}}^{\wedge{*}\,\,c}_{\{a\,b\}}=\mathrm{mathcal{J}}^{\wedge{*}\,\,c}_{\{c\}}$	$\left(\mathcal{D}_a^*+\mathcal{T}_a^*\right)\mathcal{F}^{ac}-\frac{1}{2}\mathcal{T}^{*c}_{ab}\mathcal{F}^{ab}=\mathcal{J}^c.$	$\left(\mathcal{D}_a^*+\mathcal{T}_a^*\right)\mathcal{F}^{ac}-\frac{1}{2}\mathcal{T}^{*c}_{ab}\mathcal{F}^{ab}=\mathcal{J}^c.$
1510.06699_ E00094 (102)	019	0.998 N +0	$\phi\,v^{\{b\}\{ \}^{\{0\}}}\mathrm{mathcal{D}}_{\{b\}^*}^*v_{\{a\}}=\left(\delta_{\{a\}}^b-v_{\{a\}}v^{\{b\}}\right)^0\mathrm{mathcal{D}}_{\{b\}^*}^*\phi$	$\phi v^{b\,0}\mathcal{D}_b^*v_a=\left(\delta_a^b-v_av^b\right)^0\mathcal{D}_b^*\phi.$	$\phi v^{b\,0}\mathcal{D}_b^*v_a=\left(\delta_a^b-v_av^b\right)^0\mathcal{D}_b^*\phi.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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1510.06699_ EQ0162 (178b)	030	■ 0.998 ● N -1	$\begin{aligned} & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\sigma_{ab}^c\right)+h\tau_{[ab]}^{\dagger}+\frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi=0, \\ & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\tau^{\dagger c}_{\{d\}}\right)-h\left(\sigma_{ab}^c\mathcal{R}^{\dagger ab}_{\{cd\}}-\tau^{\dagger c}_{\{b\}}\mathcal{T}^{\dagger b}_{\{cd\}}+\zeta^{\dagger c}\mathcal{H}_{cd}\right)+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_{\{d\}}^{\dagger}\phi+\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_{\{d\}}^{\dagger}\varphi=0, \\ & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\zeta^{\dagger c}\right)=0, \\ & h\tau^{\dagger c}_{\{c\}}+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi-\frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi=0, \end{aligned}$	$\begin{aligned} & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\sigma_{ab}^c\right)+h\tau_{[ab]}^{\dagger}+\frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi=0, \\ & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\tau^{\dagger c}_{\{d\}}\right)-h\left(\sigma_{ab}^c\mathcal{R}^{\dagger ab}_{\{cd\}}-\tau^{\dagger c}_{\{b\}}\mathcal{T}^{\dagger b}_{\{cd\}}+\zeta^{\dagger c}\mathcal{H}_{cd}\right)+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_{\{d\}}^{\dagger}\phi+\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_{\{d\}}^{\dagger}\varphi=0, \\ & \mathcal{D}_{\{c\}^{\dagger}}^{\{a\}}\left(h\zeta^{\dagger c}\right)=0, \\ & h\tau^{\dagger c}_{\{c\}}+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi-\frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi=0, \end{aligned}$	
1510.06699_ EQ0251 (B6)	049	■ 0.998 ● K +0	$\begin{aligned} R=& \{ \}^{\{0\}}R+\frac{1}{4}T^{\mu\nu\lambda}T_{\mu\nu\lambda}+\frac{1}{2}T^{\mu\nu\lambda}T_{\nu\mu\lambda}-T^{\mu}T_{\mu} \\ & -\frac{2}{\sqrt{-g}}\partial_{\mu}\left(\sqrt{-g}T^{\mu}\right), \end{aligned}$	$R={}^0R+\frac{1}{4}T^{\mu\nu\lambda}T_{\mu\nu\lambda}+\frac{1}{2}T^{\mu\nu\lambda}T_{\nu\mu\lambda}-T^{\mu}T_{\mu}-\frac{2}{\sqrt{-g}}\partial_{\mu}(\sqrt{-g}T^{\mu}),$	
1510.06699_EQ0104	021	■ 0.999 ● W -1	$\begin{array}{l} \mathcal{L}_{\{L\}_{\{\mathrm{M}\}}}=h^{\{-1\}}\left[\frac{1}{2}i\bar{\psi}\gamma^a\bar{\mathcal{D}}_{\{a\}}\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_{\{a\}}^*\phi\right)\left({}^0\mathcal{D}^{*a}\phi\right)-\lambda\phi^4\right. \\ \left.-a\phi^{20}\mathcal{R}^*\right],(112) \end{array}$	$\mathcal{L}_{\mathrm{M}}=h^{-1}\left[\frac{1}{2}i\bar{\psi}\gamma^a\bar{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_a^*\phi\right)\left({}^0\mathcal{D}^{*a}\phi\right)-\lambda\phi^4-a\phi^{20}\mathcal{R}^*\right],(112)$	$\mathcal{L}_{\mathrm{M}}=h^{-1}[\frac{1}{2}i\bar{\psi}\gamma^a{}^0\bar{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^*\phi)({}^0\mathcal{D}^{*a}\phi)-\lambda\phi^4-a\phi^{20}\mathcal{R}^*],(112)$
1510.06699_ EQ0171 (187)	031	■ 0.999 ● N +0	$\mathcal{L}_{\{L\}_{\{\mathrm{D}\}}}=h^{\{-1\}}L_{\{\mathrm{D}\}}=h^{\{-1\}}\left[\frac{1}{2}i\bar{\psi}\gamma^a\bar{\mathcal{D}}_{\{a\}}\psi-\mu\phi\bar{\psi}\psi\right],$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\bar{\mathcal{D}}_a^{\dagger}\psi-\mu\phi\bar{\psi}\psi\right),$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\bar{\mathcal{D}}_a^{\dagger}\psi-\mu\phi\bar{\psi}\psi),$
1510.06699_ EQ0253 (C2)	050	■ 0.999 ● N +0	$\begin{aligned} & \mathcal{R}^{ac}_{\{bc\}}-\frac{1}{2}\delta^a_b\mathcal{R}=\kappa h\tau^a_b, \\ & h b^c_{\mu}D_{\nu}\left[h^{-1}\left(h_a{}^{\mu}h_b{}^{\nu}-h_a{}^{\nu}h_b{}^{\mu}\right)\right]=2\kappa h\sigma_{ab}{}^c. \end{aligned}$	$\mathcal{R}^{ac}_{\{bc\}}-\frac{1}{2}\delta^a_b\mathcal{R}=\kappa h\tau^a_b, \\ h b^c_{\mu}D_{\nu}[h^{-1}(h_a{}^{\mu}h_b{}^{\nu}-h_a{}^{\nu}h_b{}^{\mu})]=2\kappa h\sigma_{ab}{}^c.$	
1510.06699_ EQ0144 (158)	027	■ 0.999 ● N +1	$\mathcal{A}_{\{a\}b\{c\}^{\dagger}}^{\{A\}}=\{ \}^{\{0\}}\mathcal{A}_{\{a\}b\{c\}^{\dagger}}^{\{A\}}(h,\partial h,A,V)+\mathcal{K}_{\{a\}b\{c\}^{\dagger}}^{\{A\}}(h,\partial h,A),$	$\mathcal{A}_{abc}^{\dagger}={}^0\mathcal{A}_{abc}^{\dagger}(h,\partial h,A,V)+\mathcal{K}_{abc}^{\dagger}(h,\partial h,A),$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00030 (31)	010	■ 1.000 ● N +1	$\mathrm{L}_{\left(\varphi^{\prime}\right)}\left(\mathrm{x}^{\prime}\right) ; \mathcal{D}_{\left(\varphi^{\prime}\right)}^{\ast}\left(\mathrm{x}^{\prime}\right)=\mathrm{e}^{-4 \rho(\mathrm{x})} \mathrm{L}_{\left(\varphi\right)}\left(\mathrm{x}\right) ; \mathcal{D}_{\left(\varphi\right)}^{\ast}\left(\mathrm{x}\right)$	$L_{\mathrm{M}}\left(\varphi^{\prime}\left(x^{\prime}\right) ; \mathcal{D}_a^{\ast} \varphi^{\prime}\left(x^{\prime}\right)\right)=e^{-4 \rho(x)} L_{\mathrm{M}}(\varphi(x) ; \mathcal{D}_a^{\ast} \varphi(x)) .$	$L_{\mathrm{M}}\left(\varphi^{\prime}\left(x^{\prime}\right) ; \mathcal{D}_a^{\ast} \varphi^{\prime}\left(x^{\prime}\right)\right)=e^{-4 \rho(x)} L_{\mathrm{M}}(\varphi(x) ; \mathcal{D}_a^{\ast} \varphi(x)) .$
1510.06699_ E00031 (32)	010	■ 1.000 ● N +1	$\mathrm{S}_{\left(\varphi\right)}=\int \mathrm{d}^4 x \mathrm{L}_{\left(\varphi\right)}\left(\varphi,\mathcal{D}_{\left(\varphi\right)}^{\ast} \varphi\right)$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^{\ast} \varphi) d^4 x$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^{\ast} \varphi) d^4 x,$
1510.06699_ E00034 (35)	010	■ 1.000 ● C +4	$\mathrm{D}_{\left(\mu\right)}^{\ast} \equiv \partial_{\left(\mu\right)}^{\ast}+\frac{1}{2} \mathrm{A}^{\mathrm{ab}}_{\left(\mu\right)} \Sigma_{\mathrm{ab}}=\partial_{\left(\mu\right)}^{\ast}+\frac{1}{2} \mathrm{A}^{\mathrm{ab}}_{\left(\mu\right)} \Sigma_{\mathrm{ab}},$	$D_{\mu}^{\ast} \equiv \partial_{\mu}^{\ast}+{}^0 \Gamma_{\rho \mu}^{\sigma} X_{\sigma}^{\rho}+\frac{1}{2} A_{\mu}^{a b} \Sigma_{a b}={ }^0 \nabla_{\mu}^{\ast}+\frac{1}{2} A_{\mu}^{a b} \Sigma_{a b},$	$D_{\mu}^{\ast} \equiv \partial_{\mu}^{\ast}+{}^0 \Gamma_{\rho \mu}^{\sigma} X_{\sigma}^{\rho}+\frac{1}{2} A_{\mu}^{a b} \Sigma_{a b}={ }^0 \nabla_{\mu}^{\ast}+\frac{1}{2} A_{\mu}^{a b} \Sigma_{a b},$
1510.06699_ E00035 (36)	011	■ 1.000 ● N +0	$\mathrm{D}_{\left(\mu\right)}^{\ast} \mathrm{h}_{\left\{\rho\right\}}^{\ast}=\partial_{\left(\mu\right)}^{\ast} \mathrm{h}_{\left\{\rho\right\}}^{\ast}+\frac{1}{2} \mathrm{A}^{\mathrm{ab}}_{\left(\mu\right)} \Sigma_{\mathrm{ab}} \mathrm{h}_{\left\{\rho\right\}}^{\ast}=\partial_{\left(\mu\right)}^{\ast} \mathrm{h}_{\left\{\rho\right\}}^{\ast}+\frac{1}{2} \mathrm{A}^{\mathrm{ab}}_{\left(\mu\right)} \Sigma_{\mathrm{ab}} \mathrm{h}_{\left\{\rho\right\}}^{\ast}$	$D_{\mu}^{\ast} h_a^{\nu}=\partial_{\mu}^{\ast} h_a^{\nu}+{}^0 \Gamma_{\rho \mu}^{\nu} h_a^{\rho}-A_{a \mu}^b h_b^{\nu} .$	$D_{\mu}^{\ast} h_a^{\nu}=\partial_{\mu}^{\ast} h_a^{\nu}+{}^0 \Gamma_{\rho \mu}^{\nu} h_a^{\rho}-A_{a \mu}^b h_b^{\nu} .$
1510.06699_ E00036 (37)	011	■ 1.000 ● K +0	$\left[\partial_{\left(\mu\right)}^{\ast}, \partial_{\left(\nu\right)}^{\ast}\right] \varphi=0,$	$\left[{}^0 \nabla_{\mu}, {}^0 \nabla_{\nu}\right] \varphi=0,$	$\left[{}^0 \nabla_{\mu}, {}^0 \nabla_{\nu}\right] \varphi=0,$
1510.06699_ E00037 (38)	011	■ 1.000 ● K +0	$\left[\partial_{\left(\mu\right)}^{\ast}, \partial_{\left(\nu\right)}^{\ast}\right] \varphi=\frac{1}{2} R_{\mu \nu}^{a b} \Sigma_{a b} \varphi+w H_{\mu \nu} \varphi,$	$\left[D_{\mu}^{\ast}, D_{\nu}^{\ast}\right] \varphi=\frac{1}{2} R_{\mu \nu}^{a b} \Sigma_{a b} \varphi+w H_{\mu \nu} \varphi,$	$\left[D_{\mu}^{\ast}, D_{\nu}^{\ast}\right] \varphi=\frac{1}{2} R_{\mu \nu}^{a b} \Sigma_{a b} \varphi+w H_{\mu \nu} \varphi,$
1510.06699_ E00039 (41)	012	■ 1.000 ● N +0	$\left[\partial_{\left(\mu\right)}^{\ast}, \partial_{\left(\nu\right)}^{\ast}\right] \varphi=\frac{1}{2} \mathcal{R}_{\mu \nu}^{a b} \Sigma_{a b} \varphi+w \mathcal{H}_{\mu \nu} \varphi-\mathcal{T}_{\mu \nu}^{\ast a} \mathcal{D}_a^{\ast} \varphi,$	$\left[D_c^{\ast}, D_d^{\ast}\right] \varphi=\frac{1}{2} \mathcal{R}_{c d}^{a b} \Sigma_{a b} \varphi+w \mathcal{H}_{c d} \varphi-\mathcal{T}_{c d}^{\ast a} \mathcal{D}_a^{\ast} \varphi,$	$\left[D_c^{\ast}, D_d^{\ast}\right] \varphi=\frac{1}{2} \mathcal{R}_{c d}^{a b} \Sigma_{a b} \varphi+w \mathcal{H}_{c d} \varphi-\mathcal{T}_{c d}^{\ast a} \mathcal{D}_a^{\ast} \varphi,$
1510.06699_ E00040 (42)	012	■ 1.000 ● N +0	$\mathcal{T}_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu} \mathcal{T}_{\mu \nu}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\mathcal{D}_{\mu}^{\ast} b_{\nu}^a-\mathcal{D}_{\nu}^{\ast} b_{\mu}^a\right),$	$\mathcal{T}_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu} \mathcal{T}_{\mu \nu}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\mathcal{D}_{\mu}^{\ast} b_{\nu}^a-\mathcal{D}_{\nu}^{\ast} b_{\mu}^a\right),$	$\mathcal{T}_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu} \mathcal{T}_{\mu \nu}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\mathcal{D}_{\mu}^{\ast} b_{\nu}^a-\mathcal{D}_{\nu}^{\ast} b_{\mu}^a\right),$
1510.06699_ E00041 (43)	012	■ 1.000 ● N +0	$\mathcal{T}_{b c}^{\ast a}=-b_{\mu}^a\left(\mathcal{D}_b^{\ast} h_c^{\mu}-\mathcal{D}_c^{\ast} h_b^{\mu}\right) .$	$\mathcal{T}_{b c}^{\ast a}=-b_{\mu}^a\left(\mathcal{D}_b^{\ast} h_c^{\mu}-\mathcal{D}_c^{\ast} h_b^{\mu}\right) .$	$\mathcal{T}_{b c}^{\ast a}=-b_{\mu}^a\left(\mathcal{D}_b^{\ast} h_c^{\mu}-\mathcal{D}_c^{\ast} h_b^{\mu}\right) .$
1510.06699_ E00042 (44)	012	■ 1.000 ● K +0	$\mathcal{T}_{b c}^{\ast a}=\mathcal{T}_{b c}^a+\delta_c^a \mathcal{B}_b-\delta_b^a \mathcal{B}_c,$	$\mathcal{T}_{b c}^{\ast a}=\mathcal{T}_{b c}^a+\delta_c^a \mathcal{B}_b-\delta_b^a \mathcal{B}_c,$	$\mathcal{T}_{b c}^{\ast a}=\mathcal{T}_{b c}^a+\delta_c^a \mathcal{B}_b-\delta_b^a \mathcal{B}_c,$
1510.06699_ E00043 (45)	012	■ 1.000 ● N +0	$c_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\partial_{\mu}^{\ast} b_{\nu}^a-\partial_{\nu}^{\ast} b_{\mu}^a\right),$	$c_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\partial_{\mu}^{\ast} b_{\nu}^a-\partial_{\nu}^{\ast} b_{\mu}^a\right),$	$c_{b c}^{\ast a} \equiv h_b^{\mu} h_c^{\nu}\left(\partial_{\mu}^{\ast} b_{\nu}^a-\partial_{\nu}^{\ast} b_{\mu}^a\right),$
1510.06699_ E00044 (46)	012	■ 1.000 ● N +0	$\mathcal{A}_{a b c}=\frac{1}{2}\left(c_{a b c}^{\ast}+c_{b c a}^{\ast}-c_{c a b}^{\ast}\right)-\frac{1}{2}\left(\mathcal{T}_{a b c}^{\ast}+\mathcal{T}_{b c a}^{\ast}-\mathcal{T}_{c a b}^{\ast}\right) .$	$\mathcal{A}_{a b c}=\frac{1}{2}\left(c_{a b c}^{\ast}+c_{b c a}^{\ast}-c_{c a b}^{\ast}\right)-\frac{1}{2}\left(\mathcal{T}_{a b c}^{\ast}+\mathcal{T}_{b c a}^{\ast}-\mathcal{T}_{c a b}^{\ast}\right) .$	$\mathcal{A}_{a b c}=\frac{1}{2}\left(c_{a b c}^{\ast}+c_{b c a}^{\ast}-c_{c a b}^{\ast}\right)-\frac{1}{2}\left(\mathcal{T}_{a b c}^{\ast}+\mathcal{T}_{b c a}^{\ast}-\mathcal{T}_{c a b}^{\ast}\right) .$
1510.06699_ E00045 (47)	012	■ 1.000 ● N +0	$\mathcal{A}_{a b c}={ }^0 \mathcal{A}_{a b c}^{\ast}(h, \partial h, B)+\mathcal{K}_{a b c}^{\ast}(h, \partial h, A, B),$	$\mathcal{A}_{a b c}={ }^0 \mathcal{A}_{a b c}^{\ast}(h, \partial h, B)+\mathcal{K}_{a b c}^{\ast}(h, \partial h, A, B),$	$\mathcal{A}_{a b c}={ }^0 \mathcal{A}_{a b c}^{\ast}(h, \partial h, B)+\mathcal{K}_{a b c}^{\ast}(h, \partial h, A, B),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00048 (50)	012	■ 1.000 ● W +2	$\left[\mathrm{D}_{\mathrm{a}}^{\mathrm{*}},\left[\mathrm{D}_{\mathrm{b}}^{\mathrm{*}},\mathrm{D}_{\mathrm{c}}^{\mathrm{*}}\right]\right]\varphi+\left[\mathrm{D}_{\mathrm{c}}^{\mathrm{*}},\left[\mathrm{D}_{\mathrm{a}}^{\mathrm{*}},\mathrm{D}_{\mathrm{b}}^{\mathrm{*}}\right]\right]\varphi+\left[\mathrm{D}_{\mathrm{b}}^{\mathrm{*}},\left[\mathrm{D}_{\mathrm{c}}^{\mathrm{*}},\mathrm{D}_{\mathrm{a}}^{\mathrm{*}}\right]\right]\varphi=0.$	$[\mathcal{D}_a^*,[\mathcal{D}_b^*,\mathcal{D}_c^*]]\varphi+[\mathcal{D}_c^*,[\mathcal{D}_a^*,\mathcal{D}_b^*]]\varphi+[\mathcal{D}_b^*,[\mathcal{D}_c^*,\mathcal{D}_a^*]]\varphi=0.$	
1510.06699_ E00050 (54)	013	■ 1.000 ● N +0	$\mathrm{D}_{\mathrm{a}}^{\mathrm{*}}\mathcal{R}_{bc}^{ae}-2\mathcal{D}_{\left[b\right]}^*\mathcal{R}_{\left[c\right]}^e-2\mathcal{T}_{a\left[b\right]}^{*f}\mathcal{R}_{\left[c\right]f}^{ae}-\mathcal{T}_{bc}^{*f}\mathcal{R}_{\left[f\right]}^e=0.$	$\mathcal{D}_a^*\mathcal{R}_{bc}^{ae}-2\mathcal{D}_{[b}^*\mathcal{R}_{c]}^e-2\mathcal{T}_{a[b}^{*f}\mathcal{R}_{c]f}^{ae}-\mathcal{T}_{bc}^{*f}\mathcal{R}_f^e=0.$	
1510.06699_ E00051 (55)	013	■ 1.000 ● N +0	$\mathrm{D}_{\mathrm{a}}^{\mathrm{*}}\left(\mathcal{R}_{\mathrm{c}}^a-\frac{1}{2}\delta_{\mathrm{c}}^a\mathcal{R}\right)+\mathcal{T}_{bc}^{*f}\mathcal{R}_{\left[f\right]}^b+\frac{1}{2}\mathcal{T}_{ab}^{*f}\mathcal{R}_{cf}^{ab}=0.$	$\mathcal{D}_a^*(\mathcal{R}_{\mathrm{c}}^a-\frac{1}{2}\delta_{\mathrm{c}}^a\mathcal{R})+\mathcal{T}_{bc}^{*f}\mathcal{R}_f^b+\frac{1}{2}\mathcal{T}_{ab}^{*f}\mathcal{R}_{cf}^{ab}=0.$	
1510.06699_ E00052 (56)	013	■ 1.000 ● N +0	$\mathcal{D}_a^*\mathcal{T}_{bc}^{*a}+2\mathcal{D}_{\left[b\right]}^*\mathcal{T}_{\left[c\right]}^*+\mathcal{T}_{bc}^{*e}\mathcal{T}_{\left[e\right]}^*+2\mathcal{R}_{\left[bc\right]}+2\mathcal{H}_{bc}=0.$	$\mathcal{D}_a^*\mathcal{T}_{bc}^{*a}+2\mathcal{D}_{[b}^*\mathcal{T}_{c]}^*+\mathcal{T}_{bc}^{*e}\mathcal{T}_e^*+2\mathcal{R}_{[bc]}+2\mathcal{H}_{bc}=0.$	
1510.06699_ E00053 (57)	013	■ 1.000 ● N +1	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}_{abcd},\mathcal{T}_{abc}^*,\mathcal{H}_{ab}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}(\mathcal{R}_{abcd},\mathcal{T}_{abc}^*,\mathcal{H}_{ab})\,d^4x,$	
1510.06699_ E00054 (58)	013	■ 1.000 ● N +1	$L_{\mathrm{G}}=L_{\mathcal{R}^2}+L_{\mathcal{H}^2},$	$L_{\mathrm{G}}=L_{\mathcal{R}^2}+L_{\mathcal{H}^2},$	
1510.06699_ E00055 (59)	013	■ 1.000 ● N +0	$\mathcal{R}^2-4\mathcal{R}_{ab}\mathcal{R}^{ba}+\mathcal{R}_{abcd}\mathcal{R}^{cdab}$	$\mathcal{R}^2-4\mathcal{R}_{ab}\mathcal{R}^{ba}+\mathcal{R}_{abcd}\mathcal{R}^{cdab}$	
1510.06699_ E00059 (63)	014	■ 1.000 ● W +2	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-\partial_{\mu}\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\left(\partial_{\mu}\varphi\right)}\right)=0.$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-\partial_{\mu}\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\left(\partial_{\mu}\varphi\right)}\right)=0.$	
1510.06699_ E00060 (64)	014	■ 1.000 ● N +1	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\bar{\partial}\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-D_{\mu}^*\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\left(D_{\mu}^*\varphi\right)}\right)=0$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\bar{\partial}\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-D_{\mu}^*\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\left(D_{\mu}^*\varphi\right)}\right)=0,$	
1510.06699_ E00061 (65)	014	■ 1.000 ● W +0	$\frac{\bar{\partial}L_{\mathrm{M}}}{\partial\varphi}-\mathcal{D}_a^*\left(\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^*\varphi\right)}\right)=hD_{\mu}^*\left(h^{-1}h_a{}^{\mu}\right)\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^*\varphi\right)},$	$\frac{\bar{\partial}L_{\mathrm{M}}}{\partial\varphi}-\mathcal{D}_a^*\left(\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^*\varphi\right)}\right)=hD_{\mu}^*(h^{-1}h_a{}^{\mu})\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^*\varphi\right)},$	
1510.06699_ E00062 (66)	014	■ 1.000 ● W +2	$hD_{\mu}^*\left(h^{-1}h_a{}^{\mu}\right)=b^b{}_{\mu}\left(\mathcal{D}_b^*h_a{}^{\mu}-\mathcal{D}_a^*h_b{}^{\mu}\right)=\mathcal{T}^{*b}{}_{ab}\equiv\mathcal{T}_a^*,$	$hD_{\mu}^*(h^{-1}h_a{}^{\mu})=b^b{}_{\mu}(\mathcal{D}_b^*h_a{}^{\mu}-\mathcal{D}_a^*h_b{}^{\mu})=\mathcal{T}^{*b}{}_{ab}\equiv\mathcal{T}_{a,\perp}^*,$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ EQ0069 (73)	015	■ 1.000 ● N +0	$i\,\gamma^a\left(\mathcal{D}_a^*+\frac{1}{2}\mathcal{T}_a^*\right)\psi-\mu\phi\psi=0$	$i\gamma^a\left(\mathcal{D}_a^*+\frac{1}{2}\mathcal{T}_a^*\right)\psi-\mu\phi\psi=0.$	$i\gamma^a(\mathcal{D}_a^*+\tfrac{1}{2}\mathcal{T}_a^*)\psi-\mu\phi\psi=0.$
1510.06699_ EQ0070 (74)	015	■ 1.000 ● N +0	$i\,\gamma^a\left(\mathcal{D}_a+\frac{1}{2}\mathcal{T}_a\right)\psi-\mu\phi\psi=0$	$i\gamma^a\left(\mathcal{D}_a+\frac{1}{2}\mathcal{T}_a\right)\psi-\mu\phi\psi=0.$	$i\gamma^a(\mathcal{D}_a+\tfrac{1}{2}\mathcal{T}_a)\psi-\mu\phi\psi=0.$
1510.06699_ EQ0071 (75)	016	■ 1.000 ● N +0	$i\,\gamma^a\left(\{ \}^{\{0\}}\mathcal{D}_a^*-\frac{1}{4}\mathcal{M}_{\{a\}b\}^{\{c\}}\Sigma^{bc}\right)\psi-\mu\phi\psi=0$	$i\gamma^a\left({}^0\mathcal{D}_a^*-\frac{1}{4}\mathcal{T}_{[abc]}^*\Sigma^{bc}\right)\psi-\mu\phi\psi=0.$	$i\gamma^a({}^0\mathcal{D}_a^*-\tfrac{1}{4}\mathcal{T}_{[abc]}^*\Sigma^{bc})\psi-\mu\phi\psi=0.$
1510.06699_ EQ0072 (76)	016	■ 1.000 ● K +0	$i\,\gamma^a\left(\{ \}^{\{0\}}\mathcal{D}_a-\frac{1}{4}\mathcal{M}_{\{a\}b\}^{\{c\}}\Sigma^{bc}\right)\psi-\mu\phi\psi=0$	$i\gamma^a\left({}^0\mathcal{D}_a-\frac{1}{4}\mathcal{T}_{[abc]}\Sigma^{bc}\right)\psi-\mu\phi\psi=0,$	$i\gamma^a({}^0\mathcal{D}_a-\tfrac{1}{4}\mathcal{T}_{[abc]}\Sigma^{bc})\psi-\mu\phi\psi=0,$
1510.06699_ EQ0078 (83)	017	■ 1.000 ● K +0	$\mathcal{L}_{\mathcal{M}}=\mathcal{L}_{\mathcal{G}}(\widehat{h},\widehat{A},\partial\widehat{A},\partial\widehat{B})+\mathcal{L}_{\mathcal{M}}(\widehat{\varphi},\partial\widehat{\varphi},\phi_0,0,\widehat{h},\partial\widehat{h},\widehat{A},\partial\widehat{A},\widehat{B}).$	$\mathcal{L}_{\mathcal{M}}=h^{-1}L_{\mathcal{M}}=-h^{-1}\left(\frac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}+\mathcal{J}^a\mathcal{A}_a\right),$	$\mathcal{L}_{\mathcal{M}}=h^{-1}L_{\mathcal{M}}=-h^{-1}(\tfrac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}+\mathcal{J}^a\mathcal{A}_a),$
1510.06699_ EQ0084 (90)	017	■ 1.000 ● K +0	$h\,\tau^a{}_b=\frac{1}{4}\delta^a_b\mathcal{F}^{cd}\mathcal{F}_{cd}-\mathcal{F}^{ac}\mathcal{F}_{bc},$	$h\tau^a{}_b=\frac{1}{4}\delta^a_b\mathcal{F}^{cd}\mathcal{F}_{cd}-\mathcal{F}^{ac}\mathcal{F}_{bc},$	$h\tau^a{}_b=\tfrac{1}{4}\delta^a_b\mathcal{F}^{cd}\mathcal{F}_{cd}-\mathcal{F}^{ac}\mathcal{F}_{bc},$
1510.06699_ EQ0087 (93)	018	■ 1.000 ● N +0	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(\widehat{h},\widehat{A},\partial\widehat{A},\partial\widehat{B})+\mathcal{L}_{\mathcal{M}}(\widehat{\varphi},\partial\widehat{\varphi},\phi_0,0,\widehat{h},\partial\widehat{h},\widehat{A},\partial\widehat{A},\widehat{B}).$	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(\widehat{h},\widehat{A},\partial\widehat{A},\partial\widehat{B})+\mathcal{L}_{\mathcal{M}}(\widehat{\varphi},\partial\widehat{\varphi},\phi_0,0,\widehat{h},\partial\widehat{h},\widehat{A},\partial\widehat{A},\widehat{B}).$	$\mathcal{L}_{\mathcal{T}}=\mathcal{L}_{\mathcal{G}}(\widehat{h},\widehat{A},\partial\widehat{A},\partial\widehat{B})+\mathcal{L}_{\mathcal{M}}(\widehat{\varphi},\partial\widehat{\varphi},\phi_0,0,\widehat{h},\partial\widehat{h},\widehat{A},\partial\widehat{A},\widehat{B}).$
1510.06699_ EQ0088 (91d)	018	■ 1.000 ● N +0	$\left.\frac{\partial\mathcal{L}_{\mathcal{M}}}{\partial\phi}\right _{\phi=\phi_0}=-h_a{}^\mu\left.\frac{\partial\mathcal{L}_{\mathcal{M}}}{\partial h_a{}^\mu}\right _{\phi=\phi_0}+\partial_\mu\left(\frac{\partial\mathcal{L}_{\mathcal{M}}}{\partial B_\mu}\right)_{\phi=\phi_0},$	$\phi_0\left.\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta\phi}\right _{\phi=\phi_0}=-h_a{}^\mu\left.\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta h_a{}^\mu}\right _{\phi=\phi_0}+\partial_\mu\left(\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta B_\mu}\right)_{\phi=\phi_0},$	$\phi_0\left.\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta\phi}\right _{\phi=\phi_0}=-h_a{}^\mu\left.\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta h_a{}^\mu}\right _{\phi=\phi_0}+\partial_\mu\left(\frac{\delta\mathcal{L}_{\mathcal{M}}}{\delta B_\mu}\right)_{\phi=\phi_0},$
1510.06699_ EQ0090 (96)	019	■ 1.000 ● N +2	$S=-\int d\lambda\left[p_av^a-\frac{1}{2}e\left(p_ap^a-\mu^2\phi^2\right)\right]$	$S=-\int d\lambda\left[p_av^a-\frac{1}{2}e\left(p_ap^a-\mu^2\phi^2\right)\right]$	$S=-\int d\lambda\left[p_av^a-\frac{1}{2}e\left(p_ap^a-\mu^2\phi^2\right)\right],$
1510.06699_ EQ0092 (100)	019	■ 1.000 ● K +0	$v^c\left(\mathcal{D}_c^*p_a-\mathcal{T}_{cab}^*p^b\right)=e\mu^2\phi\mathcal{D}_a^*\phi.$	$v^c\left(\mathcal{D}_c^*p_a-\mathcal{T}_{cab}^*p^b\right)=e\mu^2\phi\mathcal{D}_a^*\phi.$	$v^c(\mathcal{D}_c^*p_a-\mathcal{T}_{cab}^*p^b)=e\mu^2\phi\mathcal{D}_a^*\phi.$
1510.06699_ EQ0096 (104)	019	■ 1.000 ● N +0	$v^{\{b\}\{ \}^{\{0\}}}\mathcal{D}_{\{b\}}v_{\{a\}}=0$	$v^{b\,0}\mathcal{D}_bv_a=0.$	$v^{b\,0}\mathcal{D}_bv_a=0.$
1510.06699_ EQ0098 (106)	020	■ 1.000 ● N +1	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right]$	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right]$	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right],$
1510.06699_ EQ0099 (107)	020	■ 1.000 ● N +0	$\widehat{v}^{\{b\}\{ \}^{\{0\}}}\widehat{\mathcal{D}}_{\{b\}}\widehat{v}_{\{a\}}=0$	$\widehat{v}^{b\,0}\widehat{\mathcal{D}}_b\widehat{v}_a=0,$	$\widehat{v}^{b\,0}\widehat{\mathcal{D}}_b\widehat{v}_a=0,$
1510.06699_ EQ0100 (108)	020	■ 1.000 ● N +0	$\left[\{ \}^{\{0\}}D_{\mu}^*,\{ \}^{\{0\}}D_{\nu}^*\right]\varphi=\frac{1}{2}{}^0R^{*ab}{}_{\mu\nu}\Sigma_{ab}\varphi+wH_{\mu\nu}\varphi,$	$[{}^0D_{\mu}^*,{}^0D_{\nu}^*]\varphi=\tfrac{1}{2}{}^0R^{*ab}{}_{\mu\nu}\Sigma_{ab}\varphi+wH_{\mu\nu}\varphi,$	$[{}^0D_{\mu}^*,{}^0D_{\nu}^*]\varphi=\tfrac{1}{2}{}^0R^{*ab}{}_{\mu\nu}\Sigma_{ab}\varphi+wH_{\mu\nu}\varphi,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00138 (152)	026	■ 1.000 ● N +0	$H_{\mu\nu}^{\dagger}=\partial_{\mu}\left(V_{\nu}+\frac{1}{3}T_{\nu}\right)-\partial_{\nu}\left(V_{\mu}+\frac{1}{3}T_{\mu}\right)$	$H_{\mu\nu}^{\dagger}=\partial_{\mu}\left(V_{\nu}+\frac{1}{3}T_{\nu}\right)-\partial_{\nu}\left(V_{\mu}+\frac{1}{3}T_{\mu}\right).$	$H_{\mu\nu}^{\dagger}=\partial_{\mu}(V_{\nu}+\frac{1}{3}T_{\nu})-\partial_{\nu}(V_{\mu}+\frac{1}{3}T_{\mu}).$
1510.06699_ E00139 (153)	026	■ 1.000 ● N +1	$\left[\mathrm{d}\right]_{\mathrm{c}}^{\dagger}\mathrm{d}^{\dagger}\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}^{\dagger}_{cd}\varphi-\mathcal{T}^{\dagger a}_{cd}\mathcal{D}^{\dagger}_a\varphi,$	$\left[\mathcal{D}_c^{\dagger},\mathcal{D}_d^{\dagger}\right]\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}^{\dagger}_{cd}\varphi-\mathcal{T}^{\dagger a}_{cd}\mathcal{D}^{\dagger}_a\varphi,$	$[\mathcal{D}_c^{\dagger},\mathcal{D}_d^{\dagger}]\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}^{\dagger}_{cd}\varphi-\mathcal{T}^{\dagger a}_{cd}\mathcal{D}^{\dagger}_a\varphi,$
1510.06699_ E00140 (154)	027	■ 1.000 ● N +1	$\mathrm{t}^{\dagger}_{\mathrm{a}}\mathrm{b}^{\dagger}_{\mathrm{c}}\equiv h^{\mu}_{\mathrm{b}}h^{\nu}_{\mathrm{c}}\left(D^{\dagger}_{\mu}b^a_{\nu}-D^{\dagger}_{\nu}b^a_{\mu}\right)\equiv h^{\mu}_{\mathrm{b}}h^{\nu}_{\mathrm{c}}T^{\dagger a}_{\mu\nu}.$	$\mathcal{T}^{\dagger a}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\dagger}b^a{}_{\nu}-D_{\nu}^{\dagger}b^a{}_{\mu}\right)\equiv h_b{}^{\mu}h_c{}^{\nu}T^{\dagger a}{}_{\mu\nu}.$	$\mathcal{T}^{\dagger a}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}(D_{\mu}^{\dagger}b^a{}_{\nu}-D_{\nu}^{\dagger}b^a{}_{\mu})\equiv h_b{}^{\mu}h_c{}^{\nu}T^{\dagger a}{}_{\mu\nu}.$
1510.06699_ E00142 (156)	027	■ 1.000 ● N +1	$\mathrm{t}^{\dagger}_{\mathrm{a}}\mathrm{b}^{\dagger}_{\mathrm{c}}=\mathrm{t}^{\dagger}_{\mathrm{a}}\mathrm{b}^{\dagger}_{\mathrm{c}}+\frac{1}{3}\left(\delta^a_b\mathcal{T}_c-\delta^a_c\mathcal{T}_b\right).$	$\mathcal{T}^{\dagger a}_{bc}=\mathcal{T}^a_{bc}+\frac{1}{3}\left(\delta^a_b\mathcal{T}_c-\delta^a_c\mathcal{T}_b\right).$	$\mathcal{T}^{\dagger a}_{bc}=\mathcal{T}^a_{bc}+\frac{1}{3}(\delta^a_b\mathcal{T}_c-\delta^a_c\mathcal{T}_b).$
1510.06699_ E00143 (157)	027	■ 1.000 ● N +1	$\mathrm{t}^{\dagger}_{\mathrm{b}}\equiv \mathrm{t}^{\dagger a}_{ba}=0,$	$\mathcal{T}_b^{\dagger}\equiv\mathcal{T}^{\dagger a}{}_{ba}=0,$	$\mathcal{T}_b^{\dagger}\equiv\mathcal{T}^{\dagger a}{}_{ba}=0,$
1510.06699_ E00146 (160)	027	■ 1.000 ● N +0	$\mathrm{d}^{\dagger}_{\mathrm{a}}\varphi=\left({}^0\mathcal{D}^{\dagger}_{\mathrm{a}}+\frac{1}{2}\mathcal{K}^{\dagger bc}_{\mathrm{a}}\Sigma_{bc}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=\left({}^0\mathcal{D}_a^{\dagger}+\frac{1}{2}\mathcal{K}^{\dagger bc}{}_a\Sigma_{bc}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=\left({}^0\mathcal{D}_a^{\dagger}+\frac{1}{2}\mathcal{K}^{\dagger bc}{}_a\Sigma_{bc}\right)\varphi,$
1510.06699_ E00148 (162)	028	■ 1.000 ● N +0	$\mathrm{d}^{\dagger}_{\mathrm{a}}\mathcal{R}^{\dagger ae}_{bc}-2\mathcal{D}^{\dagger}_{[b}\mathcal{R}^{\dagger e}_{c]}\mathrm{d}^{\dagger}_{\mathrm{a}}-2\mathcal{T}^{\dagger f}_{a[b}\mathcal{R}^{\dagger ae}_{c]f}-\mathcal{T}^{\dagger f}_{bc}\mathcal{R}^{\dagger e}_{f\mathrm{a}}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{R}^{\dagger ae}{}_{bc}-2\mathcal{D}^{\dagger}_{[b}\mathcal{R}^{\dagger e}{}_{c]}\mathrm{d}^{\dagger}_{\mathrm{a}}-2\mathcal{T}^{\dagger f}{}_{a[b}\mathcal{R}^{\dagger ae}{}_{c]f}-\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger e}{}_{f\mathrm{a}}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{R}^{\dagger ae}{}_{bc}-2\mathcal{D}^{\dagger}_{[b}\mathcal{R}^{\dagger e}{}_{c]}\mathrm{d}^{\dagger}_{\mathrm{a}}-2\mathcal{T}^{\dagger f}{}_{a[b}\mathcal{R}^{\dagger ae}{}_{c]f}-\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger e}{}_{f\mathrm{a}}=0.$
1510.06699_ E00149 (163)	028	■ 1.000 ● W +2	$\mathrm{d}^{\dagger}_{\mathrm{a}}\left(\mathcal{R}^{\dagger a}_{\mathrm{c}}-\frac{1}{2}\delta^a_c\mathcal{R}^{\dagger}\right)+\mathcal{T}^{\dagger f}_{bc}\mathcal{R}^{\dagger b}_{f\mathrm{a}}+\frac{1}{2}\mathcal{T}^{\dagger f}_{ab}\mathcal{R}^{\dagger ab}_{cf}=0.$	$\mathcal{D}_a^{\dagger}\left(\mathcal{R}^{\dagger a}{}_c-\frac{1}{2}\delta^a_c\mathcal{R}^{\dagger}\right)+\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger b}{}_f+\frac{1}{2}\mathcal{T}^{\dagger f}{}_{ab}\mathcal{R}^{\dagger ab}{}_{cf}=0.$	$\mathcal{D}_a^{\dagger}(\mathcal{R}^{\dagger a}{}_c-\frac{1}{2}\delta^a_c\mathcal{R}^{\dagger})+\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger b}{}_f+\frac{1}{2}\mathcal{T}^{\dagger f}{}_{ab}\mathcal{R}^{\dagger ab}{}_{cf}=0.$
1510.06699_ E00150 (164)	028	■ 1.000 ● N +1	$\mathrm{d}^{\dagger}_{\mathrm{a}}\mathcal{T}^{\dagger a}_{bc}+2\mathcal{R}^{\dagger}_{[bc]}+2\mathcal{H}^{\dagger}_{bc}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{T}^{\dagger a}{}_{bc}+2\mathcal{R}^{\dagger}_{[bc]}+2\mathcal{H}^{\dagger}_{bc}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{T}^{\dagger a}{}_{bc}+2\mathcal{R}^{\dagger}_{[bc]}+2\mathcal{H}^{\dagger}_{bc}=0.$
1510.06699_ E00151 (165)	028	■ 1.000 ● N +1	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}^{\dagger}_{abcd},\mathcal{T}^{\dagger}_{abc},\mathcal{H}^{\dagger}_{ab}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}^{\dagger}_{abcd},\mathcal{T}^{\dagger}_{abc},\mathcal{H}^{\dagger}_{ab}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}(\mathcal{R}^{\dagger}_{abcd},\mathcal{T}^{\dagger}_{abc},\mathcal{H}^{\dagger}_{ab})d^4x,$
1510.06699_ E00156 (170)	029	■ 1.000 ● N +0	$\tau^{\dagger}_{ab}\equiv\tau_{ab}+2\sigma_{cba}\mathcal{V}^c-2\sigma_{ac}{}^c\mathcal{V}_b,$	$\tau^{\dagger}_{ab}\equiv\tau_{ab}+2\sigma_{cba}\mathcal{V}^c-2\sigma_{ac}{}^c\mathcal{V}_b,$	$\tau^{\dagger}_{ab}\equiv\tau_{ab}+2\sigma_{cba}\mathcal{V}^c-2\sigma_{ac}{}^c\mathcal{V}_b,$
1510.06699_ E00157 (171)	029	■ 1.000 ● N +0	$t^{\dagger a}_{\mathrm{b}}+\tau^{\dagger a}_{\mathrm{b}}=0.$	$t^{\dagger a}{}_b+\tau^{\dagger a}{}_b=0.$	$t^{\dagger a}{}_b+\tau^{\dagger a}{}_b=0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0158 (168a)	029	■ 1.000 ● W +0	$\begin{aligned} & \frac{\delta L_{\mathrm{M}}}{\delta \varphi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \varphi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^\dagger \varphi)} \right) = 0, \\ & \frac{\delta L_{\mathrm{M}}}{\delta \phi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \phi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \phi)} \right) = 0, \end{aligned}$	$\frac{\delta L_{\mathrm{M}}}{\delta \varphi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \varphi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^\dagger \varphi)} \right) = 0,$ $\frac{\delta L_{\mathrm{M}}}{\delta \phi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \phi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \phi)} \right) = 0,$	$\frac{\delta L_{\mathrm{M}}}{\delta \varphi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \varphi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^\dagger \varphi)} \right) = 0,$ $\frac{\delta L_{\mathrm{M}}}{\delta \phi} = \frac{\bar{\partial} L_{\mathrm{M}}}{\partial \phi} - \mathcal{D}_a^\dagger \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \phi)} \right) = 0,$
1510.06699_ EQ0160 (175)	029	■ 1.000 ● N +0	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b.$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b.$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b.$
1510.06699_ EQ0164 (180)	030	■ 1.000 ● N +0	$\mathcal{D}_a^\natural \varphi \equiv h_a{}^\mu D_\mu^\natural \varphi \equiv h_a{}^\mu \left(\partial_\mu + \frac{1}{2} A^{\dagger bc}{}_\mu \Sigma_{bc} \right) \varphi.$	$\mathcal{D}_a^\natural \varphi \equiv h_a{}^\mu D_\mu^\natural \varphi \equiv h_a{}^\mu \left(\partial_\mu + \frac{1}{2} A^{\dagger bc}{}_\mu \Sigma_{bc} \right) \varphi.$	$\mathcal{D}_a^\natural \varphi \equiv h_a{}^\mu D_\mu^\natural \varphi \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger bc}{}_\mu \Sigma_{bc}) \varphi.$
1510.06699_ EQ0166 (182)	031	■ 1.000 ● N +0	$\mathcal{D}_a^\dagger \varphi = \left(\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural \right) \varphi,$	$\mathcal{D}_a^\dagger \varphi = \left(\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural \right) \varphi,$	$\mathcal{D}_a^\dagger \varphi = (\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural) \varphi,$
1510.06699_ EQ0169 (185)	031	■ 1.000 ● N +0	$\zeta^a = 2\sigma^{ab}{}_b,$	$\zeta^a = 2\sigma^{ab}{}_b,$	$\zeta^a = 2\sigma^{ab}{}_b,$
1510.06699_ EQ0172 (191)	032	■ 1.000 ● N +0	$i\gamma^a \mathcal{D}_a^\dagger \psi - \mu \phi \psi = 0.$	$i\gamma^a \mathcal{D}_a^\dagger \psi - \mu \phi \psi = 0.$	$i\gamma^a \mathcal{D}_a^\dagger \psi - \mu \phi \psi = 0.$
1510.06699_ EQ0173 (189)	032	■ 1.000 ● N +0	$\tau^a{}_b = \frac{1}{2} i h^{-1} \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a \mathcal{L}_\mathrm{D} - 2\sigma_{cb}{}^a \mathcal{V}^c,$	$\tau^a{}_b = \frac{1}{2} i h^{-1} \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a \mathcal{L}_\mathrm{D} - 2\sigma_{cb}{}^a \mathcal{V}^c,$	$\tau^a{}_b = \frac{1}{2} i h^{-1} \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a \mathcal{L}_\mathrm{D} - 2\sigma_{cb}{}^a \mathcal{V}^c,$
1510.06699_ EQ0174 (190)	032	■ 1.000 ● N +1	$\tau'^a{}_b = \tau^a{}_b - 2\theta \sigma_{cd}{}^a \mathcal{P}^c \delta_b^d$	$\tau'^a{}_b = \tau^a{}_b - 2\theta \sigma_{cd}{}^a \mathcal{P}^c \delta_b^d$	$\tau'^a{}_b = \tau^a{}_b - 2\theta \sigma_{cd}{}^a \mathcal{P}^c \delta_b^d.$
1510.06699_ EQ0177 (193)	032	■ 1.000 ● K +0	$\mathcal{L}_\mathrm{M} = h^{-1} L_\mathrm{M} = -h^{-1} \left(\frac{1}{4} \mathcal{F}_{ab} \mathcal{F}^{ab} + \mathcal{J}^a \mathcal{A}_a \right),$	$\mathcal{L}_\mathrm{M} = h^{-1} L_\mathrm{M} = -h^{-1} \left(\frac{1}{4} \mathcal{F}_{ab} \mathcal{F}^{ab} + \mathcal{J}^a \mathcal{A}_a \right),$	$\mathcal{L}_\mathrm{M} = h^{-1} L_\mathrm{M} = -h^{-1} (\frac{1}{4} \mathcal{F}_{ab} \mathcal{F}^{ab} + \mathcal{J}^a \mathcal{A}_a),$
1510.06699_ EQ0179 (195)	033	■ 1.000 ● N +1	$\mathcal{D}_{[a}^\dagger \mathcal{F}_{bc]} - \mathcal{T}^{\dagger d}{}_{[ab} \mathcal{F}_{c]d} = 0.$	$\mathcal{D}_{[a}^\dagger \mathcal{F}_{bc]} - \mathcal{T}^{\dagger d}{}_{[ab} \mathcal{F}_{c]d} = 0.$	$\mathcal{D}_{[a}^\dagger \mathcal{F}_{bc]} - \mathcal{T}^{\dagger d}{}_{[ab} \mathcal{F}_{c]d} = 0.$
1510.06699_ EQ0181 (197)	033	■ 1.000 ● N +0	$\mathcal{D}_a^\dagger \chi = \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^\dagger \widehat{\chi},$	$\mathcal{D}_a^\dagger \chi = \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^\dagger \widehat{\chi},$	$\mathcal{D}_a^\dagger \chi = \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_a^\dagger \widehat{\chi},$
1510.06699_ EQ0186 (202)	035	■ 1.000 ● K +0	$\left[{}^0\mathcal{D}_c^\dagger, {}^0\mathcal{D}_d^\dagger \right] \varphi = \frac{1}{2} {}^0\mathcal{R}^{\dagger ab}{}_{cd} \Sigma_{ab} \varphi + w \mathcal{H}_{cd}^\dagger \varphi,$	$\left[{}^0\mathcal{D}_c^\dagger, {}^0\mathcal{D}_d^\dagger \right] \varphi = \frac{1}{2} {}^0\mathcal{R}^{\dagger ab}{}_{cd} \Sigma_{ab} \varphi + w \mathcal{H}_{cd}^\dagger \varphi,$	$[{}^0\mathcal{D}_c^\dagger, {}^0\mathcal{D}_d^\dagger] \varphi = \frac{1}{2} {}^0\mathcal{R}^{\dagger ab}{}_{cd} \Sigma_{ab} \varphi + w \mathcal{H}_{cd}^\dagger \varphi,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00193 (209)	036	■ 1.000 ● N +0	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\dagger}+D_{\mu}^{\dagger}-\partial_{\mu}^{\dagger},$	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\dagger}+D_{\mu}^{\dagger}-\partial_{\mu}^{\dagger},$	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\dagger}+D_{\mu}^{\dagger}-\partial_{\mu}^{\dagger},$
1510.06699_ E00194 (210)	036	■ 1.000 ● N +0	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-\Gamma^{\sigma}{}_{\nu\mu}e^a{}_{\sigma}+A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0,$	$\Delta_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-\Gamma^{\sigma}{}_{\nu\mu}e^a{}_{\sigma}+A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0,$	$\Delta_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-\Gamma^{\sigma}{}_{\nu\mu}e^a{}_{\sigma}+A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0,$
1510.06699_ E00195 (211)	036	■ 1.000 ● K +0	$\nabla_{\sigma}g_{\mu\nu}=2\left(V_{\sigma}+\frac{1}{3}T_{\sigma}\right)g_{\mu\nu}.$	$\nabla_{\sigma}g_{\mu\nu}=2\left(V_{\sigma}+\frac{1}{3}T_{\sigma}\right)g_{\mu\nu}.$	$\nabla_{\sigma}g_{\mu\nu}=2(V_{\sigma}+\frac{1}{3}T_{\sigma})g_{\mu\nu}.$
1510.06699_ E00197 (215)	037	■ 1.000 ● K +0	$\Gamma^{\lambda}{}_{\mu\lambda}=\Gamma^{\lambda}{}_{\lambda\mu}.$	$\Gamma^{\lambda}{}_{\mu\lambda}=\Gamma^{\lambda}{}_{\lambda\mu}.$	$\Gamma^{\lambda}{}_{\mu\lambda}=\Gamma^{\lambda}{}_{\lambda\mu}.$
1510.06699_ E00198 (216)	037	■ 1.000 ● N +0	$\widehat{R}^{\rho}{}_{\sigma\mu\nu}\equiv R^{\dagger\rho}{}_{\sigma\mu\nu}-H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma}.$	$\widehat{R}^{\rho}{}_{\sigma\mu\nu}\equiv R^{\dagger\rho}{}_{\sigma\mu\nu}-H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma}.$	$\widehat{R}^{\rho}{}_{\sigma\mu\nu}\equiv R^{\dagger\rho}{}_{\sigma\mu\nu}-H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma}.$
1510.06699_ E00199 (217)	037	■ 1.000 ● N +0	$[\nabla_{\mu}^{\dagger},\nabla_{\nu}^{\dagger}]J^{\rho}=\widehat{R}^{\rho}{}_{\sigma\mu\nu}J^{\sigma}-wH^{\dagger}{}_{\mu\nu}J^{\rho}-T^{\dagger\sigma}{}_{\mu\nu}\nabla_{\sigma}^{\dagger}V^{\rho}.$	$[\nabla_{\mu}^{\dagger},\nabla_{\nu}^{\dagger}]J^{\rho}=\widehat{R}^{\rho}{}_{\sigma\mu\nu}J^{\sigma}-wH^{\dagger}{}_{\mu\nu}J^{\rho}-T^{\dagger\sigma}{}_{\mu\nu}\nabla_{\sigma}^{\dagger}V^{\rho}.$	$[\nabla_{\mu}^{\dagger},\nabla_{\nu}^{\dagger}]J^{\rho}=\widehat{R}^{\rho}{}_{\sigma\mu\nu}J^{\sigma}-wH^{\dagger}{}_{\mu\nu}J^{\rho}-T^{\dagger\sigma}{}_{\mu\nu}\nabla_{\sigma}^{\dagger}V^{\rho}.$
1510.06699_ E00200 (218)	037	■ 1.000 ● N +0	$\Gamma^{\lambda}{}_{\mu\nu}={}^0\Gamma^{\dagger\lambda}{}_{\mu\nu}+K^{\dagger\lambda}{}_{\mu\nu},$	$\Gamma^{\lambda}{}_{\mu\nu}={}^0\Gamma^{\dagger\lambda}{}_{\mu\nu}+K^{\dagger\lambda}{}_{\mu\nu},$	$\Gamma^{\lambda}{}_{\mu\nu}={}^0\Gamma^{\dagger\lambda}{}_{\mu\nu}+K^{\dagger\lambda}{}_{\mu\nu},$
1510.06699_ E00202 (220)	037	■ 1.000 ● K +0	$K^{\dagger\lambda}{}_{\mu\nu}=-\frac{1}{2}\left(T^{\dagger\lambda}{}_{\mu\nu}-T^{\dagger}{}_{\nu}{}^{\lambda}{}_{\mu}+T^{\dagger}{}_{\mu\nu}{}^{\lambda}{}_{}\right).$	$K^{\dagger\lambda}{}_{\mu\nu}=-\frac{1}{2}\left(T^{\dagger\lambda}{}_{\mu\nu}-T^{\dagger}{}_{\nu}{}^{\lambda}{}_{\mu}+T^{\dagger}{}_{\mu\nu}{}^{\lambda}{}_{}\right).$	$K^{\dagger\lambda}{}_{\mu\nu}=-\frac{1}{2}(T^{\dagger\lambda}{}_{\mu\nu}-T^{\dagger}{}_{\nu}{}^{\lambda}{}_{\mu}+T^{\dagger}{}_{\mu\nu}{}^{\lambda}{}_{}).$
1510.06699_ E00206 (228)	038	■ 1.000 ● K +0	$\left(t_{\mathrm{R}}^{\dagger}\right)_b+\left(t_{\mathcal{H}^2}^{\dagger}\right)_b+(\tau_{\varphi}^{\dagger})_b+(\tau_{\phi}^{\dagger})_b+(\tau_{\mathcal{R}}^{\dagger})_b+(\tau_{\mathcal{T}^2}^{\dagger})_b=0,$	$\left(t_{\mathcal{R}^2}^{\dagger}\right)_b+\left(t_{\mathcal{H}^2}^{\dagger}\right)_b+(\tau_{\varphi}^{\dagger})_b+(\tau_{\phi}^{\dagger})_b+(\tau_{\mathcal{R}}^{\dagger})_b+(\tau_{\mathcal{T}^2}^{\dagger})_b=0,$	$(t_{\mathcal{R}^2}^{\dagger})_b+(t_{\mathcal{H}^2}^{\dagger})_b+(\tau_{\varphi}^{\dagger})_b+(\tau_{\phi}^{\dagger})_b+(\tau_{\mathcal{R}}^{\dagger})_b+(\tau_{\mathcal{T}^2}^{\dagger})_b=0,$
1510.06699_ E00212 (240)	039	■ 1.000 ● N +0	$(j_{\mathcal{R}^2})_a+(j_{\mathcal{H}^2})_a+(\zeta_{\varphi})_a+(\zeta_{\phi})_a+(\zeta_{\mathcal{R}})_a+(\zeta_{\mathcal{T}^2})_a=0,$	$(j_{\mathcal{R}^2})_a+(j_{\mathcal{H}^2})_a+(\zeta_{\varphi})_a+(\zeta_{\phi})_a+(\zeta_{\mathcal{R}})_a+(\zeta_{\mathcal{T}^2})_a=0,$	$(j_{\mathcal{R}^2})_a+(j_{\mathcal{H}^2})_a+(\zeta_{\varphi})_a+(\zeta_{\phi})_a+(\zeta_{\mathcal{R}})_a+(\zeta_{\mathcal{T}^2})_a=0,$
1510.06699_ E00219 (253)	041	■ 1.000 ● N +0	$\mathcal{D}'_c\varphi'=e^{(w-1)\rho}\left[\mathcal{D}_c\varphi+w\mathcal{P}_c\varphi+\theta\mathcal{P}^{[b}\delta_c^{a]}\Sigma_{ab}\varphi\right],$	$\mathcal{D}'_c\varphi'=e^{(w-1)\rho}\left[\mathcal{D}_c\varphi+w\mathcal{P}_c\varphi+\theta\mathcal{P}^{[b}\delta_c^{a]}\Sigma_{ab}\varphi\right],$	$\mathcal{D}'_c\varphi'=e^{(w-1)\rho}[\mathcal{D}_c\varphi+w\mathcal{P}_c\varphi+\theta\mathcal{P}^{[b}\delta_c^{a]}\Sigma_{ab}\varphi],$
1510.06699_ E00224 (258)	042	■ 1.000 ● N +0	$\frac{1}{2}\nu=-6a\theta=3(\theta-1)(2\beta_1+\beta_2+3\beta_3),$	$\frac{1}{2}\nu=-6a\theta=3(\theta-1)(2\beta_1+\beta_2+3\beta_3),$	$\frac{1}{2}\nu=-6a\theta=3(\theta-1)(2\beta_1+\beta_2+3\beta_3),$
1510.06699_ E00225 (259)	042	■ 1.000 ● K +0	$\mathcal{L}'_{\mathrm{M}}=\mathcal{L}_{\mathrm{M}}+6a\theta h^{-1}(\mathcal{D}_a+\mathcal{T}_a)(\phi^2\mathcal{P}^a).$	$\mathcal{L}'_{\mathrm{M}}=\mathcal{L}_{\mathrm{M}}+6a\theta h^{-1}(\mathcal{D}_a+\mathcal{T}_a)(\phi^2\mathcal{P}^a).$	$\mathcal{L}'_{\mathrm{M}}=\mathcal{L}_{\mathrm{M}}+6a\theta h^{-1}(\mathcal{D}_a+\mathcal{T}_a)(\phi^2\mathcal{P}^a).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1510.06699_ E00232 (266)	043	■ 1.000 ● K +0	$L_{\{\mathrm{G}\}=2\,\alpha\left(\{ \}^{\{0\}}\,\mathrm{cal}\{R\}_{\{a\,b\}}^{\{ \}^{\{0\}}\,\mathrm{cal}\{R\}^{\{a\,b\}}-\frac{1}{3}\{ \}^{\{0\}}\,\mathrm{cal}\{R\}^{\{2\}}\right)}$	$L_G=2\alpha\left({}^0\mathcal{R}_{ab}{}^0\mathcal{R}^{ab}-\frac{1}{3}{}^0\mathcal{R}^2\right),$	$L_G=2\alpha({}^0\mathcal{R}_{ab}{}^0\mathcal{R}^{ab}-\tfrac{1}{3}{}^0\mathcal{R}^2),$
1510.06699_ E00236 (270)	044	■ 1.000 ● N +0	$\mathrm{cal}\{L_{\{\mathrm{M}\}}^{\{\prime\}}=\mathrm{cal}\{L_{\{\mathrm{M}\}}\}-h^{\{-1\}}\left[\frac{1}{2}\nu^0\mathcal{D}_a\left(\phi^2\mathcal{P}^a\right)\right],$	$\mathcal{L}'_{\mathrm{M}}=\mathcal{L}_{\mathrm{M}}-h^{-1}\left[\frac{1}{2}\nu^0\mathcal{D}_a\left(\phi^2\mathcal{P}^a\right)\right],$	$\mathcal{L}'_{\mathrm{M}}=\mathcal{L}_{\mathrm{M}}-h^{-1}[\tfrac{1}{2}\nu^0\mathcal{D}_a(\phi^2\mathcal{P}^a)],$
1510.06699_ E00237 (A1)	047	■ 1.000 ● C +4	$L_{\{\mathrm{D}\}}=\frac{1}{2}\,i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-\left(\partial_{\mu}\bar{\psi}\right)\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\Re(i\bar{\psi}\,\partial\psi)-m\bar{\psi}\psi,$	$L_{\mathrm{D}}=\tfrac{1}{2}i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-\left(\partial_{\mu}\bar{\psi}\right)\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\Re(i\bar{\psi}\,\partial\psi)-m\bar{\psi}\psi,$	$L_{\mathrm{D}}=\tfrac{1}{2}i[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-(\partial_{\mu}\bar{\psi})\gamma^{\mu}\psi]-m\bar{\psi}\psi\equiv\Re(i\bar{\psi}\partial\psi)-m\bar{\psi}\psi,$
1510.06699_ E00245 (A9)	047	■ 1.000 ● K +0	$S=-\int\mathrm{d}\,\lambda\mathrm{bda}\left[p_{\{\mu\}}\dot{x}^{\{\mu\}}-\frac{1}{2}\mathrm{e}\left(p_{\{\mu\}}\,p^{\{\mu\}}-m^2\right)\right]$	$S=-\int d\lambda\left[p_{\mu}\dot{x}^{\mu}-\frac{1}{2}e\left(p_{\mu}p^{\mu}-m^2\right)\right]$	$S=-\int d\lambda\,[p_{\mu}\dot{x}^{\mu}-\tfrac{1}{2}e(p_{\mu}p^{\mu}-m^2)],$
1510.06699_ E00246 (B1)	048	■ 1.000 ● K +0	$L_{\{\mathrm{G}\}}=-\kappa^{\{-1\}}(\Lambda+a\,\mathrm{cal}\{R\})+L_{\{\mathrm{cal}\{R\}^{\{2\}}\}+\kappa^{\{-1\}}\,L_{\{\mathrm{cal}\{T\}^{\{2\}}\}}$	$L_G=-\kappa^{-1}(\Lambda+a\mathcal{R})+L_{\mathcal{R}^2}+\kappa^{-1}L_{\mathcal{T}^2},$	$L_G=-\kappa^{-1}(\Lambda+a\mathcal{R})+L_{\mathcal{R}^2}+\kappa^{-1}L_{\mathcal{T}^2},$
1510.06699_ E00252 (C1)	049	■ 1.000 ● K +0	$\mathrm{cal}\{L_{\{\mathrm{G}\}}\}=-\frac{1}{2}\,\kappa\mathrm{a}\,h^{\{-1\}}\,\mathrm{cal}\{R\}$	$\mathcal{L}_G=-\frac{1}{2\kappa}h^{-1}\mathcal{R},$	$\mathcal{L}_G=-\frac{1}{2\kappa}h^{-1}\mathcal{R},$
1510.06699_ E00254 (C4)	050	■ 1.000 ● N +2	$\mathrm{cal}\{T^{\{c\}}\}_{\{a\,b\}}+\delta_{\{a\}^{\{c\}}}\,\mathrm{cal}\{T\}_{\{b\}}-\delta_{\{b\}^{\{c\}}}\,\mathrm{cal}\{T\}_{\{a\}}=2\,\kappa\,h\,\sigma_{\{a\,b\}}^{\{c\}}$	$\mathcal{T}^c{}_{ab}+\delta_a^c\mathcal{T}_b-\delta_b^c\mathcal{T}_a=2\kappa h\sigma_{ab}{}^c$	$\mathcal{T}^c{}_{ab}+\delta_a^c\mathcal{T}_b-\delta_b^c\mathcal{T}_a=2\kappa h\sigma_{ab}{}^c.$
1510.06699_ E00255 (C5)	050	■ 1.000 ● N +0	$\mathrm{cal}\{T\}_{\{a\}}=-\kappa\,h\,\sigma_{\{a\,b\}}^{\{ \}^{\{b\}}}$	$\mathcal{T}_a=-\kappa h\sigma_{ab}{}^b.$	$\mathcal{T}_a=-\kappa h\sigma_{ab}{}^b.$
1510.06699_ E00256 (C6)	050	■ 1.000 ● N +0	$\mathrm{cal}\{T^{\{c\}}\}_{\{a\,b\}}=\kappa\,h\left(2\,\sigma_{\{a\,b\}}^{\{c\}}+\delta_{\{a\}^{\{c\}}}\,\sigma_{\{b\,d\}}^{\{ \}^{\{d\}}}-\delta_b^c\sigma_{ad}^{\{d\}}\right).$	$\mathcal{T}^c{}_{ab}=\kappa h\left(2\sigma_{ab}{}^c+\delta_a^c\sigma_{bd}{}^d-\delta_b^c\sigma_{ad}{}^d\right).$	$\mathcal{T}^c{}_{ab}=\kappa h(2\sigma_{ab}{}^c+\delta_a^c\sigma_{bd}{}^d-\delta_b^c\sigma_{ad}{}^d).$
1510.06699_ E00259 (C9)	050	■ 1.000 ● K +0	$\mathrm{cal}\{T\}_{\{a\}}=-\kappa\,h\,\sigma_{\{a\,b\}}^{\{ \}^{\{b\}}}=0,$	$\mathcal{T}_a=-\kappa h\sigma_{ab}{}^b=0,$	$\mathcal{T}_a=-\kappa h\sigma_{ab}{}^b=0,$
1510.06699_ E00260 (C10)	050	■ 1.000 ● K +0	$i\,\gamma^a\mathrm{cal}\{D\}_{\{a\}}\,\psi-m\psi=0,$	$i\gamma^a\mathcal{D}_a\psi-m\psi=0,$	$i\gamma^a\mathcal{D}_a\psi-m\psi=0,$
1510.06699_ E00262 (D1)	051	■ 1.000 ● K +0	$L_{\{\mathrm{G}\}}=-\frac{1}{4}\,\xi\mathcal{H}_{ab}\mathcal{H}^{ab},$	$L_G=-\frac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab},$	$L_G=-\tfrac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab},$
1510.06699_ E00267 (D6)	052	■ 1.000 ● N +0	$\xi\left[\frac{1}{2}\mathcal{T}^{*c}{}_{ab}\mathcal{H}^{ab}-(\mathcal{D}_a^*+\mathcal{T}_a^*)\,\mathcal{H}^{ac}\right]-\nu\phi\mathcal{D}^{*c}\phi=h(\zeta_{\varphi})^c$	$\xi\left[\frac{1}{2}\mathcal{T}^{*c}{}_{ab}\mathcal{H}^{ab}-(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{H}^{ac}\right]-\nu\phi\mathcal{D}^{*c}\phi=h(\zeta_{\varphi})^c$	$\xi[\tfrac{1}{2}\mathcal{T}^{*c}{}_{ab}\mathcal{H}^{ab}-(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{H}^{ac}]-\nu\phi\mathcal{D}^{*c}\phi=h(\zeta_{\varphi})^c$
1510.06699_ E00268 (D7)	052	■ 1.000 ● W +1	$\frac{\partial}{\partial\varphi}\mathrm{cal}\{L_{\{\mathrm{varphi}\}}\}\{\partial\,\varphi\}-\nu\left(\mathrm{cal}\{D\}_{\{a\}}^{\{*\}}+\mathrm{cal}\{T\}_{\{a\}}^{\{*\}}\right)\mathrm{cal}\{D\}^{\{*\,a\}}\phi-4\lambda\phi^3-a\phi\mathcal{R}=0$	$\frac{\partial L_{\varphi}}{\partial\phi}-\nu(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{D}^{*a}\phi-4\lambda\phi^3-a\phi\mathcal{R}=0$	$\frac{\partial L_{\varphi}}{\partial\phi}-\nu(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{D}^{*a}\phi-4\lambda\phi^3-a\phi\mathcal{R}=0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0269 (D8)	052	1.000 N +0	$\mathcal{R}^{\mathfrak{a}}_{\mathfrak{b}}-\frac{1}{2}\delta^{\mathfrak{a}}_{\mathfrak{b}}\Lambda=\kappa\left[h(\tau_{\varphi})^{\mathfrak{a}}_{\mathfrak{b}} _{\phi=\phi_0}+\nu\left(\mathcal{B}^{\mathfrak{a}}\mathcal{B}_{\mathfrak{b}}-\frac{1}{2}\delta^{\mathfrak{a}}_{\mathfrak{b}}\mathcal{B}^c\mathcal{B}_c\right)-\xi\left(\mathcal{H}^{ac}\mathcal{H}_{bc}-\frac{1}{4}\delta^{\mathfrak{a}}_{\mathfrak{b}}\mathcal{H}_{cd}\mathcal{H}^{cd}\right)\right].$	$\mathcal{R}^a_b-\frac{1}{2}\delta^a_b\mathcal{R}-\delta^a_b\Lambda=\kappa\left[h(\tau_{\varphi})^a_b _{\phi=\phi_0}+\nu(\mathcal{B}^a\mathcal{B}_b-\frac{1}{2}\delta^a_b\mathcal{B}^c\mathcal{B}_c)-\xi(\mathcal{H}^{ac}\mathcal{H}_{bc}-\frac{1}{4}\delta^a_b\mathcal{H}_{cd}\mathcal{H}^{cd})\right].$	
1510.06699_ EQ0270 (D9)	052	1.000 K +0	$\mathcal{T}^{\mathfrak{c}}_{\mathfrak{a}\mathfrak{b}}+\delta^{\mathfrak{c}}_{\mathfrak{a}}\mathcal{T}_{\mathfrak{b}}-\delta^{\mathfrak{c}}_{\mathfrak{b}}\mathcal{T}_{\mathfrak{a}}=\left.2\kappa h(\sigma_{\varphi})_{ab}{}^c\right _{\phi=\phi_0},$	$\mathcal{T}^c_{ab}+\delta^c_a\mathcal{T}_b-\delta^c_b\mathcal{T}_a=\left.2\kappa h(\sigma_{\varphi})_{ab}{}^c\right _{\phi=\phi_0},$	
1510.06699_ EQ0271 (D10)	052	1.000 N +0	$\xi\left[\frac{1}{2}\mathcal{T}^c_{ab}\mathcal{H}^{ab}-(\mathcal{D}_a+\mathcal{T}_a)\mathcal{H}^{ac}\right]+\nu\phi_0^2\mathcal{B}^c=h(\zeta_{\varphi})^c _{\phi=\phi_0},$	$\xi\left[\frac{1}{2}\mathcal{T}^c_{ab}\mathcal{H}^{ab}-(\mathcal{D}_a+\mathcal{T}_a)\mathcal{H}^{ac}\right]+\nu\phi_0^2\mathcal{B}^c=h(\zeta_{\varphi})^c _{\phi=\phi_0},$	
1510.06699_ EQ0272 (D11)	052	1.000 N +0	$\mathcal{R}+4\Lambda=\kappa\left[\phi_0\partial_{\phi}L_{\varphi} _{\phi=\phi_0}+\nu\phi_0^2(\mathcal{D}_a+\mathcal{T}_a+\mathcal{B}_a)\mathcal{B}^a\right]$	$\mathcal{R}+4\Lambda=\kappa\left[\phi_0\partial_{\phi}L_{\varphi} _{\phi=\phi_0}+\nu\phi_0^2(\mathcal{D}_a+\mathcal{T}_a+\mathcal{B}_a)\mathcal{B}^a\right]$	
1510.06699_ EQ0274 (D13)	053	1.000 N +0	$i\gamma^a\mathcal{D}_a\psi-\mu\phi\psi=0,$	$i\gamma^a\mathcal{D}_a\psi-\mu\phi\psi=0,$	
1510.06699_ EQ0276 (E1)	053	1.000 K +0	$A^{\dagger ab}{}_{\mu}=A^{ab}{}_{\mu}+(\mathcal{V}^ab^b{}_{\mu}-\mathcal{V}^bb^a{}_{\mu}),$	$A^{\dagger ab}{}_{\mu}=A^{ab}{}_{\mu}+(\mathcal{V}^ab^b{}_{\mu}-\mathcal{V}^bb^a{}_{\mu}),$	
1510.06699_ EQ0277 (E2)	053	1.000 N +0	$\mathcal{D}^{\natural}_{\mathfrak{a}}\varphi\equiv h_a{}^{\mu}D^{\natural}_{\mu}\varphi\equiv h_a{}^{\mu}\left(\partial_{\mu}+\frac{1}{2}A^{\dagger bc}{}_{\mu}\Sigma_{bc}\right)\varphi.$	$\mathcal{D}^{\natural}_a\varphi\equiv h_a{}^{\mu}D^{\natural}_{\mu}\varphi\equiv h_a{}^{\mu}(\partial_{\mu}+\frac{1}{2}A^{\dagger bc}{}_{\mu}\Sigma_{bc})\varphi.$	
1510.06699_ EQ0278 (E3)	054	1.000 K +0	$[D^{\natural}_{\mu},D^{\natural}_{\nu}]\varphi=\frac{1}{2}R^{\dagger ab}{}_{\mu\nu}\Sigma_{ab}\varphi,$	$[D^{\natural}_{\mu},D^{\natural}_{\nu}]\varphi=\frac{1}{2}R^{\dagger ab}{}_{\mu\nu}\Sigma_{ab}\varphi,$	
1510.06699_ EQ0279 (E4)	054	1.000 K +0	$V_{\mu}^{\prime}=V_{\mu}+\theta P_{\mu},$	$V_{\mu}^{\prime}=V_{\mu}+\theta P_{\mu},$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_TAB_001	055	—	(no LaTeX source; 1777 × 1835 px region)	—	$h_a{}^\mu$$h$$b_\mu^a$$A^{ab}{}_\mu$$B_\mu$$V_\mu$$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ $D_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$$\partial_\mu^* \equiv \partial_\mu + w B_\mu$$D_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$$\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$$D_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$$D_a^{\natural} \equiv h_a{}^\mu D_\mu^{\natural} \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\natural} b^a{}_{\nu]}$$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$$\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ $L$$\mathcal{L} \equiv h^{-1} L$$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G / \delta B_\mu$$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{ab}{}^a \mathcal{V}^c - s^a{}_c \mathcal{V}_b)$$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^a{}_c \mathcal{V}_b)$$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A</i>[†]-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced <i>A</i>[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c
1510.06699_TAB_002	055	—	(no LaTeX source; 1768 × 254 px region)	—	$h_a{}^\mu$$h$$b_\mu^a$$A^{ab}{}_\mu$$B_\mu$$V_\mu$$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ $D_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$$\partial_\mu^* \equiv \partial_\mu + w B_\mu$$D_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$$\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$$D_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$$D_a^{\natural} \equiv h_a{}^\mu D_\mu^{\natural} \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\natural} b^a{}_{\nu]}$$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$$\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A</i>[†]-field)
1510.06699_TAB_003	055	—	(no LaTeX source; 485 × 232 px region)	—	$D_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_TAB_006	055	—	(no LaTeX source; 652 × 438 px region)	—	L $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_TAB_007	000	—	$\begin{gathered} \backslash\mathrm{begin}\{\mathrm{tabular}\}\{\mathrm{ll}\}\{\backslash\mathrm{em}\mathrm{Gravitational\ gauge\ field\ variables}\}\&\backslash\backslash\ \\$\{h_a\}^{\backslash\mathrm{mu}}\\$&\&\mathrm{translational\ gauge\ field}\ (\\$\\$-\mathrm{field})\ \backslash\backslash\ \\$h\\$&\&\mathrm{determinant\ of}\ \\$\{h_a\}^{\backslash\mathrm{mu}}\\$ \\ \backslash\backslash\ \\$\{b^a\}_{\backslash\mathrm{mu}}\\$&\&\mathrm{inverse}\ \\$h\\$-\mathrm{field}\ \backslash\backslash\ \\$\{A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\\$&\&\mathrm{rotational\ gauge\ field}\ (\\$\\$-\mathrm{field})\ \backslash\backslash\ \\$B_{\backslash\mathrm{mu}}\\$&\&\mathrm{WGT\ dilational\ gauge\ field}\ (\\$\\$-\mathrm{field})\backslash\backslash\ \\$V_{\backslash\mathrm{mu}}\\$&\&\mathrm{eWGT\ dilational\ gauge\ field}\ (\\$\\$-\mathrm{field})\ \backslash\backslash\ \\$\{A^{\backslash\mathrm{dagger}\ \mathrm{ab}}\}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ \{A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}+2\eta^{\mathrm{c}\{a\}\{b\}\{b\}}_{\backslash\mathrm{mu}}\ \\$h_c\ ^{\backslash\mathrm{nu}}\ V_{\backslash\mathrm{nu}}\\$&\&\mathrm{eWGT\ extended\ rotational\ gauge\ field}\ (\\$A^{\backslash\mathrm{dagger}}\\$-\mathrm{field})\backslash\backslash \\ \&\backslash\backslash\ \{\mathrm{em}\ \mathrm{Derivative\ operators}\}\&\backslash\backslash \\ \\$\mathrm{mathcal}\{D\}_a\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}\ D_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}(\backslash\mathrm{partial}_{\backslash\mathrm{mu}}+\{\frac{1}{2}\}\{A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\Sigma_{\backslash\mathrm{ab}})\\$&\&\mathrm{PGT\ (generalised)\ covariant\ derivative}\footnotemark[1]\ \backslash\backslash\ \\$\backslash\mathrm{partial}^{\backslash\mathrm{ast}}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ \backslash\mathrm{partial}_{\backslash\mathrm{mu}}+wB_{\backslash\mathrm{mu}}\\$&\&\mathrm{WGT\ `augmented'\ partial\ derivative}\ \backslash\backslash\ \\$\mathrm{mathcal}\{D\}^{\backslash\mathrm{ast}}_a\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}\ D^{\backslash\mathrm{ast}}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}(\backslash\mathrm{partial}^{\backslash\mathrm{ast}}_{\backslash\mathrm{mu}}+\{\frac{1}{2}\}\{A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\Sigma_{\backslash\mathrm{ab}})\\$&\&\mathrm{WGT\ (generalised)\ covariant\ derivative}\footnotemark[1]\ \backslash\backslash\ \\$\backslash\mathrm{partial}^{\backslash\mathrm{dagger}}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ \backslash\mathrm{partial}_{\backslash\mathrm{mu}}-w(V_{\backslash\mathrm{mu}}+\{\frac{1}{3}\}T_{\backslash\mathrm{mu}})\\$&\&\mathrm{eWGT\ `augmented'\ partial\ derivative}\ \backslash\backslash\ \\$\mathrm{mathcal}\{D\}^{\backslash\mathrm{dagger}}_a\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}\ D^{\backslash\mathrm{dagger}}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}(\backslash\mathrm{partial}^{\backslash\mathrm{dagger}}_{\backslash\mathrm{mu}}+\{\frac{1}{2}\}\{A^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\Sigma_{\backslash\mathrm{ab}})\\$&\&\mathrm{eWGT\ (generalised)\ covariant\ derivative}\footnotemark[1]\ \backslash\backslash\ \\$\mathrm{mathcal}\{D\}^{\backslash\mathrm{natural}}_a\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}\ D^{\backslash\mathrm{natural}}_{\backslash\mathrm{mu}}\ \mathrm{equiv}\ {h_a}^{\backslash\mathrm{mu}}(\backslash\mathrm{partial}_{\backslash\mathrm{mu}}+\{\frac{1}{2}\}\{A^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\Sigma_{\backslash\mathrm{ab}})\\$&\&\mathrm{eWGT\ (generalised)\ semi-covariant\ derivative}\footnotemark[1]\ \backslash\backslash\ \&\backslash\backslash\ \{\mathrm{em}\ \mathrm{Gauge\ field\ strengths}\}\&\backslash\backslash\ \\$\{\mathrm{mathcal}\{R\}^{\backslash\mathrm{ab}}\}_{\mathrm{cd}}=\{\mathrm{mathcal}\{R\}^{\backslash\mathrm{ast}\ \mathrm{ab}}\}_{\mathrm{cd}}=2\{h_a\}^{\backslash\mathrm{mu}}\ {h_b}^{\backslash\mathrm{nu}}(\backslash\mathrm{partial}_{\backslash\backslash\mathrm{mu}}\{A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{nu}}\}+\{A^a\}_{\mathrm{c}\backslash\backslash\mathrm{mu}}\{A^{\backslash\mathrm{cb}}\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{PGT\ and\ WGT\ rotational\ gauge\ field\ strength}\footnotemark[1],\ \backslash\backslash\ \\$\{\mathrm{mathcal}\{R\}^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{ab}}\}_{\mathrm{cd}}=2\{h_a\}^{\backslash\mathrm{mu}}\ {h_b}^{\backslash\mathrm{nu}}(\backslash\mathrm{partial}_{\backslash\backslash\mathrm{mu}}\{A^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{ab}}\}_{\backslash\mathrm{nu}}\}+\{A^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{a}}\}_{\mathrm{c}\backslash\backslash\mathrm{mu}}\{A^{\backslash\mathrm{dagger}\backslash\backslash\mathrm{cb}}\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{eWGT\ rotational\ gauge\ field\ strength}\footnotemark[1],\ \backslash\backslash\ \\$\{\mathrm{mathcal}\{T\}^a\}_{\mathrm{bc}}=2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\ D_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{PGT\ translational\ gauge\ field\ strength}\footnotemark[2]\backslash\backslash\ \\$\{\backslash\mathrm{cal}\ T\}^{\backslash\mathrm{ast}\ \mathrm{a}}\}_{\mathrm{bc}}=2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\ D^{\backslash\mathrm{ast}}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{WGT\ translational\ gauge\ field\ strength}\footnotemark[2]\ \backslash\backslash\ \\$\{\backslash\mathrm{cal}\ T\}^{\backslash\mathrm{dagger}\ \mathrm{a}}\}_{\mathrm{bc}}=2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\ D^{\backslash\mathrm{dagger}}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{eWGT\ translational\ gauge\ field\ strength}\footnotemark[2]\ \backslash\backslash\ \\$\{\backslash\mathrm{cal}\ T\}^{\backslash\mathrm{natural}\ \mathrm{a}}\}_{\mathrm{bc}}=2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\ D^{\backslash\mathrm{natural}}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{eWGT\ translational\ gauge\ `semi'\ field\ strength}\footnotemark[1],\backslash\backslash\ \\$\{\backslash\mathrm{cal}\ H\}_{\mathrm{ab}}=2\{h_a\}^{\backslash\mathrm{mu}}\ {h_b}^{\backslash\mathrm{nu}}\backslash\mathrm{partial}_{\backslash\backslash\mathrm{mu}}B_{\backslash\backslash\mathrm{nu}})\\$&\&\mathrm{WGT\ dilational\ gauge\ field\ strength}\ \backslash\backslash\ \\$\{\backslash\mathrm{cal}\ H\}^{\backslash\mathrm{dagger}}_{\mathrm{ab}}=2\{h_a\}^{\backslash\mathrm{mu}}\ {h_b}^{\backslash\mathrm{nu}}\backslash\mathrm{partial}_{\backslash\backslash\mathrm{mu}}(V_{\backslash\backslash\mathrm{nu}}+\{\frac{1}{3}\}T_{\backslash\backslash\mathrm{nu}})\\$&\&\mathrm{eWGT\ dilational\ gauge\ field\ strength}\ \backslash\backslash\ \&\backslash\backslash\ \{\mathrm{em}\ \mathrm{Reduced}\ \\$A\\$-\mathrm{fields\ and\ related\ quantities}\}\&\backslash\backslash\ \\$\{c^a\}_{\mathrm{bc}}\mathrm{equiv}\ 2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\backslash\mathrm{partial}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{PGT\ Ricci\ rotation\ coefficients}\ \backslash\backslash\ \\$\{c^{\backslash\mathrm{ast}\ \mathrm{a}}\}_{\mathrm{bc}}\mathrm{equiv}\ 2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\backslash\mathrm{partial}^{\backslash\mathrm{ast}}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{WGT\ Ricci\ rotation\ coefficients}\ \backslash\backslash\ \\$\{c^{\backslash\mathrm{dagger}\ \mathrm{a}}\}_{\mathrm{bc}}\mathrm{equiv}\ 2\{h_b\}^{\backslash\mathrm{mu}}\ {h_c}^{\backslash\mathrm{nu}}\backslash\mathrm{partial}^{\backslash\mathrm{dagger}}_{\backslash\backslash\mathrm{mu}}\{b^a\}_{\backslash\mathrm{nu}})\\$&\&\mathrm{eWGT\ Ricci\ rotation\ coefficients}\ \backslash\backslash\ \\$\{^{\mathrm{0}}A^{\backslash\mathrm{ab}}\}_{\backslash\mathrm{mu}}\mathrm{equiv}\ \{\frac{1}{2}\}\{b^c\}_{\backslash\mathrm{mu}}(\{c^{\backslash\mathrm{ab}}\}_{\mathrm{c}}+\{\{c^b\}_{\mathrm{c}}\}^a-\{c_c\}^{\backslash\mathrm{ab}})\\$&\&\mathrm{PGT\ reduced}\ \\$A\\$-\mathrm{field}\ (\mathrm{when}\ \\$\{\backslash\mathrm{cal}\ T\}^a\}_{\mathrm{bc}}\ \mathrm{equiv}\ 0\\$)\backslash\backslash\ \\$\{^{\mathrm{0}}A^{\backslash\mathrm{ast}\ \mathrm{ab}}\}_{\backslash\mathrm{mu}}\mathrm{equiv}\ \{\frac{1}{2}\}\{b^c\}_{\backslash\mathrm{mu}}(\{c^{\backslash\mathrm{ast}\ \mathrm{ab}}\}_{\mathrm{c}}+\{\{c^{\backslash\mathrm{ast}\ \mathrm{b}}\}_{\mathrm{c}}\}^a-\{c^{\backslash\mathrm{ast}\ \mathrm{c}}\}^{\backslash\mathrm{ab}})\\$&\&\mathrm{WGT\ reduced}\ \\$A\\$-\mathrm{field}\ (\mathrm{when}\ \\$\{\backslash\mathrm{cal}\ T\}^{\backslash\mathrm{ast}\ \mathrm{a}}\}_{\mathrm{bc}}\ \mathrm{equiv}\ 0\\$)\backslash\backslash\ \\$\{^{\mathrm{0}}A^{\backslash\mathrm{dagger}\ \mathrm{ab}}\}_{\backslash\mathrm{mu}}\mathrm{equiv}\ \{\frac{1}{2}\}\{b^c\}_{\backslash\mathrm{mu}}\end{gathered}$	(not rendered)	—

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Image regions — rendered against scan

Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
1510.06699_PIC_0001	055	— <i>(dup)</i>	<i>tex.zip holds no images</i>	<i>(no tex.zip crop for this row)</i>	<i>(image not on disk — pass -texzip or download the CDN crop)</i>	<i>(image not on disk — pass -texzip or download the CDN crop)</i>
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